

New River Valley MPO Travel Demand Model

Model Development Technical Documentation

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New River Valley



The NRV MPO Model was created by WSP for the Virginia Department of Transportation and the New River Valley Metropolitan Planning Organization

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1 INTRODUCTION

This report describes the development of the New River Valley MPO (NRV) travel demand model. Development of the NRV model was sponsored by the Virginia Department of Transportation (VDOT) and the NRV MPO in a collaborative effort.

The main objective while developing the new NRV travel demand model was to provide the information necessary to assist in identification of transportation planning needs for corridor studies, access studies and the long-range transportation plan.

The NRV model is a traditional four step model that includes trip generation, trip distribution, mode choice and traffic assignment. Additional components that are included in the model include a household disaggregation model which develops household distributions required by the trip generation as well as a separate commercial vehicle model, an external model and a highway project network manager. The model includes a Time of Day structure that includes four periods: AM peak (7AM to 8:15AM), midday (8:15AM to 4:30PM), PM peak (4:30PM to 5:30PM) and overnight. The NRV model also includes a feedback loop, where congested travel time skims leaving the highway assignment module are passed back into the trip distribution and mode choice modules. It is based on the Virginia Transportation Modeling (VTM) Policies and Procedures Manual, Version 2.00 (2014)¹; hereafter referred to as the VDOT Guidelines.

The NRV Model v3 model has an overall Root Mean Square Error (RMSE) of 33% and is below the acceptable standards in all volume groups and facility types.

The NRV model is developed in TransCAD v8 build 222180 and has been calibrated to a base year of 2017. It includes a forecast year scenario for 2045. The model was tested successfully with various future year scenarios in order to assure the model was sensitive to changes in transportation supply and demand. An existing plus committed network and long range planning network for 2045 are also included.

The model set and Users' Guide are available on a GitHub site. Access should be requested through VDOT. This documentation contains information about the development, calibration, and validation of the model.

¹ Since this model is in a different software, references to attributes, column names, etc may be different than the most recent version of the users guide.

DOCUMENTATION ACRONYMS

TAZ	Traffic Analysis Zones
NHTS	National Household Travel Survey
MPO	Metropolitan Planning Organization
ACS	American Community Survey
VDOT	Virginia Department of Transportation
NAICS	North American Industrial Classification System
UDA	Urban Development Area
SE	Socio-economic
HCM	Highway Capacity Manual
LOS	Level of Service
FFS	Free Flow Speed
AAWDT	Average Annual Weekday Traffic
SF1	U.S. Census Summary File 1
API	Application Programming Interface
CTPP	Census Transportation Planning Package
HBW	Home-Based Work
NHBW	Non Home-Based Work
HBSchool	Home-Based School
NHBO	Non Home-Based Other
HBShop	Home-based Shopping
HBO	Home-Based Other
HBU	Home-Based University
MNL	Multinomial Logit Model
PA	Production-Attraction
TLF	Trip Length Frequency
FF	Friction Factors
EE	External-External
IE	Internal-External
EI	External-Internal
IPF	Iterative Proportional Fitting
PCE	Passenger Car Equivalency Factor
TOD	Time of Day
CBD	Central Business District
CV	Commercial Vehicles
SUT	Single Unit Truck
MUT	Multi Unit Truck
OD	Origin-Destination
VMT	Vehicle Miles Traveled
RMSE	Root Mean Square Error
EC	Existing Plus Committed

2 TRAFFIC ANALYSIS ZONES AND DATA

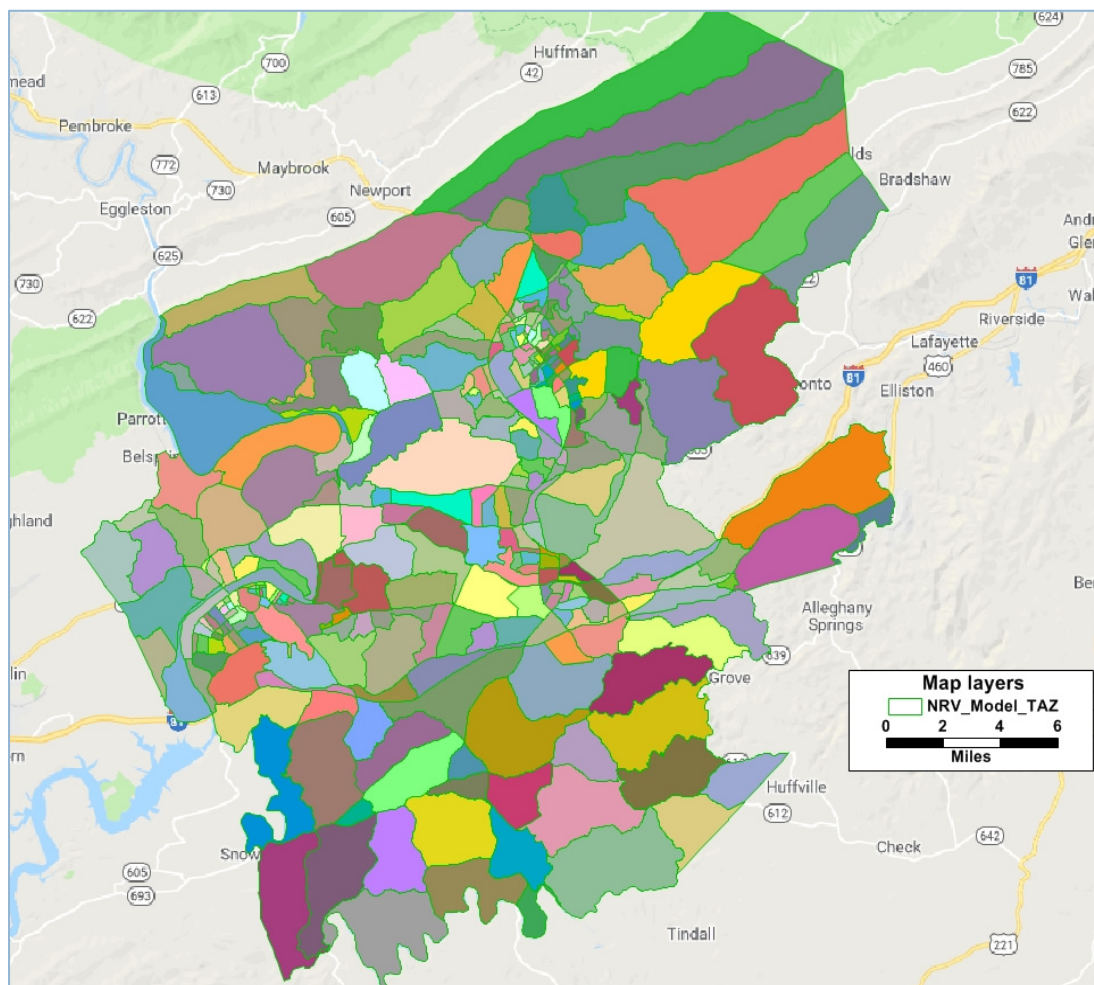
2.1 Transportation Analysis Zones (TAZ)

The NRV model area covers the Towns of Blacksburg and Christiansburg, the City of Radford, and parts of Montgomery and Pulaski counties. Figure 1 provides a map of the TAZs defined for the NRV model area. The TAZs were developed by VDOT staff and are based on the 2010 census boundaries. They adhere to the guidance that they:

- Be comprised entirely of whole 2010 Census Blocks.
- Fit entirely within the boundary of a single 2010 Census TAZ

The final TAZ structure includes 310 internal zones.

Figure 1: NRV Zone System



2.1 Socio-economic (SE) Data

The base year for the NRV model is 2016. Table 1 shows the socioeconomic variables in the model as well as their value in 2016.

Table 1: Socioeconomic Variables in 2016

Field Name	Value	Description
ID	NA	TAZ Number
Households	45,556	Occupied housing units
HHPopulation	112,581	Population living in households
GQPopulation	12,075	Not used by model. Population living in group quarters. Left in data to make the difference between HHPopulation & TotalPopulation explicit.
Workers	51,604	Employed persons
Vehicles	134,047	Vehicles owned by households
MedianIncome	NA	Median household income
Industry	6,466	Industrial Employment NAICS 11, 21, 22, 23, 31-33, 42, 48, 49, 56
Retail	7,944	Retail Employment NAICS 44, 45
Service	5,913	Service Employment NAICS 61, 62, 71, 72, 81
Office	25,312	Office Employment NAICS 51, 52, 53, 54, 55, 56, 92
Ammo	1,243	Control for Employment on the Ammo Base
K12Enrollment	8,321	K-12 School Enrollment The number of students attending
VTEnrollOn	9,000	On-campus enrollment for VT
VTEnrollOff	25,440	Off-campus enrollment for VT This is used for HBU trip attractions, and is evenly distributed across the VT zones.
VTStudOff	25,435	Off-campus student households for VT
RadEnrollOn	3,000	On-campus enrollment for Radford
RadEnrollOff	6,401	Off-campus enrollment for Radford
RadStudOff	6,370	Off-campus student households for Radford

Initial household and population data for the region was based on 2010 census allocation and using the Weldon Cooper's Virginia Population Estimates 2010-2016 to calculate the annual growth rate of population between 2010 and 2016. The growth value for Montgomery County(.67%/year) was utilized since this county is the main county in the modeling region and every TAZ was grown from the 2010 to 2016 value using the same rate². When compared to the most recent American Community Survey (ACS) the allocation of the MPO provided TAZ data for population and households was inconsistent with the ACS allocation. For the model development, additional data for vehicles, income and worker were needed. This data was not available in the original MPO dataset. To maintain consistency in datasets

² A document entitled "Method & Data source 06052017.docx" was the basis for this original development and can be obtained through NRV Regional Commission

used for the model development and given that the ACS was a recent dataset, it was decided to re-develop population and housing information pivoting from the ACS.

The American Community Survey (ACS) data provides data that is more current than the decennial Census, but it does so by doing a continuous sample of households. As a result, of a smaller sample size, data is not available at the block geography. At that small geography, the standard errors in the estimates are too high. Block groups are the lowest geography available from the ACS.

Therefore, additional steps were required to populate the model's traffic analysis zone (TAZ) system with current SE data, and these steps require block, block group, and TAZ layers. The map below shows the relationship between the three layers.

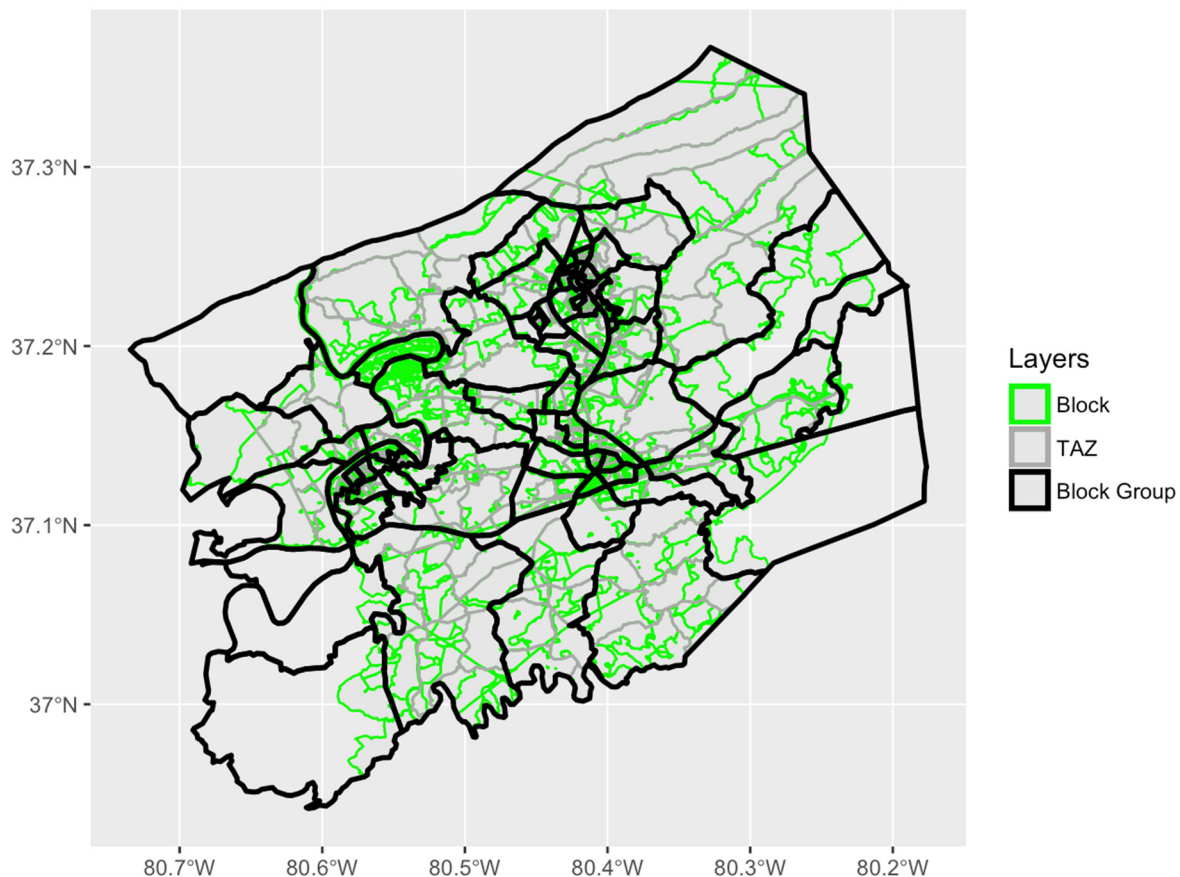


Figure 2: Spatial Relationship of NRV TAZs to Census Blocks and Block Groups

The general steps to create TAZ-level household data was:

1. Determine 2010 households in each TAZ by aggregating SF1 block data.
2. Within each block group, calculate the percent of 2010 households in each TAZ.
3. Apply those percentages to the latest ACS blockgroup household data to get 2015
4. Leverage Weldon Cooper annual growth (.6674%) to grow each TAZ to 2017
5. The households were then adjusted/scaled to maintain the same persons per household

The process is the same for the population in households and group quarters.

Income

Multiple components of the model depend on measures of income. Due to the skewed nature of incomes, the median is a better approximation of the average than the mean. Income information is only available at the block group geography level. Therefore, all TAZs within a block group have the same median income. In addition, income measures were suppressed by the Census for 15 block groups. For these, the regional median income of \$45,607 was used.

Workers

Estimates of workers are only available from the ACS at the tract level. Without more-disaggregate data to inform the process, the following steps are taken to assign workers to TAZs:

1. Calculate the average workers per household for the ACS tract.
2. Use that average for each TAZ within the tract.
3. In this way, workers are allocated within each tract proportionally based on the number of households.

Also, it is important to note that ACS tables that count total workers cannot be used, as they include workers living in group quarters. Instead, total workers in households must be imputed from tables that count households by number of workers.

Vehicles

Vehicle data is available from the ACS at the block group level. In addition, total vehicles can be retrieved for just households. Like workers, however, the lack of information at the block level means that all TAZs within the same block group are assigned the same number of vehicles per household. Also, some block groups had suppressed vehicle information and the regional average was used.

Employment

Base year employment by TAZ and employment sector was derived from a proprietary database of employers in Montgomery and Pulaski Counties and the City of Radford produced by *InfoUSA* and provided by VDOT. The database includes business name, street address, geographic coordinates, number of employees, and an eight-digit North American Industrial Classification System (NAICS) code, indicating the type of business. The *InfoUSA* database was examined for duplicates and other types of error, and cleaned.

Each employer was assigned an employment sector to which the model applies a specific trip attraction rate by matching its three-digit NAICS code with one of five sectors defined for modeling purposes: high-traffic retail, industrial, office, retail and service. Employers not classified were removed. Table 2 explains the NAICS groupings for the NRV model employment data.

Each employer was assigned a TAZ through its geographic coordinates, and total employment was summed for each TAZ by sector. 2017 employment data was provided by local staff. The calculated annual growth rate was 2.87%. Because the base year of the model is 2016, the forecast had to be scaled back by one year. To keep the distribution of employment constant, total employment was scaled, and then the 2017 ratios of employment by type were applied.

Table 2: NAICS Model Categories

Employment Category	NAICS 3-digit codes for Category
Industry	111-115, 211-213, 221, 236-238, 311-339, 424, 481-484, 486, 488, 491-493, 562
Retail	423, 441-444, 446, 448-453
High-traffic Retail	445, 447, 722
Service	485, 487, 532, 541, 561, 611, 621-624, 711-713, 721, 811- 814, 922, 923
Office	425, 454, 511-519, 521-525, 531, 533, 551, 921, 924-928

Area Type

The VDOT Guidelines provide an area type classification table mapping levels of employment density and population density to five area types. The types are

1. CBD - note that this is reserved for manually defined areas.
2. Urban
3. Exurban (dense suburban)
4. Suburban
5. Rural

Because the "CBD" area type is intended for more dense regions than exist in the NRV area, we do not provide logic to handle it. The example classification methodology in the area type methodology in the VDOT Guidelines results in area types that are relative to each other and not absolute for the region; that is, changing activity levels for one zone can change the area type definitions *in other zones*.

As an alternative, a different methodology was applied. The model calculates an area index that is calibrated to base year conditions; as shown in the formula below.

$$index_z = \sum_{i \in I} i \in I \frac{HH_i + \alpha EMP_i}{Area_i} / |I|$$

Where

- I is the set of zones *near* to z , defined as within one and a half miles. This value can be adjusted with the Radius key in the model catalog.
- HH_i is the number of households in the zone.
- EMP_i is the number of jobs in the zone.
- α is an optional constant relating the relative importance of employment to households. In this case, we set the value of α to 1.
- $Area_i$ is the area of the TAZ in acres.

The densities are grouped into area types using the breakpoints below:

Area Type	Density
Urban	2000 units/area
Suburban	500 units/area
Rural	0 units/area

A smoothing process evens out area types to remove artifacts from the previous process. For example, a single zone in a downtown area may have low density and be classified as rural - even if it is surrounded by urban zones. After the smoothing process, it will be classified as urban.

The index is a spatial moving average of the household and employment density allows for changing area type dynamics as households and employment changes.

Links are then assigned the same area type as the nearest zone.

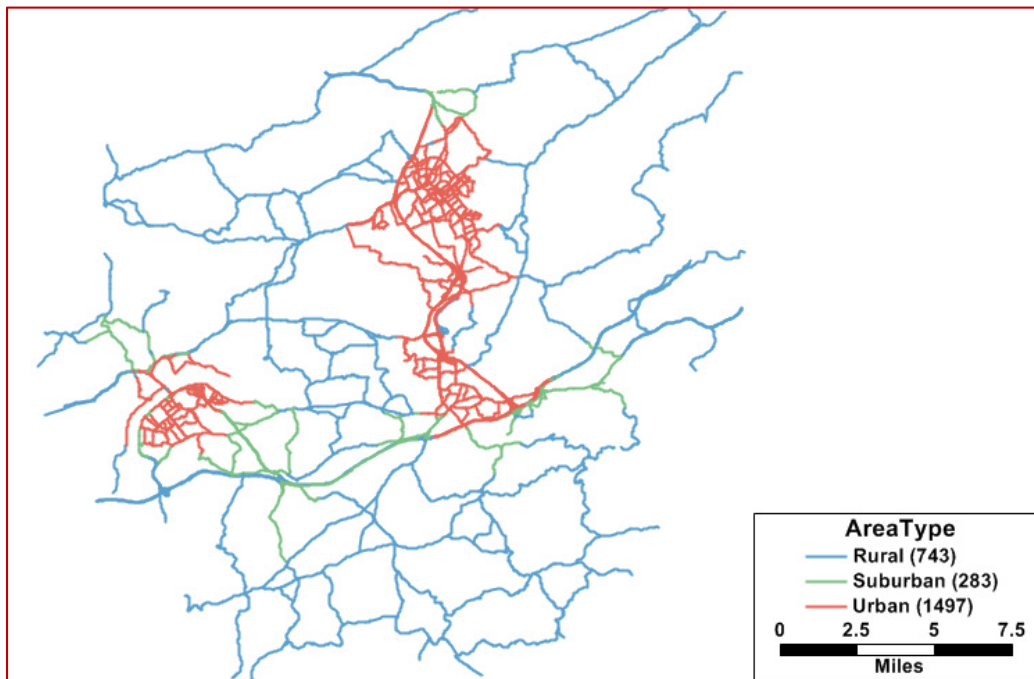


Figure 3: Zones by Calculated Area Type

3 HIGHWAY NETWORK

A TransCAD highway network that is representative of the highway network in the NRV region was developed and reviewed for attribute accuracy, including coding of speeds, lanes, and facility type.

3.1 Network Attributes

The fields on the network were organized and renamed to conform to the required fields for the implementation of the generic GT tools model³ for the region. Fields on the starting network include those listed in Table 3. VDOT staff reviewed the values of these variables; additionally, thematic maps and aerial imagery were used to verify or correct the link attributes. Appendix A outlines the checks and validation performed on the network.

Table 3: Network Attributes

Field	Description
Name	Name of road
HCMTType	HCM Facility Type
PostedSpeed	Posted Speed Limit
HCMMedian	Binary variable for whether the road has a restrictive median
ABLanes	Number of lanes in direction of link topology
BALanes	Number of lanes against direction of link topology
Terrain	Surrounding terrain
HOV	Restrictions on number of passengers per vehicle to use link
Walk	1: Walking allowed 2: Not allowed
Drive	1: Driving allowed 2: Not allowed
Transit	1: Transit allowed 2: Not allowed
CountID	Count station ID (corresponds to VDOT TMS ID)
DailyCount	Traffic count registered at count station
SUTCOUNT	Single unit truck percentage applied to DailyCount
MUTCount	Multi-unit truck percentage applied to DailyCount
Screenline	Screenline number
FutureLink	If marked, link does not exist today
p1ID	Future project ID number
p1HCMTType	Future project HCM Facility Type
p1PostedSpeed	Future project Posted Speed
p1ABLanes	Future project lanes in direction of link topology
p1BALanes	Future project lanes against direction of link topology
p1HCMMedian	Future project binary variable for whether the road has a restrictive median
p1Terrain	Future project surrounding terrain

³ GTtools is a TransCAD based generic modeling platform that is intended to be easily transferred, implemented and specified for any region. WSP wrote this implementation.

Appendix B shows a complete list of network attributes(both inputs and model generated), route attributes and socio-economic attributes used by the model

3.1.1 FACILITY TYPES

The NRV model utilizes facility type definitions to assist in capacity development and the model highway assignment. They are shown below in Table 4. VDOT staff coded the initial facility types in the network and they were reviewed for reasonableness against aerial imagery.

Table 4: Facility Type Definitions

HCMTYPE	NAME
Freeway	Freeway
Ramp	Freeway Ramp
MLHighway	Multi-lane highway
TLHighway	Two lane highway
PrArterial	Principal arterial
MinArterial	Minor arterial
Collector	Collector
Local	Local
CC	Centroid Connector

3.1.2 FREE FLOW SPEED

Several approaches to developing free flow speeds were analyzed including algorithms to adapt the Highway Capacity Manual (HCM) speed calculations, use of existing speed tables from other models, and an evaluation of INRIX speed data on the major facilities. Consequently, the NRV model uses the posted speed limit as a pivot point to develop the initial free flow speed. Based on the area type, facility type and the posted speed a free flow speed is developed as shown in Table 5.

Table 5: Free Flow Speed Calculation

AreaType	HCMTYPE	ModifyPosted	AreaType	HCMTYPE	ModifyPosted
Urban	Freeway	2	Suburban	MinArterial	-3
Urban	MLHighway	0	Suburban	Collector	-3
Urban	TLHighway	0	Suburban	Local	-4
Urban	PrArterial	-3	Rural	Freeway	7
Urban	MinArterial	-5	Rural	MLHighway	5
Urban	Collector	-5	Rural	TLHighway	5
Urban	Local	-6	Rural	PrArterial	0
Suburban	Freeway	3	Rural	MinArterial	0
Suburban	MLHighway	3	Rural	Collector	3
Suburban	TLHighway	3	Rural	Local	0
Suburban	PrArterial	-2	All	Centroids	0

3.1.3 LINK CAPACITIES

The model uses Level of Service E (LOS E) capacities for model implementation of congestion and volume delay. Per lane link capacities by facility type and land use density (LUD) are attached to roadway links from a capacity lookup table. Land use density is derived from the standard area types defined in previous sections.

For planning level purposes of evaluating congestion and to better understand roadway deficiencies, the model also calculates a LOS D capacity. This value is listed as Capacity D in Table 6 and can be used for long range and project planning mapping and calculations. The values used in highway assignment are the Capacity E values in the table.

Table 6: Link Capacity Table

Facility Type	Area Type	Capacity D (per hr/per lane)	Capacity E (per hr/per lane)
Freeway	Urban	1800	2000
Freeway	Suburban	1800	2000
Freeway	Rural	1800	2000
Freeway	Urban	1710	1900
Freeway	Suburban	1445	1700
Freeway	Rural	1530	1800
Multi-Lane Highway	Urban	1440	1600
Multi-Lane Highway	Suburban	1148	1350
Multi-Lane Highway	Rural	1105	1300
Two-Lane Highway	Urban	1140	1200
Two-Lane Highway	Suburban	990	1100
Two-Lane Highway	Rural	900	1000
Principal Arterial	Urban	1440	1600
Principal Arterial	Suburban	1148	1350
Principal Arterial	Rural	1105	1300
Minor Arterial	Urban	1140	1200
Minor Arterial	Suburban	990	1100
Minor Arterial	Rural	900	1000
Collector	Urban	998	1050
Collector	Suburban	743	825
Collector	Rural	630	700
Local	Urban	903	950
Local	Suburban	495	550
Local	Rural	450	500
CC	Urban	10000	10000
CC	Suburban	10000	10000
CC	Rural	10000	10000

3.1.4 TRAFFIC COUNTS

VDOT provided annualized 2015 traffic count data at 203 locations within the model area to support the model development effort. Effort to revise and clean up duplicate counts and split counts by direction occurred as a part of the count process. Counts were selected from this universe of counts to support the creation of external stations, screenlines, as well as overall highway assignment validation. Counts for external stations were broken into auto and truck to ensure proper external travel in the region. The counts are available in the network database as the DailyCount field and the count location is identified as CountID. Figure 4 shows the count locations in the model highlighted by count volume.

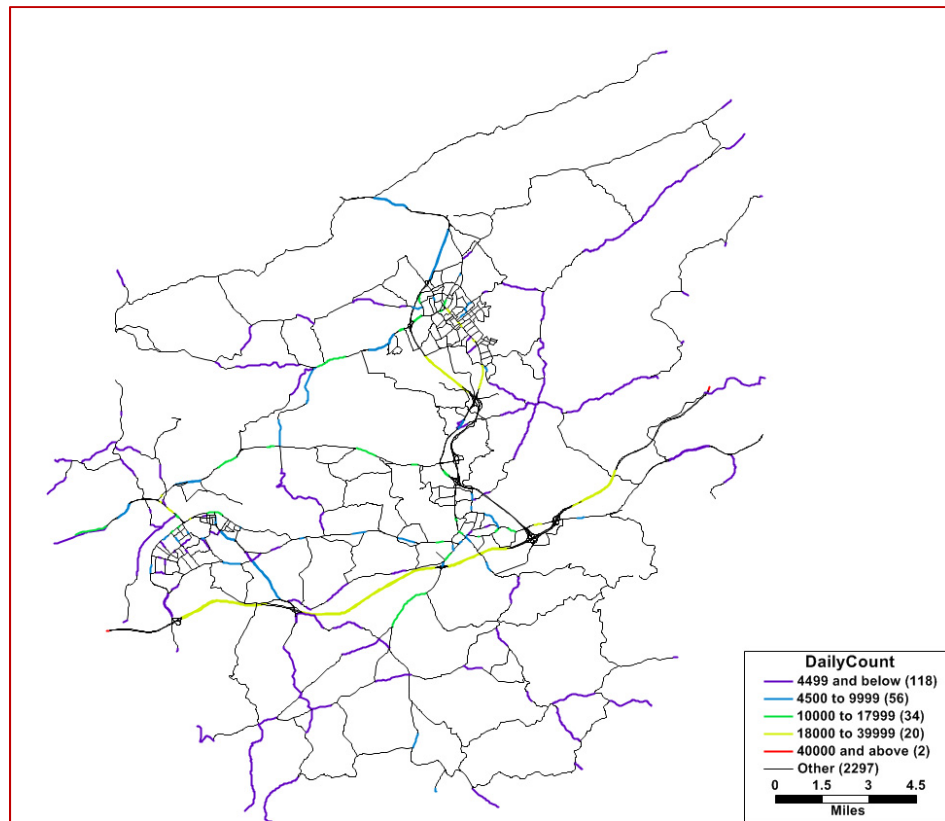


Figure 4: NRV Count Locations

4 TRANSIT NETWORKS AND SKIMS

There can be numerous different public transportation services available to travelers depending on the physical and operating characteristics, and the way the vehicle is accessed by the traveler. These characteristics are represented in a model via the transit network, which consists of transit routes or lines, modes, operators, fares, speeds and access connectors. These elements are discussed in this section.

4.1.1 ROUTES

Table 7 shows the list of transit routes which are part of the NRV base year transit network. Most routes are coded as two one-way routes to allow for skimming, loading and reporting by direction and time of day. In some instance based on the operation of the route and to reduce transfers some routes were coded as loops. Information regarding the alignment, stop locations and headways of the transit lines were developed from the NRV Public Transportation website (<https://www.NRVva.gov/public-transportation>).

Service frequencies, or headways, vary by the time of day. A transit route's headway is the time gap between two consecutive runs at a specific point, and is typically shorter in the peak period than in the off-peak period. Each route has an average headway for the peak period and another for the off-peak period. Route level peak and off-peak period average headways are also shown in Table 7.

Figure 5 shows a map of the coded transit networks in the model. All the routes described in Table 7 were coded in the model.

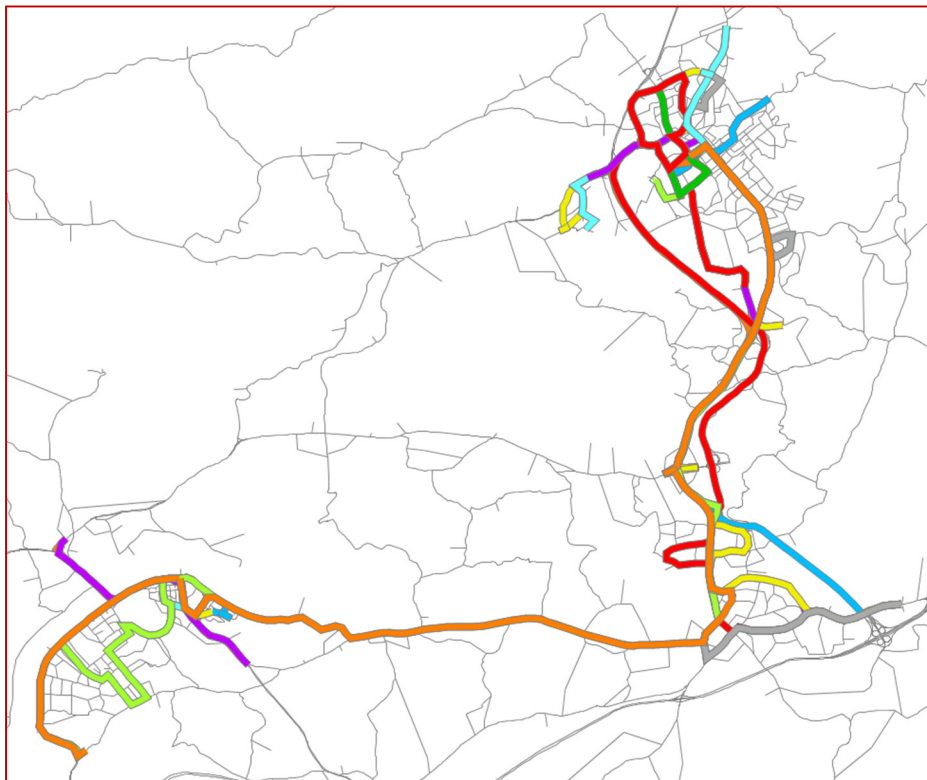


Figure 5: NRV Base Year Transit Routes

Table 7: Base Year Transit Routes and Attributes

Route Name	Mode	Fare	Headways			
			AM	MD	PM	NT
CRC Full S	2	0.5	15	15	15	
CRC Full N	2	0.5	15	15	15	
CRC Evening S	2	0.5				30
CRC Evening N	2	0.5				30
Hethwood B East	2	0.5	15	15	15	15
Hethwood B West	2	0.5	15	15	15	15
Hethwood A East	2	0.5	10	10	10	30
Hethwood A West	2	0.5	10	10	10	30
Hokie Express Weekday Loop	2	0.5	10	10	10	30
240 Commuter S	3	0.5				180
Harding Avenue W	2	0.5	15	15	15	15
Carpenter Boulevard E	2	0.5	30	30	30	
240 Commuter N	3	0.5	180			
Harding Avenue E	2	0.5	15	15	15	15
Patrick Henry Loop	2	0.5	10	10	15	
Progress Street Loop	2	0.5	10	10	10	30
Two Town Trolley South	3	0.5	60	60	60	60
Two Town Trolley North	3	0.5	60	60	60	60
UMall Shuttle Loop	2	0.5		15	15	15
Explorer Blue Loop North	2	0.5	120	120	120	
Explorer Blue Loop South	2	0.5	120	120	120	
Explorer Gold Loop North	2	0.5	120	120		
Explorer Gold Loop South	2	0.5	120	120		
Explorer Green Loop North	2	0.5			180	
Explorer Green Loop South	2	0.5			180	
Radford 10 South	2	1	10	10	15	30
Radford 10 North	2	1	10	10	15	30
Radford 20 South	2	1	60	60	60	60
Radford 20 North	2	1	60	60	60	60
Radford 30 South	2	1	60	60	60	60
Radford 30 North	2	1	60	60	60	60
Radford 50 East	2	1	10	10	10	20
Radford 50 West	2	1	10	10	10	20
Radford 60 South	2	1	15	15	15	30
Radford 60 North	2	1	15	15	15	30
Radford 31 North	2	1	60	60	60	60
Radford 31 South	2	1	60	60	60	60
Carpenter Boulevard W	2	0.5	30	30	30	
Toms Creek Loop	2	0.5	10	10	10	30

University City Blvd South	2	0.5	10	10	10	30
Main Street Weekday North	2	0.5	15	15	15	
Main Street Weekday South	2	0.5	15	15	15	
Main Street Evening North	2	0.5				30
Main Street Evening South	2	0.5				30
Radford 40 East	3	1			60	60
Radford 40 West	3	1			60	60

4.1.2 MODES

Only one transit mode (local bus) was available in the base year in the NRV region. All the local bus routes were coded as mode code 2 if they were Blacksburg Transit or mode 3 if Radford transit. The region may have a desire to evaluate premium transit service in the future, therefore, the model nesting structure was designed to include a premium transit mode. Should the premium mode need to be used for testing future transit alternatives, the constants and coefficients for this mode will need to be further evaluated since the values were borrowed from another region.

4.1.3 OPERATORS AND FARES

Transit operators are the agencies responsible for providing transit service and typically set the fare policy for their service. Given there are two operators in the region two fare system exists. Local bus boarding fares are designated in the model and displayed in Table 7.

4.1.4 SPEEDS

Travel times for transit vehicles that operate in mixed-flow traffic are typically a function of auto travel time on the same roadway. For NRV it was decided instead to use dwell times to better estimate and control the route travel times. It is difficult to get exact run times on every route in a small network but most of the route runs times compare favorably.

4.1.5 WALK ACCESS CONNECTORS

A walk-to-transit path from an origin zone to a transit stop is represented in the network by a straight line walk access connector from the origin zone centroid to nearby stops. The distance on the access link is derived from accumulated times and distances along the roadway network links from the zone centroid to transit stops. For an access link to be created, the stop must be within a 0.6 mile radius of the centroid. Access links are created to the five closest stops within the 0.06 mile tolerance. The walk time is computed from the accumulated roadway distance and an assumed walking speed of 2.5 miles per hour. All walk access connectors are two-way legs to allow for both access and egress.

Figure 6 shows examples of walk access connectors generated in the model.

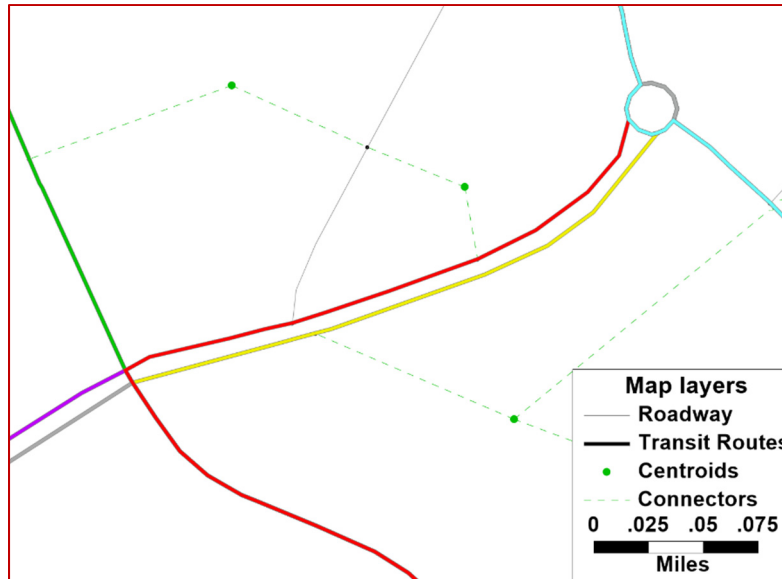


Figure 6: Walk Access Connectors in NRV

4.2 Transit Paths and Skim Tables

This section provides a description of transit path parameters, procedures and final skim matrix tables.

4.2.1 WAIT TIMES

Wait time curves are used to relate the frequency of transit services to the time spent waiting at a transit stop or station. Wait time is assumed to be half of the headway time. A minimum wait time of two minutes and a maximum wait time of 30 minutes are used in the model. The maximum wait time is capped at 30 minutes because the transit rider is usually aware of the approximate schedules and therefore would not typically wait longer than 30 minutes. The same assumptions are used for the first wait time, at the beginning of a trip, and for transfer wait times. The wait curve used in the NRV model is shown in Figure 7.

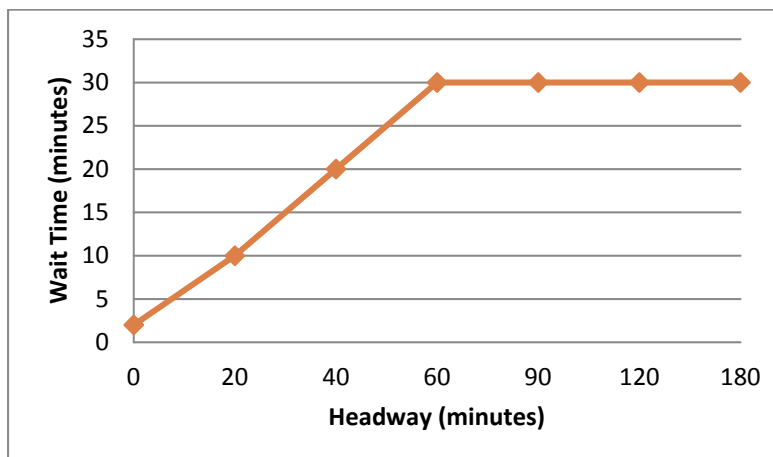


Figure 7: Wait Time Curve

Two types of transit paths are built in the NRV model including walk-to-bus and walk-to-premium paths. Given that only local bus exists in 2015, both the walk-to-bus paths and walk-to-premium paths are developed using the local bus connectors and parameters. This serves as a place holder in the scripting for creating premium bus paths in the future. Table 8 shows the transit and non-transit modes used in the two transit paths. Drive-to-bus alternatives are not modeled given the small market in NRV and lack of transit on-board survey data.

Table 8: Transit Path Types and Modes Used

Mode	Mode ID	Paths			
		Local Bus	Express Bus	BRT	Commuter Rail
Non-Transit	1	1	1	1	1
Local Bus	2	2	1	1	1
Express Bus	3		2	1	1
Bus Rapid Transit	4			2	1
Commuter Rail	5				2
		1: mode is allowed 2: mode is required			

Table 9 shows the commonly accepted transit path weights and parameters used in the transit path building procedures for NRV.

Table 9: Transit Path Weights and Parameters

Transit Path Component	Weight
Fare	1
Travel Time	1
Wait Time	2
Transfer Time	2
Dwell Time	2

4.2.2 TRANSIT SKIMS

Skims or travel time matrices contain information about the minimum impedance paths between all TAZ pairs in a network. Also referred to as the skim tree matrix or travel time matrix, these matrices are used in trip distribution, mode split, and the trip assignment steps. For transit, there are many variables that affect the travel time experienced by transit users. Each variable is converted into an equivalent minute value and summed together to get the total transit travel time skim.

The individual values of the travel components are stored in output skim files and are used in the mode choice model. The values stored in skim files represent the actual values of the attributes, and not the weighted values used in the path building step. Table 10 shows the transit skim matrix tables and their

description. The matrices listed in **Table 10** are repeated for every combination of access mode and primary transit mode (e.g. walk-to-local-bus) and time of day.

Table 10: Transit Skim Matrix Tables

Matrix Name	Description
gen_cost	Generalized Cost
fare	Fare
ivt_transit	In-Vehicle Time
initial_wait	Initial Wait Time
xfer_wait_time	Transfer Wait Time
initial_penalty_time	Initial Penalty Time
xfer_penalty_time	Transfer Penalty Time
xfer_walk_time	Transfer Walk Time
access_walk_time	Access Walk Time
egress_walk_time	Egress Walk Time
access_drive_time	Access Drive Time
egress_drive_time	Egress Walk Time
dwell_time	Dwell Time
total_time	Total Time
iv_cost	In-Vehicle Cost
initial_wait_cost	Initial Wait Cost
xfer_wait_cost	Transfer Wait Cost
initial_penalty_cost	Initial Penalty Cost
xfer_penalty_cost	Transfer Penalty Cost
xfer_walk_cost	Transfer Walk Cost
access_walk_cost	Access Walk Cost
egress_walk_cost	Egress Walk Cost
access_drive_cost	Access Drive Cost
egress_drive_cost	Egress Walk Cost
dwell_cost	Dwell Cost
num_xfers	Number of Transfers
iv_dist	In-Vehicle Distance
access_drive_dist	Access Drive Distance
egress_drive_dist	Egress Drive Distance

5 HOUSEHOLD CLASSIFICATION AND JOINT DISTRIBUTION

A cross-classification approach to trip production models requires the use of a household sub-model to disaggregate the household attribute data into the proper format required by the cross-classification model. The NRV model uses three auxiliary sub-models to forecast at the TAZ level, distributions of households by household size, by number of workers, and by number of vehicles⁴. These sub-models were developed based on observed distributions of these attributes among households in the NRV region. An additional sub-model then calculates the joint distribution of these variables (i.e. the number of two-person households with one worker owning one vehicle) via iterative proportional fitting (IPF) beginning with a seed table based on data from the region.

The disaggregate model requires two key pieces of information to perform this task:

- Marginal targets
 - Estimated from Census data
- Initial/Seed distribution of households
 - Taken directly from Census data

5.1 Marginal Targets

Each marginal target is a distribution of households by one attribute (e.g. size). In the case of size, the distribution of households should change as the average household size in a geography increases (more households with multiple members). Marginal estimation, in general, estimates the relationship between aggregate metrics, like household size, and the shape of a one-dimensional distribution, like households by size.

In NRV, the marginal distribution curves estimated are:

- Size (1-4+)
- Income
 - Low: \$0 - \$24,999
 - Medium-Low: \$25,000 - \$49,999
 - Medium-High: \$50,000 - \$99,999
 - High: \$100,000 and above
- Workers (0-3+)
- Vehicles (0-3+)

⁴ Curves for income were developed but ultimately not used in the model given the lack of available data

The Census data is taken from different data products and geographies, depending on availability. For example, the data needed to estimate the worker curve is only available from the American Community Survey (ACS) product known as the Census Transportation Planning Package (CTPP). Vehicle and income information is available from the ACS 5-year file, and size information is available from multiple products.

5.1.1 HOUSEHOLD SIZE

Household size marginal tables for 1, 2, 3, and 4+ persons within the household were retrieved using the Census API, as well as the population in households and the total number of households.

From these data, the average number of persons per household in each block and the share of households of each size was calculated. The mean share of each size was subsequently calculated for bins of average persons per household. For example, the average share of one-person households for all zones with an average household size of 2.04 people is 0.349. This bin-level data was smoothed into a continuous and interpolated curve by fitting a multi-degree polynomial curve. Figure 8 shows the raw block-level data as well as the fitted curves for the household size by average persons per household.

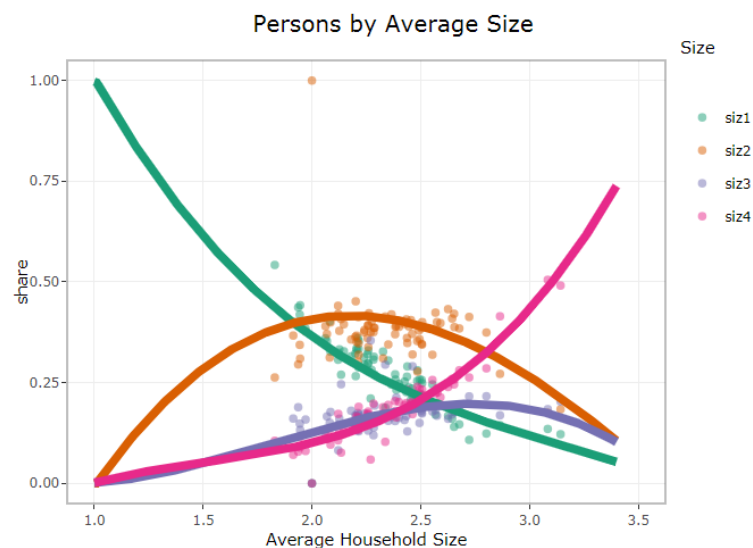


Figure 8: Household Size by Average Persons per Household

The curves above were used to construct a look-up table for the proportion of households at each level over the range of the data: from 1 to 3.4 average persons per household on a 0.01 step interval. At the edges of average household size, manual edits were required to adjust curve behavior. These were required due to the low number of observations in these ranges. This same treatment was applied to create all marginal tables.

Table 11 shows a segment⁵ of rows from the look-up table the model uses to dis-aggregate the zonal data into households by size. All other markets follow the same structure as shown in this table.

Table 11: Household Size Lookup Table, Sample Rows

Average Household Size	1-person	2-person	3-person	4-person
2.00	0.3641	0.4079	0.1280	0.1000
2.01	0.3603	0.4089	0.1295	0.1013
2.02	0.3565	0.4097	0.1311	0.1027
2.03	0.3527	0.4106	0.1326	0.1041
2.04	0.3490	0.4114	0.1341	0.1055
2.05	0.3453	0.4121	0.1356	0.1070
2.06	0.3417	0.4127	0.1371	0.1085

5.1.2 HOUSEHOLD WORKERS

The total number of workers in a TAZ is an input into the travel demand model and is used to calculate the number of workers per household in order to estimate the number of 0, 1, 2, and 3+ worker households using the household worker sub-model.

Similar to the household size disaggregation model, CTPP data tables were used to calculate the share of households by number of workers at different levels of zonal average worker per household. This informed the creation of continuous smooth disaggregation curves.

Figure 9 shows only the raw data from the CTPP TAZs in NRV as well as the fitted curves.

⁵ The full table used by the model can be found in the master model folder
/master/generation/disagg_hh_size.csv

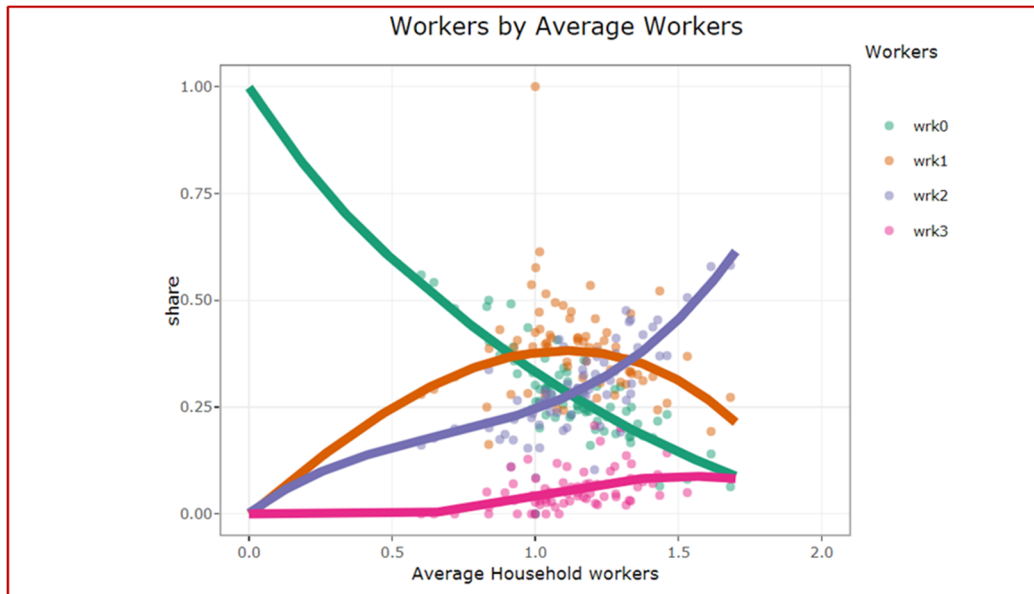


Figure 9: Household Workers by Average Workers per Household

The fitted mean value curves were used to construct a look-up table for the proportion of households in each bin over the range of the data: from 0 to 1.75 average workers per household on a 0.01 step interval. This table⁶ was manually edited before placement in the model to ensure that behavior along the edges of the data is reasonable and intuitive.

⁶ The full table used by the model can be found in the master model folder `/master/generation/disagg_hh_worker.csv`

5.1.3 HOUSEHOLD VEHICLES

Similar to the household size disaggregation model, CTPP data tables were used to calculate the share of households by number of available vehicles at different levels of zonal average vehicles per household. This informed the creation of continuous smooth disaggregation curves. Figure 10 shows the raw data from CTPP TAZ data in NRV as well as the fitted curves. Because the zonal averages observed in the CTPP cover a relatively narrow range of vehicles per household, additional information based on experience and intuitiveness was used to extrapolate the curves to the required range.

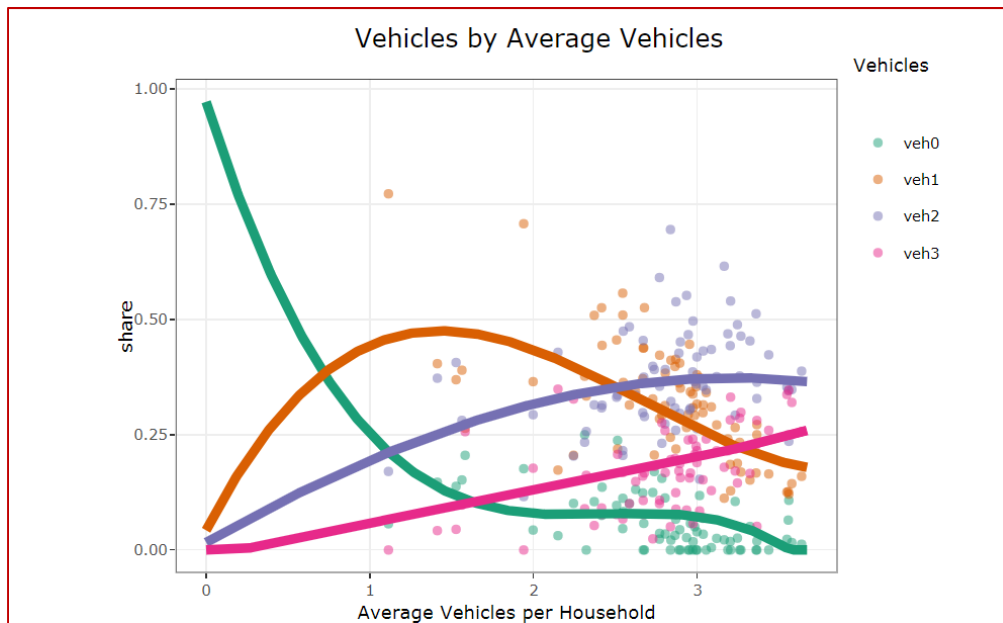


Figure 10: Household Vehicles by Average Vehicles per Household

The fitted mean value curves were used to construct a look-up table for the proportion of households in each bin over the range of the data: from 0 to 3.6 average vehicles per household on a 0.01 step interval. The final table⁷ was manually edited before placement in the model to ensure that behavior along the edges of the data is reasonable and intuitive.

⁷ The full table used by the model can be found in the master model folder `/master/generation/disagg_hh_vehicle.csv`

5.2 Joint Distribution

The household classification models are *marginal* models; they predict the unconditional proportion of 1, 2, 3, and 4+ person households in a zone based on the average household size for the zone (and the corresponding marginals for number of vehicles per household and number of workers per household). Cross-classification trip production models require a *joint* distribution of these attributes, for example the number of three-person households with one worker that owns two vehicles.

The NRV model constructs the joint distribution of household attributes via iterative proportional fitting (IPF), a process that adjusts an initial joint distribution (or seed matrix) to match the marginal totals predicted by the household classification models. The model uses a regional seed matrix from Table A112305 of the CTPP. It iterates over the seed matrix to match the NRV Household marginals for size, workers and vehicles. Iterations continue until all estimated marginals are matched to NRV developed marginals within a tolerance of ± 0.01 or until a maximum of 50 iterations. Two joint distributions are used for trip generation. The joint distribution of households by size and by number of vehicles is used for generation of non-work purposes. The joint distribution of households by number of workers and number of vehicles is used for work related purposes.

5.3 Model Validation

After applying the curves estimated, the results of the disaggregate model are checked against the estimation data. In the tables that follow, the first 4 rows are the regional totals and then each county individually.

In addition, the 2010 SF1, 2010-2015 ACS, and the 2015 NRVMPD model SE datasets have slightly different total households. This is expected given the sources and years. In the validation tables, the disaggregate models are validated based on marginal percentage distributions.

The “Observed” column shows the regional distribution of households by number of persons in the 2010 SF1 census data. The “Modeled” column shows the results from the NRVMPD classification model.

Table 12: Household Size Disaggregation Comparison

County	Size	Observed	Modeled	Percent Difference
ALL	1	27.20%	24.40%	-2.80%
ALL	2	36.60%	35.70%	-0.90%
ALL	3	17.40%	16.20%	-1.20%
ALL	4	18.80%	23.70%	4.90%
Radford	1	31.40%	21.50%	-9.90%
Radford	2	33.40%	35.00%	1.60%
Radford	3	17.40%	16.80%	-0.60%
Radford	4	17.80%	26.70%	8.90%
Pulaski	1	27.90%	28.60%	0.70%
Pulaski	2	40.70%	40.40%	-0.30%
Pulaski	3	15.00%	15.90%	0.90%
Pulaski	4	16.50%	15.10%	-1.40%
Montgomery	1	26.40%	24.40%	-2.00%
Montgomery	2	36.60%	35.40%	-1.30%
Montgomery	3	17.70%	16.10%	-1.60%
Montgomery	4	19.30%	24.20%	4.90%

Table 13: Vehicle Disaggregation Comparison

County	Vehicles	Observed	Modeled	Percent Difference
ALL	0	5.80%	6.50%	0.70%
ALL	1	31.50%	28.00%	-3.50%
ALL	2	36.80%	45.80%	9.10%
ALL	3	25.90%	19.70%	-6.20%
Radford	0	6.80%	7.50%	0.70%
Radford	1	36.40%	29.30%	-7.10%
Radford	2	32.60%	44.50%	11.90%
Radford	3	24.20%	18.70%	-5.50%
Pulaski	0	2.80%	7.30%	4.40%
Pulaski	1	29.80%	26.40%	-3.40%
Pulaski	2	38.70%	46.40%	7.60%
Pulaski	3	28.70%	20.00%	-8.70%
Montgomery	0	6.10%	6.20%	0.20%
Montgomery	1	31.00%	28.00%	-3.00%
Montgomery	2	37.20%	46.00%	8.80%
Montgomery	3	25.80%	19.80%	-6.00%

Table 14: Worker Disaggregation Comparison

County	Workers	Observed	Modeled	Percent Difference
ALL	0	27.90%	27.10%	-0.82%
ALL	1	37.30%	37.60%	0.24%
ALL	2	29.20%	29.10%	-0.06%
ALL	3	5.60%	6.20%	0.64%
Radford	0	6.80%	7.50%	0.70%
Radford	1	36.40%	29.30%	-7.10%
Radford	2	32.60%	44.50%	11.90%
Radford	3	24.20%	18.70%	-5.50%
Pulaski	0	33.20%	31.10%	-2.08%
Pulaski	1	37.20%	38.00%	0.80%
Pulaski	2	24.50%	25.80%	1.27%
Pulaski	3	5.10%	5.10%	0.02%
Montgomery	0	25.60%	26.70%	1.09%
Montgomery	1	37.40%	37.50%	0.13%
Montgomery	2	31.20%	29.50%	-1.76%
Montgomery	3	5.80%	6.30%	0.54%

6 TRIP PRODUCTION

This section describes the development and validation of the trip production model. The forecasting of productions is the first step in the traditional 4-step process. The trip production model applies estimated trip rates to the results from the household joint disaggregation for each TAZ. The outputs from the trip production model are fed into the destination choice models.

The trip production model for NRV follows a cross-classification approach, which applies average trip rates for households by household stratification. Households are stratified by number of workers and auto ownership for all work trip purposes, and by number of persons and auto ownership for non-work purposes. Trip production rates were estimated from the 2009 NHTS collected on behalf of VDOT. Because of the sparseness of the NHTS sample in the NRV region, manual adjustment to rates was required.

6.1 Trip Rate Estimation

The 2009 NHTS allowed for states and Metropolitan Planning Organizations (MPOs) to request *add-on* samples for detailed regional analysis. Virginia participated in this program as a means to obtain travel survey data for small and medium-sized communities in the state. This is one of the biggest advantages of the NHTS, in that it provides a mechanism for states to obtain household travel survey for smaller communities and MPOs as a part of a larger national survey. This common format data supports the development of statewide models, and facilitates the analysis of survey data across multiple regions.

The following list of initial trip purposes resulted from the processing of the NHTS data:

- HBW: home-based work
- HBSC: home-based school
- HBSH: home-based shopping
- HBO: home-based other
- NHBW: non-home-based work
- NHBO: non-home-based other

The expansion factors provided with the NHTS are based on their sampling method, which is related to telephone land lines by telephone exchange areas. For the purposes of travel forecasting the expansion should target the number of households within each county by the characteristics that are relevant to the travel model – household size, number of vehicles per household, and number of workers per household. Thus the NHTS was re-expanded within each county to targets based on CTPP totals for households by these key characteristics (size, vehicles, and workers).

Table 15 shows how many trips of each type were observed in the survey.

Table 15: NHTS Trips by Purpose

Count of Trips by Purpose		
purpose	samples	weighted
HBW	3,189	35,273
HBO	5,872	54,328
HBSC	918	11,503
HBSH	4,567	41,226
NHBW	1,635	16,882
NHBO	4,265	38,659

6.2 Work Trip Purposes

For work trips, households are classified by the number of workers and the number of vehicles available in a household. Table 16 shows the raw number of survey responses available in each of the stratifications that resulted from the joint disaggregation of households in the NHTS data sample.

Most categories have a sufficient number of responses to estimate a statistically robust mean, with the exception of the bottom-left corner of the table. This is not unreasonable, as few households with many workers are likely to own zero vehicles in rural and semi-urban Virginia. However, the sparseness of these data need to be considered in estimating the trip rates.

Table 16: Number of NHTS Responses by Work Classification

workers	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
worker 0	52	442	382	176
worker 1	21	257	406	367
worker 2	5	35	439	511
worker 3	1	NA	11	130

6.2.1 WORK INITIAL TRIP RATES

Table 17 shows the initial home-based work trip rates estimated from the weighted surveys and Table 18 shows the rates for non-home-based work trips. The initial rates are reasonable in that households make more trips either as the number of workers increases, or the household has more vehicles available. Households with no workers make no work trips. On the other hand, small sample sizes mean that there missing cells or cells estimated on a single observation. These were manually smoothed.

Table 17: HBW Trip Rates by Workers and Vehicles

workers	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
0-worker	0.000	0.006	0.003	0.140
1-worker	0.972	0.999	1.001	0.937
2-worker	0.512	2.544	1.920	1.905
3-worker	3.000	NA	1.726	2.702

Table 18: NHBW Trip Rates by Workers and Vehicles

workers	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
0-worker	0.000	0.006	0.003	0.068
1-worker	0.156	0.457	0.481	0.423
2-worker	0.256	0.909	0.883	1.106
3-worker	1.000	NA	1.327	1.214

6.2.2 WORK FINAL TRIP RATES

The final trip rates were obtained following an iterative process of reviewing logical progression left to right and top to bottom. The initial rates in adjacent table cells were combined if there were insufficient survey data records to estimate robust statistics and the newly estimated rates were manually smoothed to provide reasonable continuity across the table dimensions.

This first manual adjustment was to set the trip rate for 0-worker households to zero. While it is possible for an unemployed person to make a work trip (an “odd job”), this behavior cannot be accurately captured by the model or survey.

Households with 1 worker and 1 or more vehicles were set to the average of this category, which is justified because the number of trips does not increase monotonically as the number of vehicles increases, the trip rates are all close together, and there is little psychological justification for why owning additional vehicles would cause someone to make more HBW trips.

Households with 2 or more workers are likely to make about twice as many trips as households with 1 worker. The only rate adjusted manually in this row was the 1-vehicle rate, because it was higher than the rate for households with more vehicles. This rate was adjusted to 1.9 based on experience with other models. Rates for households with 3 or more vehicles were left alone because the 2 or more worker classification would reasonably see households with 3 or more vehicles have a higher rate.

Adjustments to NHBW trips followed the same logic as HBW, with 0 worker households set to trip rates of zero. The rate for 1 worker households with 1 or more vehicles was set to the average trip rate for that classification for the same reasons provided in the HBW discussion. No other adjustments were made to the rates.

Table 19 and Table 20 display the rates used in the model for the work purposes.

Table 19: HBW Trip Rates by Workers and Vehicles: Final

HBW	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
1-worker	0.8	1.1	1.1	1.1
2-worker	1.0	1.7	2.1	2.1
3+ worker	1.2	1.9	2.4	2.8

Table 20: NHBW Trip Rates by Workers and Vehicles: Final

NHBW	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
1-worker	0.25	0.55	0.55	0.550
2-worker	0.35	0.60	1.00	1.134
3+ worker	0.60	0.80	1.15	1.280

6.3 Non-Work Trip Purposes

For non-work trips, the NRV model households are classified by household size and number of vehicles available. Table 21 shows the raw number of survey responses available in each category in the combined NHTS data sample.

Most categories have a sufficient number of responses to estimate a statistically robust mean, with the exception of the bottom-left corner of the table.

Table 21: Number of NHTS Responses by Non-work Classification

size	vehicle 0	vehicle 1	vehicle 2	vehicle 3
1-person	52	480	140	51
2-person	16	197	779	619
3-person	3	35	130	268
4-person	8	22	189	246

6.3.1 NON-WORK INITIAL TRIP RATES

Table 22 - Table 25 show the initial trip rates for non-work trip purposes that were estimated from the combined survey dataset. These rates are again largely reasonable, though there are some inconsistencies that raise concern. For instance, the estimated rates generally show households with no vehicles making more trips than households with a single vehicle at many different household sizes. This is likely due to sample sizes and the overall sparseness of the survey.

Table 22: HBO Trip Rates by Household Size and Vehicles

size	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
1-person	1.166	0.921	0.871	0.682
2-person	1.417	1.803	1.683	1.278
3-person	1.654	2.344	2.586	1.666
4-person	1.429	0.812	3.656	3.462

Table 23: NHBO Trip Rates by Household Size and Vehicles

size	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
1-person	0.344	0.912	0.617	0.333
2-person	0.306	1.503	1.238	1.193
3-person	0.551	1.380	1.289	1.551
4-person	0.494	1.250	1.976	2.065

Table 24: HBSHOP Trip Rates by Household Size and Vehicles

size	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
1-person	1.061	0.918	0.788	0.617
2-person	1.123	1.385	1.487	1.427
3-person	2.075	1.493	1.386	1.504
4-person	1.325	1.079	1.598	1.971

Table 25: HBSchool Trip Rates by Household Size and Vehicles

size	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
1-person	0.000	0.002	0.005	0.000
2-person	0.000	0.088	0.035	0.028
3-person	0.762	0.704	0.497	0.482
4-person	1.687	1.321	1.582	1.444

6.3.2 SMOOTHED TRIP RATES

The final trip rates were obtained following the same basic iterative process as the work trip purposes. If the initial comparison to the purpose-level trip tables was unsatisfactory, the rates were adjusted on a class-by-class basis; if the comparison was satisfactory, the smoothed values were used. Table 26 - Table 29 contain the non-work trip rates obtained in this process.

HBO trips were adjusted to ensure that households with more vehicles make at least as many trips as households with fewer vehicles (for a given household size) and that larger households make at least as many trips as smaller households (for a given number of vehicles). Achieving this required manually adjusting the rate for 1-person households with no vehicles and all 2-person households.

Table 26: HBO Trip Rates by Household Size and Vehicles, Smoothed

HBO	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
1-person	0.700	1.000	1.000	1.000
2-person	1.417	1.803	2.100	2.100
3-person	1.654	2.344	2.586	3.100
4-person	1.800	2.500	3.000	3.462

The NHBO trip rates were adjusted to match the same patterns as prior non-work trips, i.e. that trip rates should increase as the household size and number of vehicles increases.

Table 27: NHBO Trip Rates by Household Size and Vehicles, Smoothed

NHBO	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
1-person	0.420	1.021	1.021	1.021
2-person	0.854	1.556	1.556	1.556
3-person	0.976	1.586	1.708	1.892
4-person	1.220	1.830	2.411	2.519

HBSH trips estimated so that trips increase as the number of vehicles and household size increase.

Table 28: HBSHOP Trip Rates by Household Size and Vehicles, Smoothed

HBSHOP	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
1-person	0.897	0.897	0.897	0.897
2-person	1.123	1.385	1.469	1.469
3-person	1.200	1.493	1.600	1.800
4-person	1.325	1.600	1.700	1.968

Rates were assumed to be stable for households of a given size on the assumption that such all households of a given size are equally likely to have a child needing to go to school and that they must make that trip regardless of the family's access to a vehicle. It might be speculated that a household with more vehicles would be likely take more trips, e.g., additional after school activity trips, but the raw trip rate data did not bear this hypothesis out and so it was dropped.

Table 29: HBSchool Trip Rates by Household Size and Vehicles, Smoothed

HBSchool	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
1-person	0.000	0.000	0.000	0.000
2-person	0.038	0.038	0.038	0.038
3-person	0.600	0.600	0.750	0.750
4-person	1.411	1.411	1.550	1.750

6.4 Model Validation

When the initial production rates are applied in the model to the base year SE data, the results can be compared back to the survey totals to measure model performance; however, a direct comparison would be misleading for three reasons.

First, the model SE data contains 45556 households, while the weighted survey only contains 30722; the model is 1.48 times larger than the weighted survey. This difference arises because the household weights are calculated when the NHTS survey is first processed. In the generation step, trips with invalid trip purposes or from houses with incomplete records are removed. There are 350 trips with invalid "from" purposes, 137 trips with invalid "to" purposes, and 2399 trips from households with incomplete records.

Second, the household survey was expanded to 2010-2015 ACS data, which is an average of the households across all five years. The model data is meant to represent the actual number of households in 2016. The traffic counts used for model validation were also collected in 2015.

For these two reasons, all trips in the survey are scaled up to match total model trips.

Third, paper surveys like the NHTS have consistent problems with underreporting, particularly for non-home based trips (i.e. NHBW and NHBO). Model efforts in Cleveland, OH and Nashville, TN used both trip recall surveys and GPS location based data collection. Reports in trip recall surveys consistently underreported non-home based trips by 20-30%. In order to correct for this bias in the data, the summed non-home based trips from the survey were increased by 22% and then used as a target in calibration adjustment.

After these adjustments, the Table 30 shows the comparison between model and survey.

Table 30: Model trip productions, Final

Purpose	Original Survey Total	Adjusted Survey Total	Model Trips	Difference
HBW	35273	52305	52305	0
HBO	54328	80560	80560	0
HBSC	11503	17057	17057	0
HBSH	41226	61132	61132	0
NHBW	16882	30182	30183	1
NHBO	38659	69227	69227	0
Total	197871	310463	310464	1

6.4.1 FINAL NRV MODEL TRIP RATES

The final, calibrated trip rates after the adjustments are shown in Table 31 through Table 35.

Table 31: HBW Trip Rates by Household Size and Vehicles, Final

HBW	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
0-worker	0.000	0.000	0.000	0.000
1-worker	0.838	1.152	1.152	1.152
2-worker	1.048	1.781	2.200	2.200
3-worker	1.257	1.991	2.515	2.934

Table 32: HBO Trip Rates by Household Size and Vehicles, Final

HBO	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
1-person	0.604	0.863	0.863	0.863
2-person	1.223	1.556	1.813	1.813
3-person	1.429	2.024	2.233	2.677
4-person	1.554	2.159	2.590	2.989

Table 33: NHBO Trip Rates by Household Size and Vehicles, Final

NHBO	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
1-person	0.406	0.986	0.986	0.986
2-person	0.825	1.503	1.503	1.503
3-person	0.943	1.532	1.650	1.828
4-person	1.178	1.768	2.329	2.434

Table 34: HBSHOP Trip Rates by Household Size and Vehicles, Final

HBSHOP	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
1-person	0.866	0.866	0.866	0.866
2-person	1.083	1.336	1.417	1.417
3-person	1.158	1.440	1.544	1.737
4-person	1.278	1.544	1.640	1.899

Table 35: HBSchool Trip Rates by Household Size and Vehicles, Final

HBSchool	0 Vehicle	1 Vehicle	2 Vehicle	3+ Vehicle
1-person	0.000	0.000	0.000	0.000
2-person	0.029	0.029	0.029	0.029
3-person	0.455	0.455	0.568	0.568
4-person	1.070	1.070	1.175	1.327

6.5 Attraction Model

Attraction models convert zonal employment into an attractiveness score used in distribution models. The distribution model balances the attraction score of the zone against the travel time and distance required to reach it.

There were not enough records in the NHTS Virginia add-on for NRV to support the estimation of an attraction model. Instead, rates were borrowed from the travel model developed for NRV, VA. That model documentation goes into more detail, but in summary, the statewide model was used to develop attraction rates using NHTS samples from multiple MPOs. The NRV samples were included in this process, and thus inform the final average rates.

The borrowed rates used in NRV are shown in Table 36.

Table 36: Initial Borrowed Attraction Rates

Model	Households	Industry	Retail	Office	Service	K12Enrollment	VTEnrollOn	VTEnrollOff	RadEnrollOn	RadEnrollOff
HBW	0	2.16	0.74	0.62	2.16	0.95	6.2	6.2	6.2	6.2
HBO	1.39	1.21	1.36	1.21	1.21	0	0	0	0	0
HBSH	0	0	4.1	0	0	0	0	0	0	0
HBSC	0	0	0	0	0	1	0	0	0	0
NHBW	0.53	0.55	5.1	0.55	0.55	0	0	0	0	0

The NRV model did not use market stratification by income and auto sufficiency. The rates were adjusted to translate rates for HBW, for example, into rates for HBW_v0 and the other four markets. The final rates are shown in Table 37.

Table 37: Final Attraction Rates

Model	Households	Industry	Retail	Office	Service	K12Enrollment	VTEnrollOn	VTEnrollOff	RadEnrollOn	RadEnrollOff
HBWa_v0	0	2.16	0.74	0.62	2.16	0.95	6.2	6.2	6.2	6.2
HBWa_ilvi	0	2.16	0.74	0.62	2.16	0.95	6.2	6.2	6.2	6.2
HBWa_ilvs	0	2.16	0.74	0.62	2.16	0.95	6.2	6.2	6.2	6.2
HBWa_ihvi	0	2.16	0.74	0.62	2.16	0.95	6.2	6.2	6.2	6.2
HBWa_ihvs	0	2.16	0.74	0.62	2.16	0.95	6.2	6.2	6.2	6.2
HBOa_v0	1.39	1.21	1.36	1.21	1.21	0	0	0	0	0
HBOa_ilvi	1.39	1.21	1.36	1.21	1.21	0	0	0	0	0
HBOa_ilvs	1.39	1.21	1.36	1.21	1.21	0	0	0	0	0
HBOa_ihvi	1.39	1.21	1.36	1.21	1.21	0	0	0	0	0
HBOa_ihvs	1.39	1.21	1.36	1.21	1.21	0	0	0	0	0
HBSHa	0	0	4.1	0	0	0	0	0	0	0
HBSCa	0	0	0	0	0	1	0	0	0	0
NHBWa	0.53	0.55	5.1	0.55	0.55	0	0	0	0	0
NHBOa	0.53	0.55	5.1	0.55	0.55	0	0	0	0	0

6.6 Balancing

Doubly constrained destination choice models, and all gravity models, require that productions and attractions are equal at the regional level. Because of this, each trip purpose has either its productions or attractions factored before distribution.

It is important to check that this factor is reasonable - roughly between 0.9 and 1.1. For any model, extreme values (e.g., 0.2 or 4.0) could signal issues with the underlying estimation.

For doubly-constrained trip purposes, the size of the factor is critical to proper model behavior. Factors far outside the 0.9 to 1.0 range noticeably impact the model's sensitivity to changes in land use.

6.6.1 SENSITIVITY EXAMPLE

Consider a hypothetical model with two zones and the following attributes:

Table 38: Example Table

zone	households	employment	hbw_p	hbw_a
1	100	75	100	75
2	150	25	150	25

With many more productions than attractions, the balance factor used to scale up attractions is 2.5. After balancing, the attractions per zone are shown in Table 39.

Table 39: Balanced Table 1

zone	households	employment	hbw_p	hbw_a
1	100	75	100	187.5
2	150	25	150	62.5

When doubly constrained, zone 2 will receive 62.5 trips. With an original attraction rate of 1 trip per employee, we would expect adding 25 more employees to zone 2 would cause the trip attractions to increase by 25. This is the case before balancing as shown in Table 40.

Table 40: Increase Zone 2 Employment

zone	households	employment	hbw_p	hbw_a
1	100	75	100	75
2	150	50	150	50

After balancing, the effect is different from expected. Adding 25 more employees to zone 2 caused an increase of nearly 40 trips. A large balance factor drastically increased the model’s sensitivity to changes in employment. Conversely, a very small balance factor will decrease the model’s sensitivity as seen in Table 41.

Table 41: Balanced Table 2

zone	households	employment	hbw_p	hbw_a
1	100	75	100	150
2	150	50	150	100

Attention must be paid to the scale of the balance factor for doubly-constrained purposes in both the base and future-year scenarios. The model automatically writes out a report displaying these factors (balance_report.csv).

After adjustment, the base year factors for the NRV model are all near 1.0.

7 DESTINATION CHOICE MODEL

The NRV model uses a destination choice model to perform trip distribution in the region. The destination choice model replaces the gravity model used in previous versions of the model. There are several advantages for implementing a destination choice model; a destination choice model is a logit model that allows for the consideration of a greater number of independent variables for estimating trip distribution, including the LogSum variable output from the mode choice model. The NRV model uses logsums reported by the mode choice model as an accessibility metric. Conceptually, a logsum is simply an average of the travel options from one zone to another weighted by a traveler’s preferences, value of time, etc. The inclusion of the LogSum variable makes the distribution of trips sensitive to transit, unlike the gravity model. This greater sensitivity improves the resulting trip tables and overall model performance.

The utility coefficients for logsums are well established and rarely modified. Instead, calibration involves the modification of distance polynomial terms to adjust model performance to match observed travel patterns.

One of the main hurdles in the NRV MPO was the lack of adequate survey records. In generation, this was solved by using records from similar MPOs across the state of Virginia. This is not possible with the destination choice model; model travel skims aren’t available for the other MPOs. Table 42 shows the samples for a fully-stratified set of destination choice models. Most have too few samples to be calibrated. It’s important to note that low-income market segments are under-represented in the survey.

Table 42: Stratified Samples

purpose	segment	AM	MD	NT	PM
HBW	ihvi	15	55	22	14
HBW	ihvs	63	91	45	40
HBW	ilvi	1	1	1	1
HBW	ilvs	1	3	0	4
HBW	v0	2	0	1	1
HBO	ihvi	24	267	114	27
HBO	ihvs	50	369	229	92
HBO	ilvs	0	0	3	1
HBO	v0	1	13	3	1
HBSC	ihvi	10	22	3	1
HBSC	ihvs	19	29	8	2
HBSC	ilvi	0	1	1	0
HBSC	v0	0	6	2	0
NHB	ihvi	5	131	28	19
NHB	ihvs	13	289	54	46
NHB	ilvi	0	1	0	0
NHB	ilvs	0	4	0	0
NHB	v0	0	8	0	0

Table 43 shows the samples for the final models to be calibrated. Due to the lack of samples, market segment and time of day stratifications were collapsed completely.

Table 43: Collapsed Samples

purpose	segment	tod	samples
HBW	all	all	361
HBO	all	all	1194
NHB	all	all	598

Due to limitations in survey data sample size the non-home-based work (NHBW) and non-home-based other (NHBO) were combined into one NHB trip purpose.

The destination choice formulation is not used to distribute home-based K-12 school (HBSC) trips; instead, a traditional gravity model formulation is used with destinations controlled to student school enrollment by TAZ. The destination choice formulation is not used to distribute home-based university (HBU) trips – all the university trips are attracted to the university TAZ.

7.1 Model Calibration

The calibration of the destination choice model involves making small incremental adjustments to the distance coefficients to better match observed trip patterns. The model calibration first focuses on matching first-order calibration targets for trip length frequency and average trip lengths by trip purpose. Often segmented distance terms are needed to match the short distance portion of the observed trip length frequency curve. The distance cap may also be adjusted during model calibration to ensure that the model reproduces the tail (longer trips) of the trip length frequency distribution.

The NRV destination choice models were calibrated to reproduce observed trip patterns of trip length frequency distributions.

A trip length frequency distribution (TLFD) plot is simply a histogram of trips by length (time and distance). The observed trip length information comes from the household survey where small sample size results in lumpy observed data. During calibration, it is important not to over-fit the model to this noise.

For each combination of purpose, segment, and time period below, four metrics are presented:

- Trip length frequency of model and survey (distance and time)
- Diagnostic chart of distance polynomial calibration
- Average trip length of model and survey (distance and time)
- Intrazonal percentage of model and survey

Table 44 provides a summary of the model calibration results by trip purpose. The close match between the calibrated trip length profile and the observed trip length profile can be observed in the figures that follow in the next section. The percentage of total trips that are attracted to the production zone (intra-zonal trips) is also compared between the model and the observed data. They also match very closely. The model predicted average trip distance is within 10% of the observed distances for all the trip purposes.

Table 44: Goodness of Fit Measures

Purpose	%Intra-zonal trips		Average trip length (miles)		Average trip length (min)	
	Survey	Model	Survey	Model	Survey	Model
HBW	0.28	0.24	5.88	5.97	10.13	11.93
HBO	11.33	12.31	3.94	4.31	7.33	8.01
NHB	7.86	8.15	3.2	3.53	6.15	7.07

NHTS data was processed into trip length frequency. For all purpose/segment/period combinations, the model is accurately re-creating the shape of the survey TLFD, average trip length, and intrazonal percentages.

Figure 11 through Figure 16 show the trip length frequency distribution for the model output and for the survey data. The final frequencies resulted from final adjustments and smoothing of the functions.

Figure 11: HBW Trip Length Frequency Distribution (Dist)

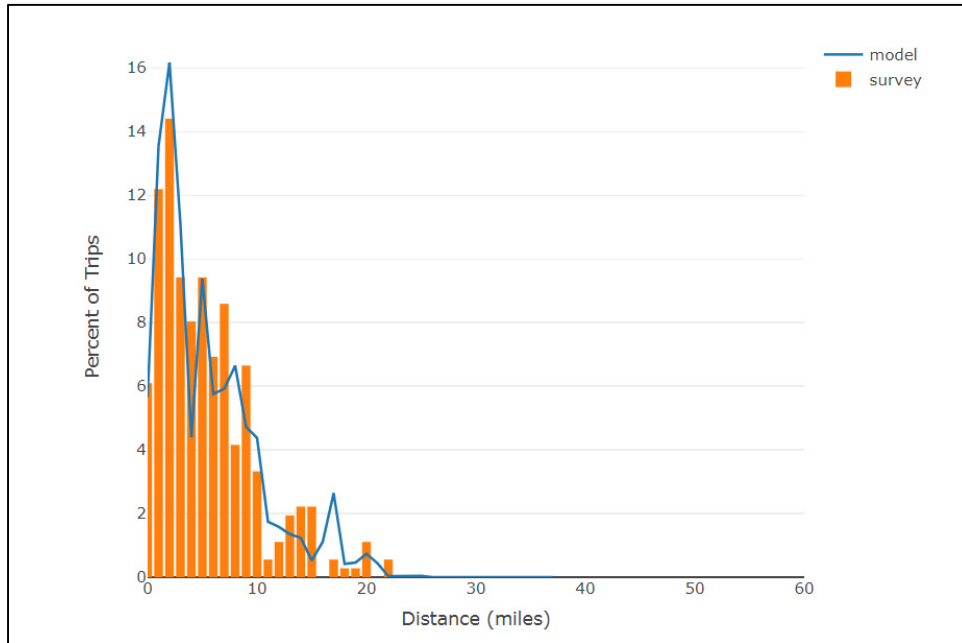


Figure 12: HBO Trip Length Frequency Distribution (Dist)

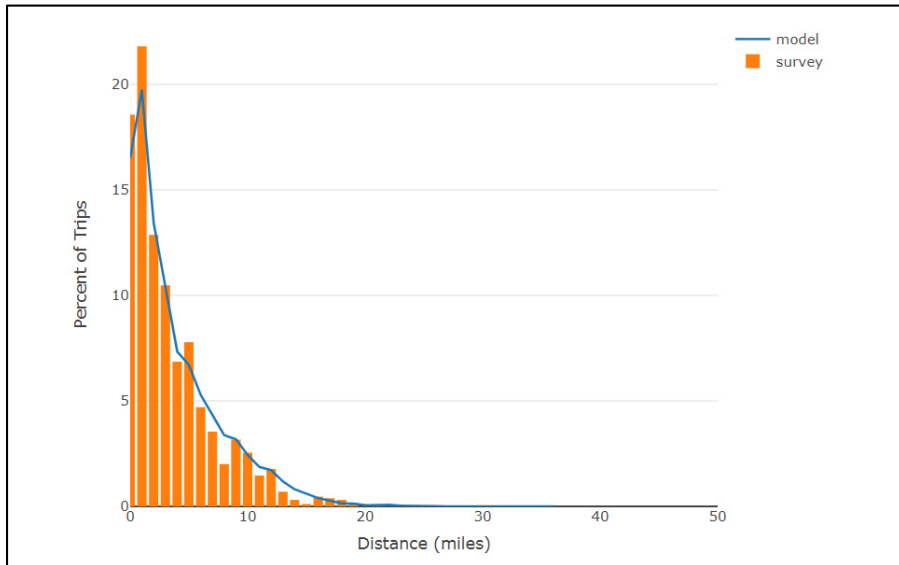


Figure 13: NHB Trip Length Frequency Distribution (Dist)

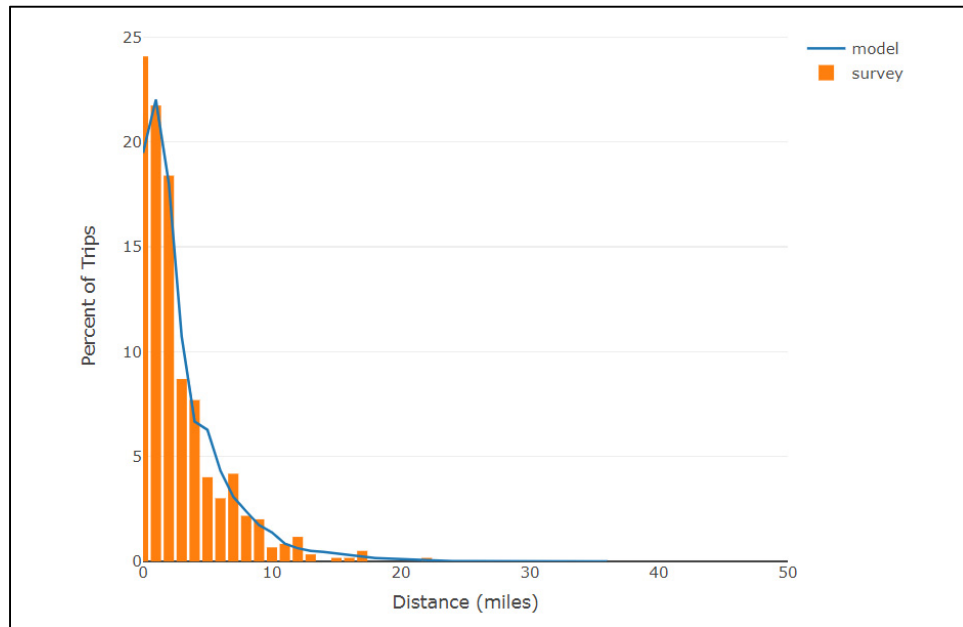


Figure 14: HBW Trip Length Frequency Distribution (Time)

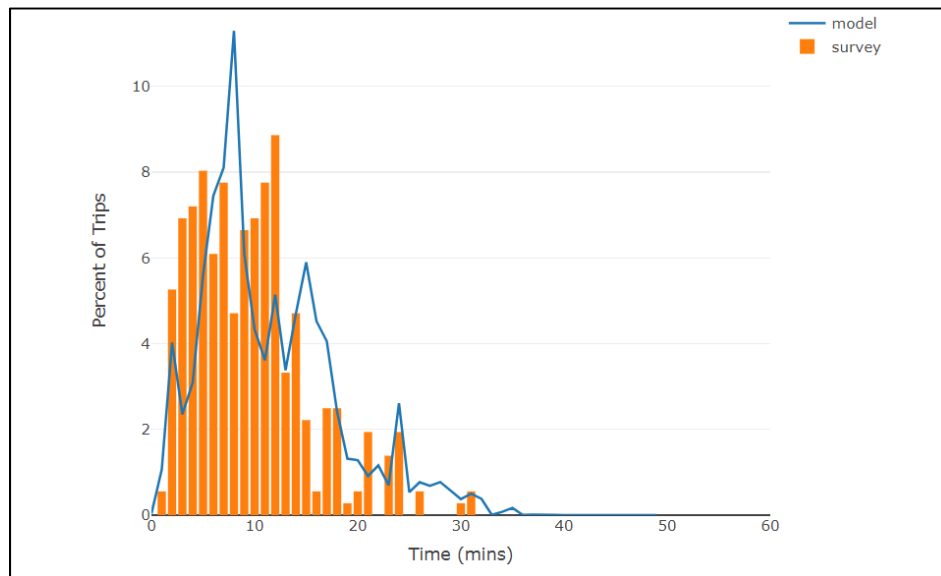


Figure 15: HBO Trip Length Frequency Distribution (Time)

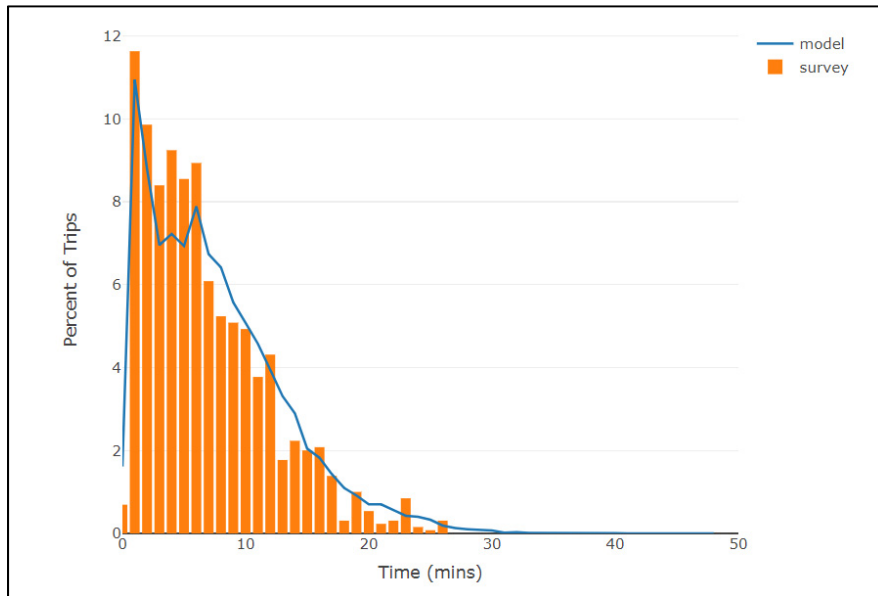
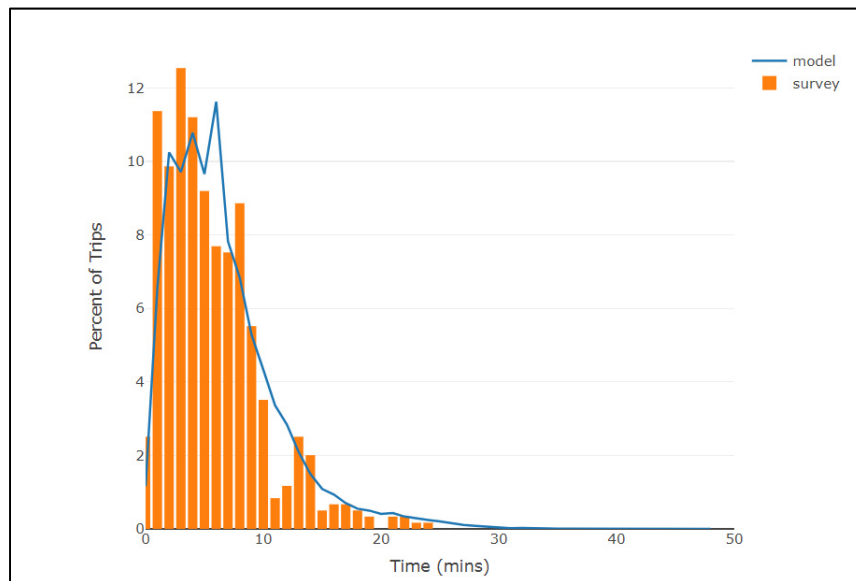


Figure 16: NHB Trip Length Frequency Distribution (Time)



8 HOME-BASED UNIVERSITY TRIPS

The original intent for the NRV model was to develop a sub-model for University travel that was based on data from a University survey in the region combined with the 2018 transit on-board survey dataset. Several attempts to collect student data from the university including the geo-coded location of beds on-campus, the classroom seats/hours by location and the addresses of students was not successful. Only addresses of students were available for the model development.

In addition, the on-board survey data did not collect the exact location details of trips and therefore was not directly useable in its format for this model development. The mode choice of students is discussed in the mode choice section of the document.

The development of the student model combines the datasets for both campuses to estimate the student travel. This means that Virginia Tech patterns will be the driving force for the student sub-model.

Evaluation of the student travel survey resulted in the breakdown of trips as shown in Table 45.

Table 45: Student Trips by Purpose

Purpose	Drive Alone	Shared Ride 2	Shared Ride 3+	Bus	Bike	Walk
HBW	1,870	40	30	690	300	860
HBU	10,590	430	330	10,090	1,790	16,760
HBO	12,470	1,300	990	2,310	540	14,860
NHB	7,440	940	690	1,890	1,450	28,740

8.1 Market Segmentation

Market segmentation is the process of grouping observed students into markets that behave in similar ways. The travel model is then better-able to predict behavior for individual markets than it would be for all students combined.

For travel, the most important distinction between students is whether they live on or off campus.

Data provided locally about the student breakdown of on and off campus is:

- VT on campus: 32,304
- VT off campus: 2,136
- Radford on campus: 3,000
- Radford off campus: 6,401

From the student travel survey, Figure 17 shows a count of the survey samples collected for each. There were more off-campus data samples collected, which is opposite of the actual data for the campuses. The information from the surveys had to be used carefully for the sub-model.

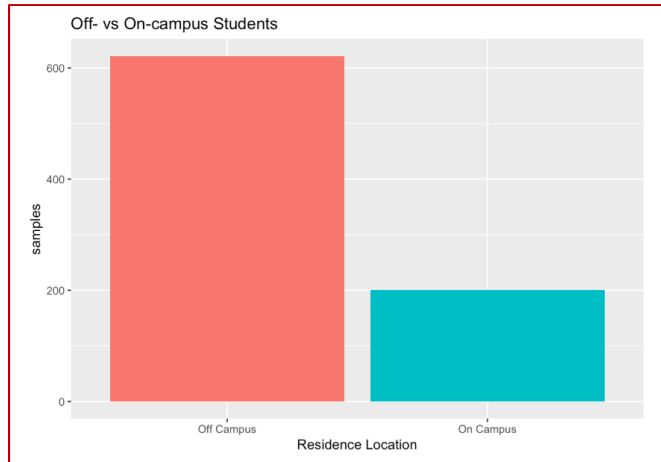


Figure 17: Sample of on and off campus students

8.2 Trip Generation

8.2.1 PRODUCTIONS

The table below shows the production rates for VT students estimated from the VT student survey.

Table 46: VT Production Rates

on_campus	HBO	HBU	HBW	NHB
No	1.35	1.78	0.24	2.05
Yes	2.49	2.65	0.10	2.12

Students living in off-campus households are in the sampling frame of the household survey. As a consequence, the resident trip generation model already accounts for their work, other, and non-home-based trips. On the other hand, the NHTS struggles to collect information on home-based university trips, observing no university trips in the last survey. The HBU rate from the student survey is used instead.

On-campus students live in group quarters, and as a result, are not captured by the household survey at all. The VT student survey provides estimates for all of their trip production rates.

Attraction rates are borrowed from low-income resident households with insufficient vehicles. In addition, on-campus enrollment is included as an attractor.

8.3 Time of Day

University trips have different time of day profiles from other trip purposes due to the scheduled nature of daily classes. Table 47 shows the breakdown by purpose. This information is used in the model to distinguish travel throughout the day by students instead of using the distribution of residents.

Table 47: Distribution of HBU Trips by Time of Day

Purpose	Period of Day	%
HBO	AM	4
HBO	MD	35.1
HBO	NT	54.5
HBO	PM	6.4
HBU	AM	10.1
HBU	MD	64.5
HBU	NT	18.2
HBU	PM	7.2
HBW	AM	8.3
HBW	MD	43.6
HBW	NT	32.7
HBW	PM	15.4
NHB	AM	1.4
NHB	MD	66.5
NHB	NT	25.9
NHB	PM	6.2

8.4 Trip Distribution

8.4.1 OFF-CAMPUS

Both Radford and Virginia Tech universities provided an address file for off- campus student housing. Some addresses were not local to the region⁸ and therefore were dropped. The addresses that were acceptable were used as model inputs, which removes the need for a distribution model. Instead, the addresses were simply geocoded to the TAZ layer and summarized by TAZ.

⁸ Most likely parents addresses or where bills are sent.

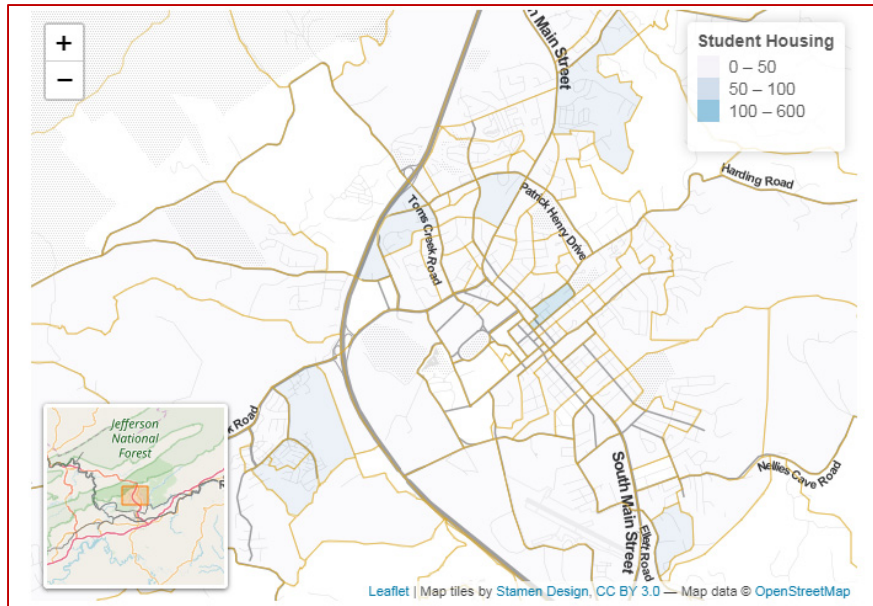


Figure 18: Va Tech Off-campus Address Geocoding

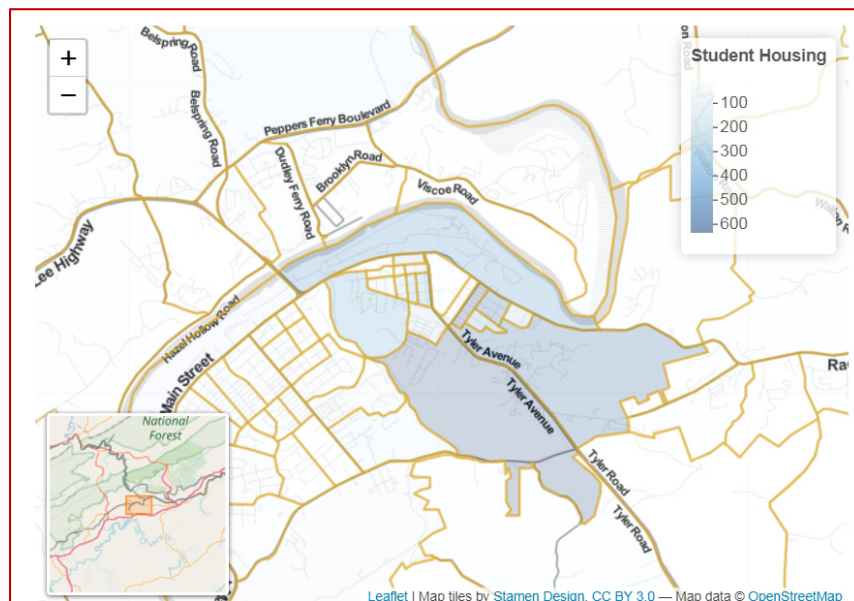


Figure 19: Radford Off-Campus Address Geocoding

8.4.2 ON-CAMPUS

The trips made by on-campus students use a gravity model for distribution. Due to issues with the VT university survey, it was not possible to use it for calibration (inadequate address information for trips).

Instead manual calibration was performed to make sure that most trips were in and around campus and as logical as possible given the limited datasets. **Figure 20** and **Figure 21** show the TLFD for the student purpose.

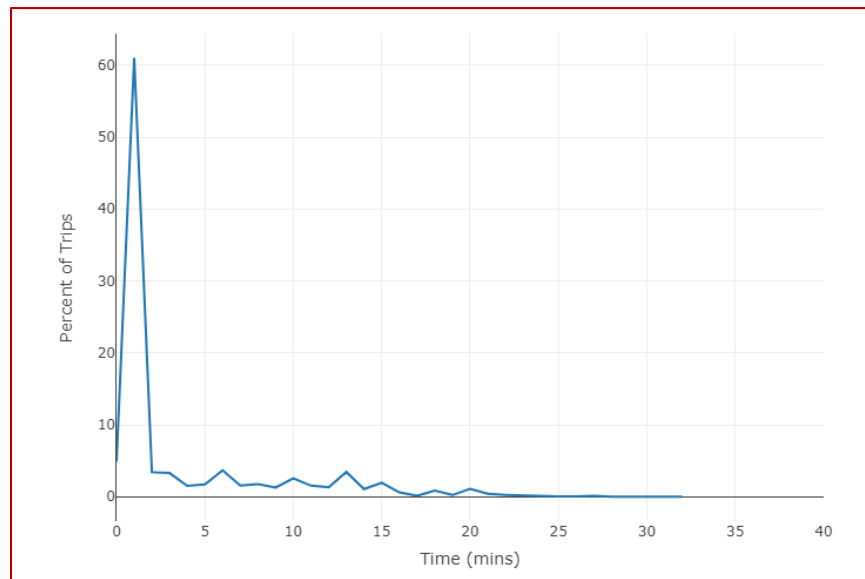


Figure 20: Student TLFD (time)

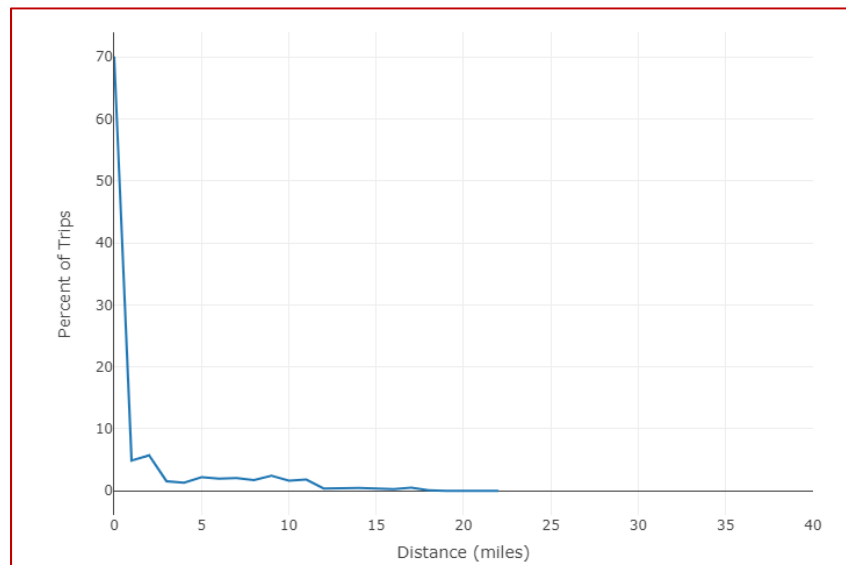


Figure 21: Student TLFD (distance)

8.5 Directionality

The directionality step of the model converts the modal trip tables from production-attraction (PA) format into origin-destination (OD) format. This is only done for auto modes (drive alone and shared ride 2 and 3).

The table below shows, for each time period, the proportion of HBU auto trips traveling from home to university. As expected, the AM period is predominantly to-university while the PM and NT periods are the reverse.

Table 48: Student Directionality by Purpose and TOD

Purpose	Period	Percent
sHBO	AM	0.665
sHBO	MD	0.452
sHBO	NT	0.407
sHBO	PM	0.619
HBU	AM	0.952
HBU	MD	0.643
HBU	NT	0.388
HBU	PM	0.314
sHBW	AM	0.892
sHBW	MD	0.649
sHBW	NT	0.351
sHBW	PM	0.092
sNHB	AM	0.500
sNHB	MD	0.500
sNHB	NT	0.500
sNHB	PM	0.500

8.6 Occupancy

For auto modes, occupancy rates are used to convert person trips (used in destination and mode choice models) into vehicle trips for network assignment. These rates are also specified by time of day, but in the case of HBU trips, they do not change.

Table 49: Occupancy Rates

Purpose	Mode	AM	MD	NT	PM
HBU	da	1.00	1.00	1.00	1.00
HBU	sr2	2.00	2.00	2.00	2.00
HBU	sr3	3.00	3.00	3.00	3.00
sHBO	da	1.00	1.00	1.00	1.00
sHBO	sr2	2.00	2.00	2.00	2.00
sHBO	sr3	4.00	3.50	3.70	3.71
sHBW	da	1.00	1.00	1.00	1.00
sHBW	sr2	2.00	2.00	2.00	2.00
sHBW	sr3	3.00	3.00	3.00	3.00
sNHB	da	1.00	1.00	1.00	1.00
sNHB	sr2	2.00	2.00	2.00	2.00
sNHB	sr3	3.74	4.33	3.58	3.50

9 EXTERNAL MODELS

Trips in the NRVMPD model can be classified in three, distinct categories:

- Internal-to-Internal (II)
 - These trips start and end within the model region.
- External-to-External (EE)
 - These trips pass completely through the model region.
- Internal-External and External-Internal (IEEI)
 - An IEEI trip has one trip end within the model region and one outside.

The primary source of information for II trips is the household survey, but the survey contains no information on EE trips, and very little information about IE/EI trips. This is one of the primary reasons for purchasing and processing mobile phone data. Once processed, the mobile phone data is used to develop an observed EE trip table and multiple parameters for modeling IEEI trips.

Figure 22 and Table 50 provide general information about the external stations' location and attributes. Subsequent sections add estimated station attributes.

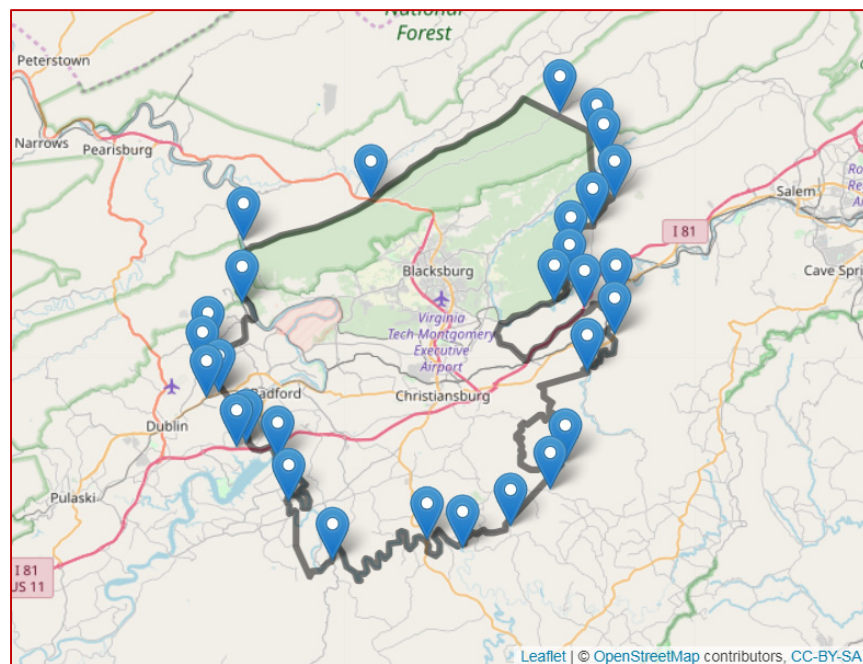


Figure 22: External Station Locations for NRV

Table 50: External Station Attributes

Station ID	EE %	Yearly Assumed Growth	AWDT 2016
340	0.587	1.0146	47549
342	0.15	1.0094	1838
345	0.533	1.0115	43522
348	0.15	1.0099	751
349	0.15	1.0047	1887
351	0.15	1.0018	13928
352	0.15	1.0047	774
356	0.15	1.0153	744
358	0.15	1.0045	462
360	0.15	1.0054	2375
362	0.023	1.0104	907
377	0.023	1.004	5215
381	0.023	1.0048	176
384	0.023	1.0081	693
387	0.023	1.0079	1303
390	0.023	1.0192	208
422	0.15	1.0047	405
427	0.635	1.0159	790
428	0.525	1.0099	1800
432	0.697	1.0127	5800
435	0.15	1.0047	345
437	0.014	1.0047	228
439	0.014	1.0098	13206
442	1.243	1.031	250
460	0.014	1.0047	291
466	1.762	1.0596	200
467	1.468	1.0352	200
468	0.624	1.0056	350
470	0.15	1.0151	509

9.1 Development of EE Trip Table

The disaggregation process described in the mobile phone processing step results in a trip table with trips from every zone to every other zone, including the external stations. An external station is defined as the point on a modeled roadway as it crosses the planning boundary (i.e. edge of the model boundary). For ease of terminology, external stations are often referred to as “external zones” even though they actually represent a point location. This trip table can be used to create one of the required model inputs: the through trip table. The through trip table contains all of the trips from each external station to every other external station. Before it can be used, the raw trip table must be balanced in order to meet the standard assumption that trips entering the region are equal to trip exiting the region on a daily basis. This is accomplished by adding the transpose of the raw table and dividing by two.

$$(EE+EET)/2$$

9.1.1 CALCULATE EE PERCENT AT STATIONS

For future-year applications of the model, the base EE trip table must grow to accommodate the growth in average weekday traffic (AWDT) at the external stations. Recall that the AWDT includes both EE trips and IEEI trips, and the base year split between these two is informed by the observed data. For future year applications, the simplifying assumption is that the split between EE trips and IEEI trips at each station will remain constant, even as the overall AWDT increases or decreases.

It should be noted that, due to the dis-aggregation process, all external stations within the same mobile district start with the same EE percentage. Manual adjustments are made to that base assumption to ensure certain logical features are present (e.g. that interstates have a higher percentage of EE trips than local streets).

9.2 IEEI Trip Model

Trips that enter or leave the model region are collectively referred to as IEEI trips.

9.2.1 TRIP GENERATION

In model application, the number of IEEI trips is controlled by the traffic count where the IEEI trips produced at a given external station is equal to the AWDT minus the EE trips. IEEI trips are attracted to internal zones based on the socio-economic data in the zone. The SE data in the zone is converted into attractions based on a regression formula, which is estimated from the observed mobile phone data. The IEEI regression model estimation file is created by summarizing IEEI trip attractions by TAZ and joining to each record the SE data for each TAZ. Sample estimation data for the NRVMPPO region is shown in Table 51.

Table 51: Sample IEEI Estimation Data

Zone	Trips			SE Data		
	EI	IE	Total	Households	Industry	Nonindustry
1	481	513	994	0	97	1008
2	1005	1073	2079	0	0	2310
3	251	268	519	0	0	577
4	516	551	1066	1	8	1176
5	121	130	251	0	0	279
6	966	1031	1997	0	0	2219

Linear regression is used to determine the relationship between the SE data and the number of IEEI trips for each zone. The estimated model coefficients are shown below. Estimated separately, the individual coefficients for non-industrial employment types are not statistically significant. Grouping those categories together allows for all employment to generate IEEI attractions at a reasonable rate, see Table 52.

IEEI attractions have a strong geographical component, with trips being more likely to employment at the edges of the region. This distance component is handled by the distribution model. In effect, the role of the IEEI attraction model is to balance, through the use of different coefficients, the attractiveness of different employment types *before* considering their location.

Table 52: IEEI Estimation Results

	Coefficient (standard error)
Intercept	-34.6610 (21.1542)
Households	1.5273 (0.0909) ^{***}
Industry	0.5633 (0.1283) ^{***}
Non-Industry	1.0438 (0.0446) ^{***}
R²	0.7467
Adj. R²	0.7440
Num. obs.	284
RMSE	241.7564
^{***} p < 0.001, ^{**} p < 0.01, [*] p < 0.05	

As with the other attraction models, the intercept term is included during estimation to achieve the proper model sensitivity to changes in employment, but it is not used during application. Instead, the attractions predicted at employment locations are balanced to match productions.

9.3 Reasonableness Check

The table below lists rates from other models that used the same regression equation. While the absolute values of the coefficients are larger, the relative size matches closely with those observed in the Triangle (NC) 1997 survey. Households and Non-industrial employment have larger co-efficients than the industrial employment.

Variable	FBRMPO	TRM 97 OD Survey
Households	0.24	0.25
Industry	0.37	0.06
Non-Industry	0.18	0.27

9.4 Non-Resident NHB Trips

Some of the people traveling into the region will make additional trips before returning home, just like residents. The traditional IEEI model does not capture this travel. The non-home-based non-resident (NHBNR) model fills in this gap.

The standard non-resident model applies a trip rate to the IEEI attractions estimated during trip generation. This approach leads to NHBNR trips spread throughout the model region - even in areas with no IEEI travel.

The NRV model waits until the IEEI trips have been distributed. After distribution, all IEEI trips originate at the external station and travel to the internal zones. IEEI trips are assumed to be an even split between EI and IE. EI travelers are assumed to make NHB trips at the same rate as residents. The percentage of NHB trips is calculated using the formula below:

$$\text{Percent} = (\text{NHBr} / \text{HBr}) * 100$$

Where:

- NHBr is the total number of resident non-home-based trips and
- HBR is the total number of resident home-based trips.

For residents, 39% of trips are non-home-based.

Figure 23 illustrates the density of NHBNR productions in the model. It follows a logical pattern of concentration around likely non-resident destinations.

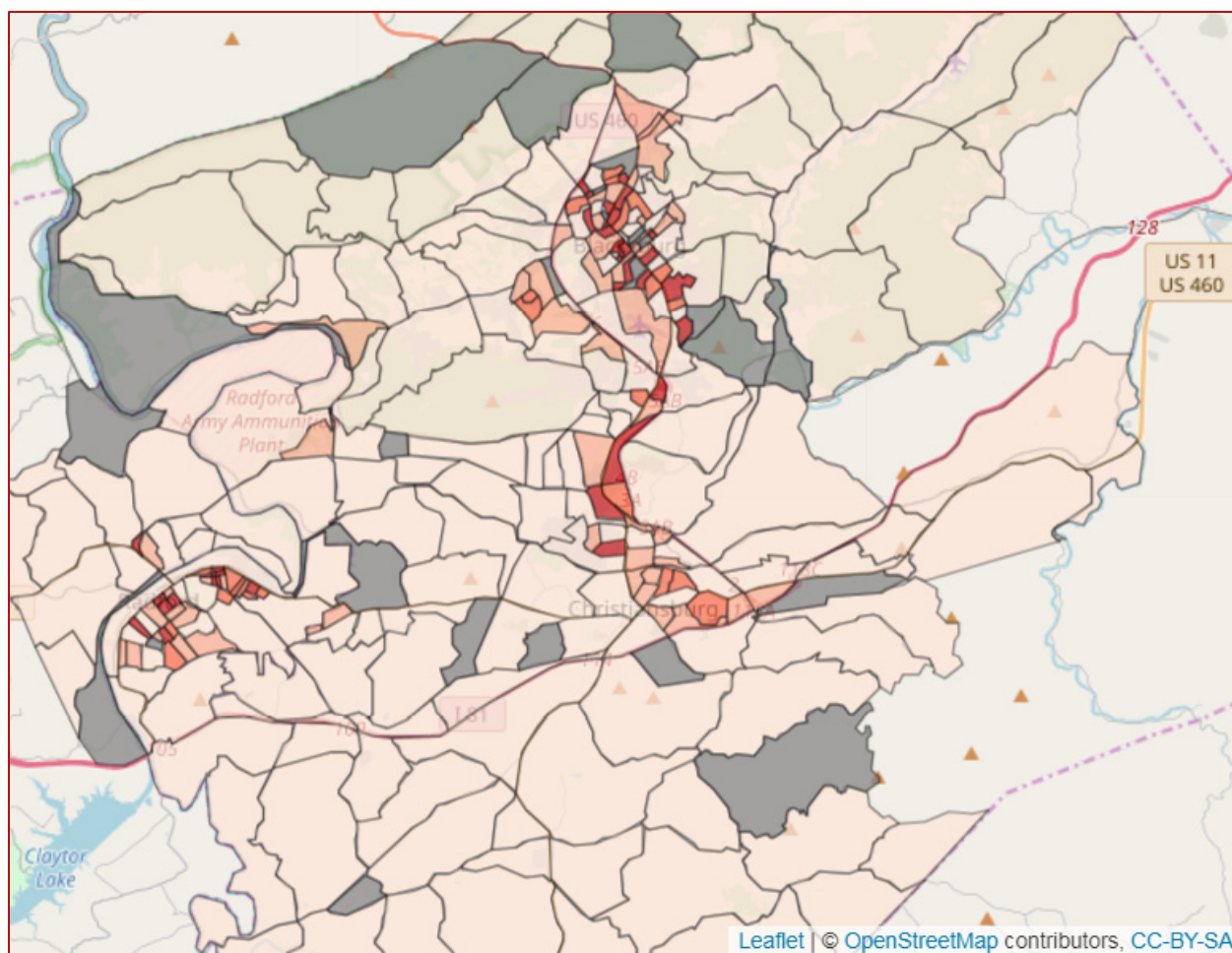


Figure 23: NHBNR Assumed Productions

NHBNR attractions are set to the same values used for resident attractions, which assumes that they are likely to be attracted to the same places once inside the region.

9.5 External/Internal Trip Distribution

The processed mobile phone data by time of day was used to calibrate the trip distribution model for the IEEI trips.

The trip distribution model for the IEEI trips takes the form of a gravity model. The gravity model links the trip productions and attractions from the trip generation step using the midday travel time from the highway network.

Trip distribution models are calibrated through the implementation and adjustment of friction factors. Friction factors can take on several forms, the NRVMP model uses the gamma function. The parameters in the friction factor function are adjusted until the estimated trip length distribution in minutes closely matches the observed trip length distribution in minutes, and the average travel time results from the model are within +/- 5% of the observed data. One of the biggest benefits of mobile

phone data is that it provides a data source for calibrating the IEEI trip distribution model. Once the mobile data is processed from district to model zones, this information can be paired with zone to zone travel time or distance in order to visualize the trip length distribution for the IEEI trips.

9.6 Calibration

Table 53 highlights the comparison for the peak period of observed and estimated trip lengths. Observed data was not available for the off-peak. The final estimated trip length distribution as compared to observed trip length distribution is shown in Figure 24 and Figure 25 for the peak (AM and PM) considering both travel time and distance.

Table 53: IEEI Trip Length Results

Period	Distance Observed (miles)	Distance Model (miles)	Time Observed (min)	Time Model (min)
Peak	13.76	13.04	16.85	16.48
OffPeak	NA	1054	NA	12.96

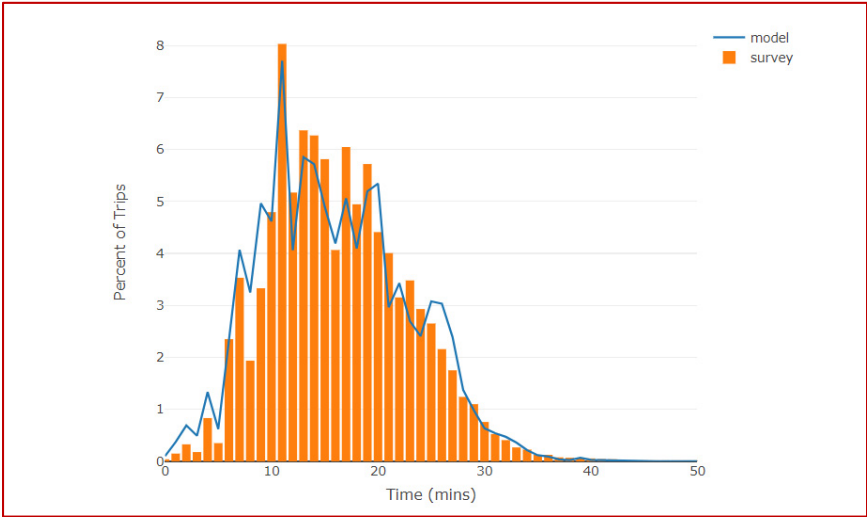


Figure 24: Peak IEEI TLFD (Time)

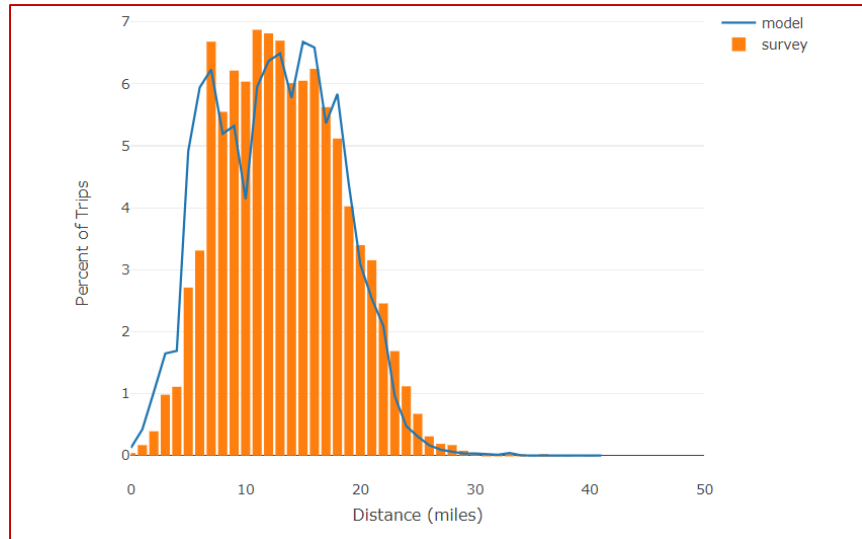


Figure 25: Peak IEEI TLFD (Dist)

Graphs of off-peak (MD and NT) TLFD are provided in Figure 26 and Figure 27 but do not include observed data for the comparison. The same distribution model for resident NHB trips is applied for NHBNR.

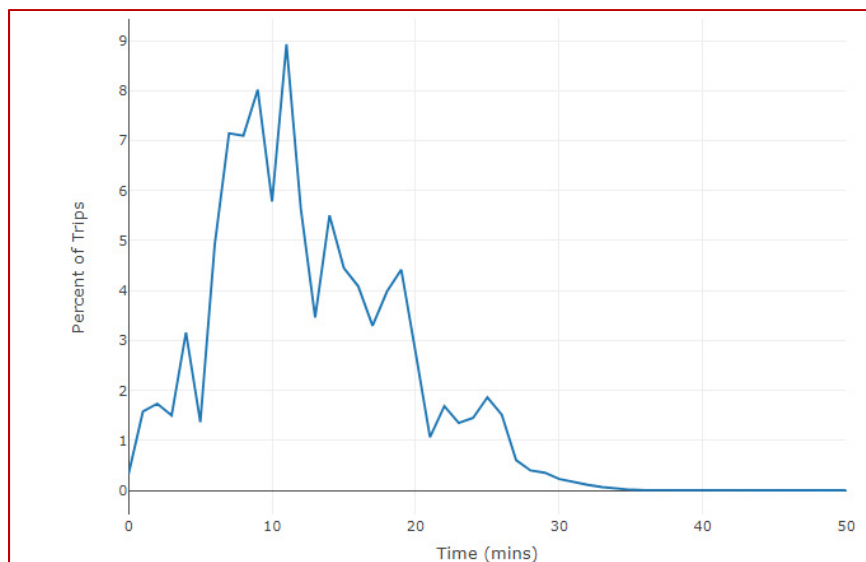


Figure 26: Off-Peak IEEI TLF (Time)

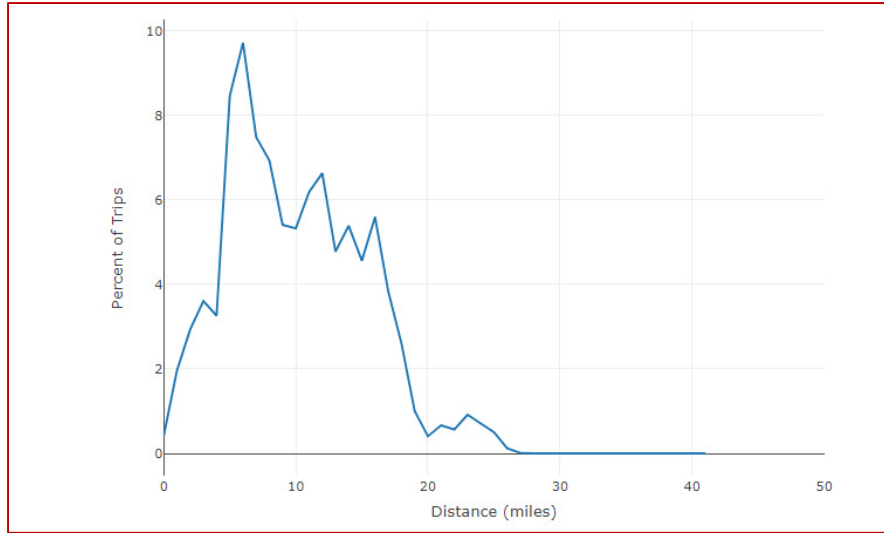


Figure 27: Off-Peak IEEI TLFD (Dist)

10 MODE CHOICE

10.1 Mode Choice Model Description

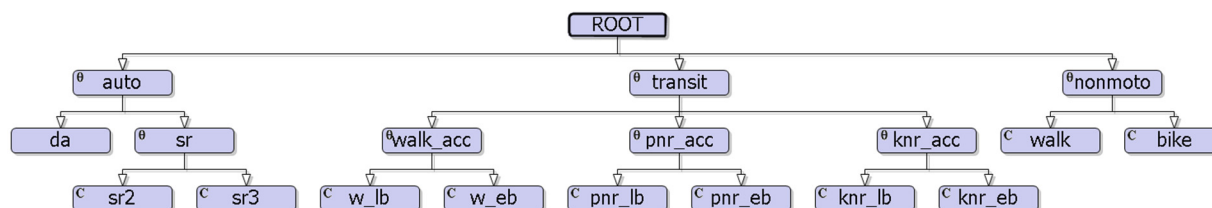
Mode choice models are mathematical expressions that are used to estimate the modal shares of the travel market given the time and cost characteristics of the various competing modes, the demographic and socio-economic characteristics of the residents, and the un-included attributes of the modes represented in the model. Mode choice models are designed to be an integral link in the travel demand chain, with possible feedback mechanisms to a number of related model components such as trip distribution.

The mode choice model is calibrated using the person trip tables by trip purpose from the trip distribution model. Calibration requires the modification of alternative specific constants to try to match estimated trips to observed trips by purpose and mode. Adjustments to path-building or coefficients may be required. It is an iterative process that considers transit network assumptions, paths, and other inputs.

10.1.1 BASIC LOGIT MODEL MATHEMATICS

The mode choice structure for the NRV mode choice model is a simple nested model that includes sub-modes beneath each primary mode choice (Figure 28). The NRV model includes a 'nest' for premium transit for possible future use.

Figure 28: NRV Nested Mode Choice Model



10.1.2 MATHEMATICAL FORMULATION FOR LOGIT MODELS

The standard logit formulation can be expressed as:

$$P_i = \frac{e^{U_i}}{\sum_k e^{U_i}}$$

Where:

- P_i = is the probability of a traveler choosing mode i
- U_i = is a linear function of the attributes of mode i that describe its attractiveness
 $\sum e^{U_i}$ = is the summation of the linear functions of the attributes over all the i
- k = alternatives (k) for which a choice is feasible

The utility expression for each available mode (i) is specified as a linear function which incorporates a range of variable types, including time, cost, locational measures, and the socio-economic characteristics of the traveler. For example:

$$U_i = \beta_1 * Time_i + \beta_2 * Cost_i + \beta_3 * Location_{var} + \beta_4 * SE + \beta_0$$

Where:

- U_i = is the utility for mode i
- β_0 = is a constant specific to mode i that captures the overall effect of any significant variables that are missing or unexplained in the expression (e.g., comfort, convenience, safety)
- β_1 = is a set coefficients describing the level-of-service (in travel time) provided by mode i (e.g., in-vehicle time, wait time, walk time)
- β_2 = is a set of coefficients describing travel cost, (e.g., transit fare, automobile operating cost, parking costs)
- β_3 = is a set of coefficients describing the specific attributes of the trip interchange (e.g., Central Business District destination, park and ride lot use)
- β_4 = is a set of coefficients describing the influence of each socio-economic characteristic of the traveler (e.g., income group, auto ownership)

The travel time variables are typically disaggregated into in-vehicle and out-of-vehicle time at a minimum. Out-of-vehicle time is further stratified into walk time, initial wait, and transfer wait time – the latter two categories being applicable to the transit modes only. Similarly, travel cost is often disaggregated into the more general out-of-pocket cost (i.e., automobile operating cost and transit fare) and destination parking cost.

Locational variables in utility expressions are used to reflect a set of unique geographically based characteristics, such as a Central Business District (CBD). Alternatively, these geographic attributes may

be represented in the form of land use variables such as employment and/or population density. A wide variety of variables are possible in the socio-economic category (SE) including variables that measure the relative wealth of the trip maker (income or auto sufficiency) or reflect other household characteristics (i.e., workers per household). Finally, an alternative specific constant reflects the unexplained behavior, or the un-included attributes of that mode. The individual coefficients associated with each variable reflect the relative importance of each attribute.

In the nested model structure shown in Figure 28 the formulation employs three multinomial logit models, one for the primary choice of mode among non-motorized, transit and auto, a second level choice among auto sub-modes (drive-alone and shared-ride) and another second level choice among transit modes (local and premium). In application, the model independently addresses the auto sub-mode and transit sub-mode choice first. For example, the auto sub-mode is expressed as:

$$P_{DA} = \frac{e^{U_{DA}}}{e^{U_{DA}} + e^{U_{SR}}} \quad P_{lb} = \frac{e^{U_{lb}}}{e^{U_{lb}} + e^{U_{pb}}}$$

A composite of the utilities of the auto sub-mode and transit sub-mode choices then represent auto and transit respectively in the upper tier of the model structure. This composite measure is the natural logarithm of the denominator of the logit model, often termed the "LogSum". The LogSum term is effectively the total utility provided by the sub-modes of a particular primary mode.

For example, a LogSum is calculated for each of the second level nests as:

$$LogSum_A = -\ln[e^{U_{DA}} + e^{U_{SR}}] \quad LogSum_T = -\ln[e^{U_{lb}} + e^{U_{pb}}]$$

The LogSum terms for the auto and transit sub-mode choice then appear in the utility expression for the primary mode level. An example for the transit next is shown as:

$$P_T = \frac{e^{\beta_T * LogSum_T}}{e^{\beta_T * LogSum_T} + e^{\beta_A * LogSum_A}}$$

The value of the LogSum coefficients in the upper tier of the model (i.e., auto versus transit), is an indicator of the degree to which the lower level choices form a sub-choice that is distinct from the primary mode alternatives. A value of 1.0 indicates that the lower level modes are not a sub-choice but rather are full options equally competitive with the primary modes. In this instance, these lower level choices can be simplified or included directly in the upper level. A value of 0.0 would indicate that the

lower level choices are perfect substitutes for each other. Values between 0.0 and 1.0 indicate the extent to which the lower level choices represent a sub-choice.

For the non-motorized nest, there are no sub-modes so the utility for this nest is computed just like the drive alone or shared ride utility (i.e. the lower nest).

10.1.3 MARKET SEGMENTATION CONSIDERATIONS

Traditionally, a larger number of trip purposes are maintained in the trip generation and trip distribution models than in mode choice. Common practice has been to merge the subset of non-work purposes into a single purpose, resulting in three basic purposes for mode choice modeling - Home-Based Work, Home-Based Non-Work, and Non-Home Based. This simplification stems from the notion that household and individual travel behavior properties, as translated into elasticities, are relatively similar when considering the choice of mode. This is the approach used for the NRV model with the addition of a university purpose and a K12 School purpose. The mode choice for school trips does not include transit, but it does include non-motorized, drive alone and shared ride. Typically, school bus trips are not modeled in the mode choice model, as its utility is hard to estimate and calibrate and thus a portion of school trips are reduced to account for school bus mode.

Although the existing transit service in the NRV region only includes local bus, the model design includes an option for evaluating a premium transit mode, such as express bus.

10.1.4 ASSERTED MODEL COEFFICIENTS

Appendix C presents the set of mode choice coefficients for the mode choice models. These coefficients are transferred from other models, and represent generally accepted industry standard coefficients for regions without sufficient data for a mode choice calibration. Logical and consistent nesting coefficients are applied at each level of the nest.

10.2 Mode Choice Calibration and Results

Model calibration is the process of establishing proper values for the alternative specific constants. A critical part of this process is to verify that the calibrated constants tell a coherent and plausible story about travel behavior, and not just reproduce the base year modal shares. The identification of constants must follow from a good understanding of the expected effect of non-included attributes, and the calibrated values must be logical and consistent with best practices.

Mode choice model calibration is done by comparing model outputs to survey data. As in this model, it is usually the case that multiple surveys are required. This is primarily due to two factors:

- Household surveys are usually a small sample size (< 2% of the population).
- Non-auto modes are usually a small percentage of travel (< 5% of trips).

These two factors together make it difficult for a general household survey to accurately measure transit ridership. On-board surveys (surveys given directly to transit riders “on-board” the transit route) are used to address this problem.

In NRV, university student trips are also important to get right. These are only partially captured by the household survey (details below). A travel survey of university students is used to get better observed data. In total, four surveys are combined to generate an estimate of total trips by purpose, income group, and mode for the region:

1. Household Survey(NHTS)
2. Virginia Tech On-Board Survey
3. Radford On-Board Survey
4. Virginia Tech Student Survey

The following acronyms are used in this section:

- HBW: home-based work
- HBU: home-based university
- HBSC: home-based school (K12)
- HBO: home-based other
- NHB: non-home-based
- il: low income
- ih: high income

10.3 Household Survey

10.3.1 SCHOOL BUSES

The travel model does not account for school busses. Instead, a fixed percentage of trips are removed for each purpose and market. The percentages in Table 54 are calculated from the survey. School bus trips are then removed from household survey before target creation.

Table 54: Surveyed school bus use

purpose	market	percent
HBO	ilvi	2.69
HBO	ilvs	3.42
HBO	ihvi	1.22
HBO	ihvs	1.30
HBSC	all	50.42

10.4 Initial MC Targets

While the market segmentation in the model includes purpose, auto sufficiency, and income, the on-board survey did not ask about auto-ownership or household size. As a result, the model can only be calibrated by purpose, income group, and mode. Target creation starts by summarizing the household survey by these groups as shown in Table 55.

Table 55: Target Trips by Market (percent of row total)

purpose	income	da	sr2	sr3	bus	bike	walk
HBW	il	41.0%	14.0%	3.0%	18.0%	13.0%	11.0%
HBW	ih	82.0%	9.0%	3.0%	0.0%	1.0%	5.0%
HBO	il	17.0%	24.0%	12.0%	16.0%	3.0%	27.0%
HBO	ih	49.0%	26.0%	14.0%	1.0%	0.0%	11.0%
HBSC	all	21.0%	35.0%	36.0%	0.0%	0%	8.0%
NHB	all	56.0%	24.0%	14.0%	1.0%	0.0%	5.0%

10.5 Student Survey

Students living on campus are in group quarters and are not part of the household survey's sampling frame. Therefore, these trips must be added, increasing the total observed trips.

Off-campus students live in households, which means the household survey accurately captures their trip making for work, other, and non-home-based trips; however, the NHTS does not capture any HBU trips. As a result, off-campus HBU trips are also added to the survey.

In summary, university trips added to household survey totals:

- Off-campus students: HBU
- On-campus students: all trips
-

Without income data from the university survey, the on-board survey is used to get a rough estimate by looking at the income breakdown of HBU trips. This percentage is used to split on-campus HBW and HBO trips into resident trip purposes by market segment. The HBU purpose for off-campus students is not stratified by market segment in the model.

10.6 On-board Surveys

The biggest challenge with both Radford and VT on-board surveys was the absence of adequate trip purpose information; the surveys did not ask about both ends of the trip. This meant that a trip could be classified as “work” or “university”, but not home-based or non-home-based because both ends of the trip locations need to be identified. This was addressed by using a previous on-board survey from 2011 that did ask about both ends. The percentage distribution of trips by purpose was assumed to be the same in 2016.

10.7 Replacing Transit Trips

Due to the small sample size of the household survey and the small relative size of the transit market, the household survey cannot be used to reliably measure transit trips. Instead, any transit trips in the household survey are removed and replaced with trips from the expanded on-board surveys. This would change the total trips by market, and must be corrected for by adjusting the non-transit trips from the household survey.

The student trips were not adjusted in this manner for two main reasons. Transit trips are a larger percentage of student travel and can be better-captured by their survey. In addition, issues with both the VT student and on-board surveys made it impossible to combine them without drastically changing the mode share to unreasonable levels of transit ridership.

As a final check, the trip totals by market and income group are compared before and after the transit modification to ensure they have not changed. Table 56 provides that check.

Table 56: Trip conservation check

purpose	income	before	after	difference
HBW	il	3761	3761	0
HBW	ih	47941	47941	0
HBO	il	10488	10488	0
HBO	ih	128609	128609	0
HBSC	all	8436	8436	0
NHB	all	97727	97727	0

10.8 Calibration Results

The model was run iteratively. At each iteration, the trips by sub-market were compared to the targets calculated above. The alternative-specific constants for each alternative (other than the reference) were adjusted according to the following formula:

$$Adjustment = \ln\left(\frac{target}{model_{result}}\right) * dampening\ factor$$

Where:

- Adjustment is the value to be added to the ASC,
- target is the observed number of trips,
- model_result is the estimated number of trips,
- dampening_factor is a value between 0 and 1 to slow the rate of adjustment.

After the ASCs are adjusted, the model is run again and a new adjustment is calculated. This process continues until the model results are sufficiently close to the observed trips by mode.

Table 57 shows the model results after this process for the HBW purpose, but all purposes match closely.

Table 57: Model Transit Target Comparison (HBW)

purpose	mode	target	model
HBW	da	79.7%	79.7%
HBW	sr2	9.7%	9.7%
HBW	sr3	3.3%	3.3%
HBW	bus	0.1%	0.1%
HBW	bike	1.9%	1.9%
HBW	walk	5.3%	5.3%

10.9 Validation

The NRV model generates 24,100 boardings compared to the target boardings of 24,700. Given the large share of university students that use the systems in the region and the difficulty in obtaining accurate data for student household locations and trip making, it is difficult to get route level ridership more reasonable than what is presented in Table 58. During calibration some routes were adjusted to make them loops instead of directional routes and slight adjustments to dwell times to better reflect local boardings.

Table 58: Route-Level Validation

route	model	target
240	640	500
CARPENTER BOULEVARD	0	350
CRC SHUTTLE	160	740
EXPLORER	780	110
HARDING AVE	590	510
HETHWOOD A	2000	2290
HETHWOOD B	1870	1620
HOKIE EXPRESS	290	1800
MAIN STREET NORTH	500	1390
MAIN STREET SOUTH	740	1390
PATRICK HENRY	1310	1860
PROGRESS STREET	1100	2690
Route 10	1910	960
Route 20	370	200
Route 30	60	110
Route 31	30	20
Route 40	500	90
Route 50	2710	430
Route 60	1510	370
TOMS CREEK	3120	3140
TWO TOWN TROLLEY	890	170
UNIVERSITY CITY BLVD	1220	2180
UNIVERSITY MALL SHUTTLE	1830	1720

11 COMMERCIAL VEHICLES

In an urban area, commercial vehicles can account for increased congestion on major facilities. In most instances the travel behavior of commercial vehicles is much different from person trip behavior, including differences in trip rates and trip distribution. In an effort to better capture the commercial vehicle element of traffic in the NRV model, a separate commercial vehicle model was developed. The model includes commercial vehicle trip estimation for three commercial vehicle types: commercial autos and vans (CV), single unit trucks (SUT), and multi-unit trucks (MUT).

11.1 Trip Generation

In the absence of field collected commercial vehicle data the trip generation approach follows that outlined in the Federal Highways 2007 Quick Response Freight Manual II (QRFM)⁹. In this case, the method assumes that productions equal attractions and one set of rates is provided. The employment rates implemented for NRV were adapted from the Triangle Cordon Survey (Raleigh-Durham, NC). The applied rates for CV, SUT, and MUT are summarized in Table 59.

Table 59: Commercial Vehicle Trip Rates Applied to Household and Employment Variables

Vehicle Type	HH	Industry	Retail ¹⁰	Service ¹¹	Office
Commercial Vehicles (CV)	0.25	1.02	0.89	0.44	0.044
Single Unit Trucks (SUT)	0.10	0.27	0.25	0.07	0.07
Multi-Unit Trucks (MUT)	0.03	0.14	0.08	0.01	0.01

The results from the application of the above rates to NRV shown in Table 60.

Table 60: Estimated Commercial Vehicle Trips (Productions and Attractions)

Vehicle Type	NRV	
	Trips	Percent of Total
Autos/Vans (CV)	40,000	74%
Pickups (SUT)	11,000	20%
Trucks (MUT)	3,000	6%
Total	54,000	100%

⁹ Publication No. FHWA-HOP-08-010

¹⁰ Includes retail special generators

¹¹ Includes special generators for hospitals, university/college, and airport

11.2 Trip Distribution

Commercial vehicles (CVs), single-unit trucks (SUT), and multi-unit trucks (MUT) are distributed using a doubly-constrained gravity model. In short, CVs are attracted to zones proportional to their employment and households, but inversely proportional to the separation between zones.

Double constraint means that the result of the distribution model will match both the productions and attractions estimated for each zone. This is the logical approach for the commercial vehicle model where by design the productions are equal to the attractions.

Commercial vehicle data was not available for model development. Instead, the initial rates were borrowed from Asheville, NC and then adjusted for reasonableness.

$$F(t_{ij}) = a * t_{ij}^{-b} * e^{-c(t_{ij})}$$

Where:

F = Friction Factor;

t = travel impedance (general cost);

i,j = the from and to zone, respectively;

a, b, c = model parameters; and

e = base of natural logarithms.

The gamma function, and the final parameters are shown Table 61. Note that TransCAD formulates the gamma function with negative coefficients for b and c. As a result, positive values indicate an aversion to travel impedance.

Table 61: Final CV Trip Distribution Gamma Coefficients

Vehicle Type	Alpha	Beta	Epsilon
Autos/Vans (CV)	5	1	0.1
Pickups (SUT)	5	0.75	0.1
Trucks (MUT)	5	0.6	0.1

The CV models are calibrated by closely matching estimated average travel time and distance for each vehicle type against the observed values. In addition to matching estimated to observed average travel time and distance, it is also important to review the trip length distributions for both time and distance to make sure the shape of the curves logically follow the shape of the observed data without over-fitting. Unlike the resident and external trip models, no survey data is available for making direct comparisons of the commercial vehicle model. As a result, the models are simply checked for reasonableness and compared to secondary data sources. For example, one of the primary reasonableness checks is to make sure that MUT trips have longer average trip lengths than CV or SUT, as experience shows us that these vehicles have the longest travel time and distances of the three vehicle classes.

12 TIME OF DAY

The TOD model estimates the hourly and time period distribution of daily travel. Using diurnal factors, the time of day model transfers the 24-hour mode choice trip tables in production-attraction (PA) format into hourly trip tables in origin-destination (OD) format. The trip assignment model uses these tables to make time period trip assignments.

The diurnal factors consist of each period's share of the daily travel, and are based upon the NHTS data. NHTS data for the NRV region alone was used for the TOD model. These diurnal factors are segmented by trip purpose. The time of day model also uses separate diurnal factors and directional split factors for commercial vehicles and external trips. The factors for commercial vehicles were borrowed from the Triad region of North Carolina, while the external trip factors were informed mobile phone data.

Peak period determination is based on an analysis of the trips in motion throughout the day. First, the day is broken up into 15-minute increments. A trip from the household survey is said to be "in motion" if any portion of the trip occurs within the 15-minute bin. As a result, a single trip can be counted in multiple bins. Determining the peak period considers the distribution of all trips as well as the distribution of just HBW trips. Figure 29 shows these two distributions, which look as expected with the AM peak being shorter and more condensed than the PM. Travelers have much more flexibility in their travel choices in the PM.

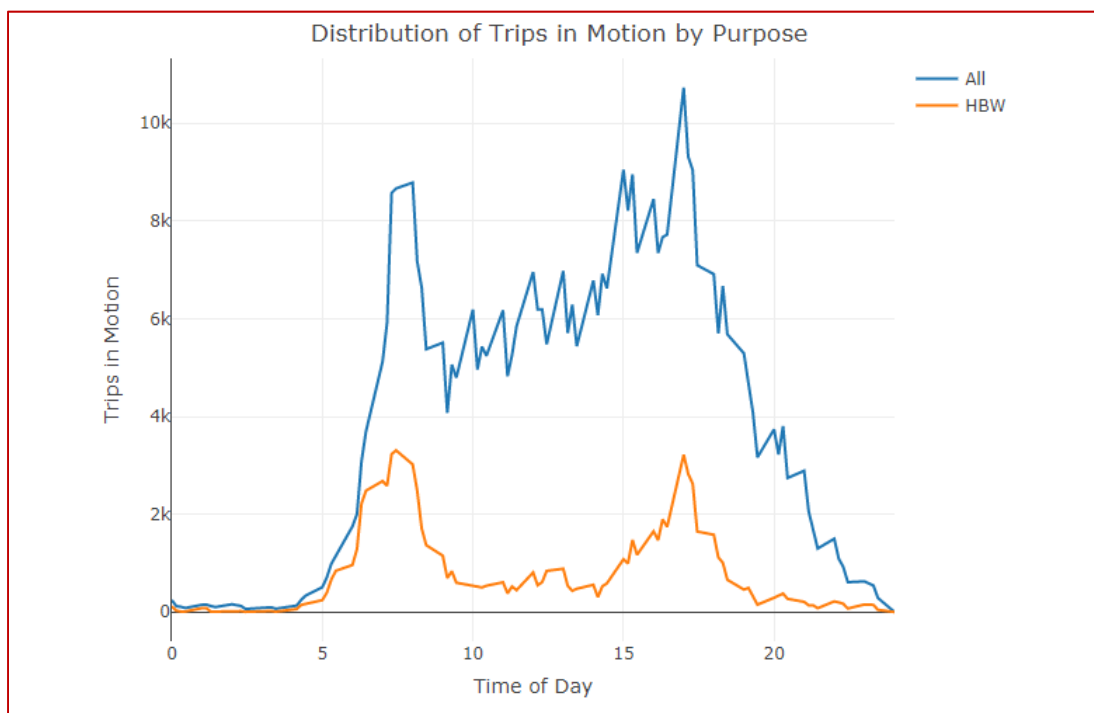


Figure 29: Trips in Motion

In Table 62 and Table 63, the AM and PM peak hours by purpose are presented to determine the peak hour for both HBW and all trips. At a minimum, the AM and PM peak periods should contain the respective peak hour for HBW trips and all trips. Figure 30 shows that the peak hour for HBW and for all trips lines up very closely.

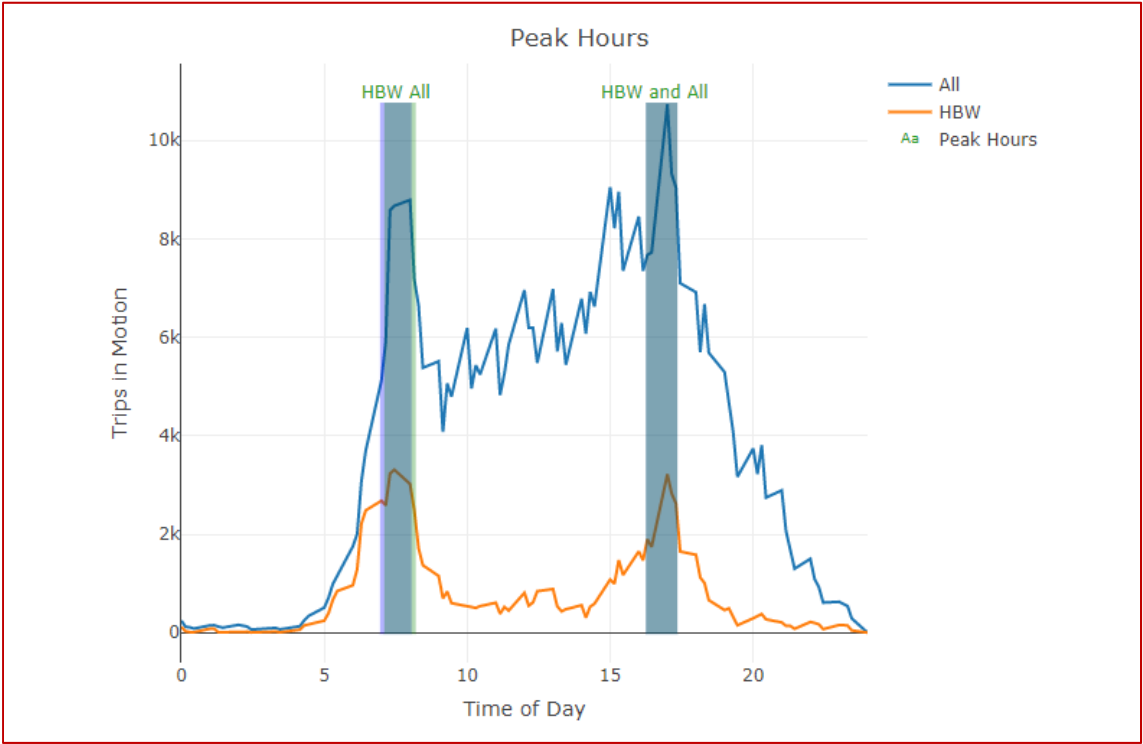


Figure 30: Determination of HBW Period

Table 62: AM Peak Hour by Purpose

Trip Purpose	From	To
HBW	07:00:00	08:00:00
All	07:15:00	08:15:00

Table 63: PM Peak Hour by Purpose

Trip Purpose	From	To
HBW	16:30:00	17:30:00
All	16:30:00	17:30:00

12.1 Final Period Definitions

The final period definitions are shown in the table below including the mid-day (MD) period. The remaining hours of the day are captured in the night (NT) period. In the chart, the final period is represented by the gray rectangle.

Table 64: Final Period Definitions

Period	From	To	Hours
AM	07:00:00	08:15:00	1.25
MD	08:15:00	16:30:00	8.25
PM	16:30:00	17:30:00	1.00
NT	17:30:00	07:00:00	13.50

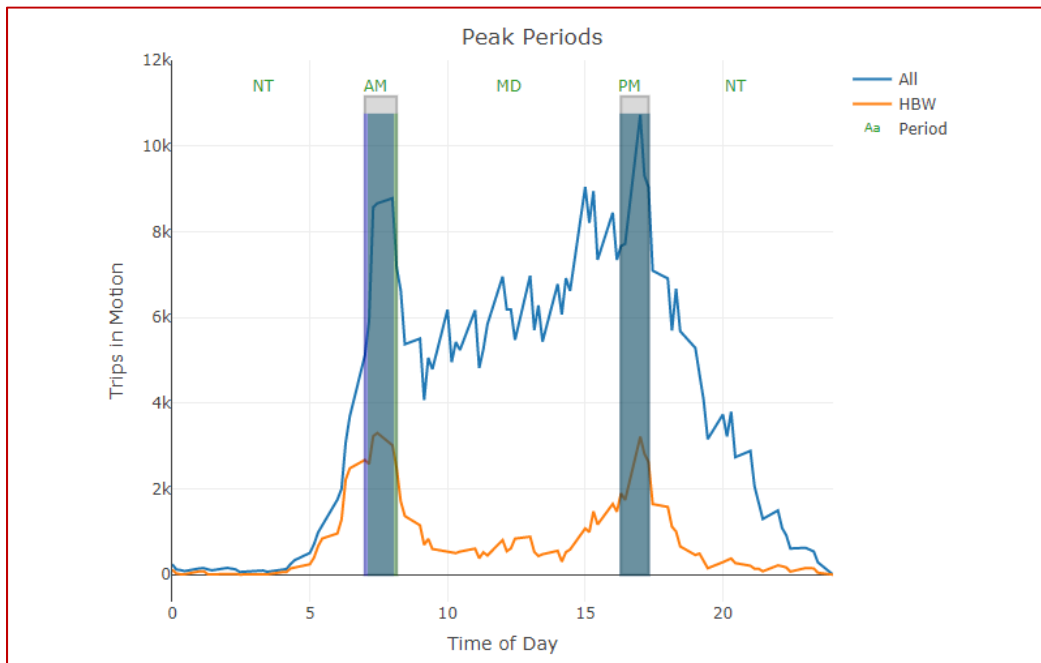


Figure 31: NRV Final Period Definition

12.2 Time of Day Factors

The percent of trips produced by purpose and time of day must be determined for each period in the model. These factors are calculated based on the departure time of each trip rather than using the trips in motion table. This ensures that each trip is only counted once, regardless of duration, and that it is included in the right period. The factors, as shown in Table 65 are used to divide the daily person trips created during trip generation into time of day (by purpose). IEEI trips are assumed to behave like HBO trips.

Table 65: Time of Day Factors by Purpose

Purpose	AM	MD	PM	NT
HBW	18.1%	38.8%	12.9%	30.2%
HBSC	38.3%	48.4%	2.6%	10.7%
HBSH	2.4%	58.9%	9.0%	29.7%
HBO	8.1%	48.1%	8.9%	34.9%
NHBW	4.9%	78.0%	8.6%	8.5%
NHBO	3.2%	71.3%	8.9%	16.6%

12.3 Directionality Factors

For most steps in a trip-based model, the trips are said to be in “Production/Attraction” (PA) format. This simply means that all home-based trips start at home and end somewhere else. This simplification of reality is done for a number of reasons in the model, but is not actually how travel occurs. As a result, before highway trip assignment can take place, the PA format must be converted to Origin/Destination” (OD) format. In this format, a trip from work to home starts at work and ends at home.

This conversion is accomplished using factors stratified by time of day and purpose. As an example, the majority of HBW trips in the AM start at home and end at work. As a result, PA-to-OD factor (“PA factor” for short) would be above 50% ($> .5$). In the PM period, this trend is usually reversed, and the PA factor would be less than 50% ($< .5$).

These factors can be calculated directly from the survey by comparing the number of home-based trips that start at home to the number home-based trips that start elsewhere. By definition, non-home-based trips are the same in either format; their PA factors are set to .5. The same treatment is applied to commercial vehicles, trucks, and external trips.

Finally, directionality factors are applied after the distribution and modal models. At that point, the purposes are collapsed due to sample size limitations in the household survey. As a result, the factors

must be estimated using the matching, more-aggregate purposes. Table 66 are the final NRV directionality factors.

Table 66: PA Factors by Purpose and Time of Day

Purpose	AM	MD	PM	NT
HBW	1.00	0.47	0.07	0.62
HBO	0.75	0.54	0.44	0.42
HBSC	1.00	0.24	0.30	0.64
NHB	0.50	0.50	0.50	0.50

12.4 Vehicle Occupancy Factors

In addition to applying directionality factors before assignment, the person trip matrices are converted to vehicles using occupancy factors. These factors are calculated from the household survey, which includes information on the party size of each trip. Table 67 are the occupancy factors used in NRV.

Table 67: Occupancy Factors by Purpose and Time of Day

Purpose	Mode	AM	MD	PM	NT
HBW	Drive Alone	1	1	1	1
HBW	Shared Ride 2	2	2	2	2
HBW	Shared Ride 3+	3.32	3.21	3.73	3.1
HBO	Drive Alone	1	1	1	1
HBO	Shared Ride 2	2	2	2	2
HBO	Shared Ride 3+	3.32	3.62	3.49	3.57
HBSC	Drive Alone	1	1	1	1
HBSC	Shared Ride 2	2	2	2	2
HBSC	Shared Ride 3+	3.47	3.47	3.69	3.49
NHB	Drive Alone	1	1	1	1
NHB	Shared Ride 2	2	2	2	2
NHB	Shared Ride 3+	3.38	3.49	3.41	3.6

12.5 Period Capacity Factors

In aggregate, period-based assignment regimes, hourly capacities must be converted to period capacities in a manner that accurately captures the balance of congestion. In the real world, the hourly capacity of a road is generally the same throughout the day; it is the demand that changes. This change in demand causes congestion levels to change throughout the day. Consider the stylized example Table 68a of a real world street segment.

Table 68: Example of Real-World Behavior

Time	Capacity	Demand	Percent
6-7	1050	500	47.62
7-8	1050	900	85.71
8-9	1050	500	47.62

In this representation, no one experiences congestion. In the real world, nearly half the vehicles in the period experience congested conditions between 7:00 am and 8:00 am. This discrepancy can cause numerous errors in the model, including:

- Longer trip lengths
- Incorrect route choices
- Lower diversion to transit
- Incorrect link assignment

The period capacity factor (PCF) corrects for this, and is the inverse of a similar concept from traffic engineering: the peak hour factor (PHF). In traffic engineering, the PHF adjusts hourly volume based on the highest 15-minute volume within the hour. While the PHF is used to adjust demand given a fixed capacity, the PCF adjusts period capacity based on the highest-volume hour of the period.

Consider our example of a three-hour period from 6:00 am to 9:00 am. If the actual demand was evenly distributed across the period, the PHF would be .33. The PCF, the inverse of the PHF, would be 3. In this scenario, the hourly capacity would be multiplied by 3, which would lead to the capacity shown in the table above. Instead, using the actual demand from our example, the PHF would be:

$$\text{PHF} = 900 / (500 + 900 + 500) = 0.474$$

Taking the inverse gives a PCF of:

$$\text{PCF} = 1 / .474 = 2.11$$

Finally, the period capacity that most-accurately reflects congestion would be:

$$\text{Period Capacity} = 1050 * 2.11 = 2215$$

That capacity is shown relative to the period demand in Figure 32. This relationship is a better representation of real world conditions.

From this example the levels of congestion would be very different depending on the factor used. The PCF of 2.11 represents the more accurate level of congestion and is therefore the way capacity is calculated for NRV.

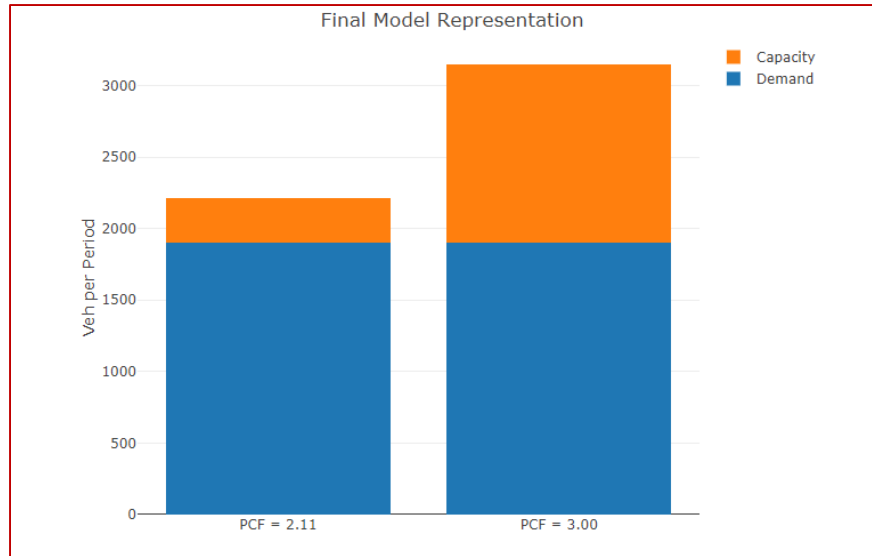


Figure 32: Graph of Final Model Example

12.6 Calculation from Survey

The PCFs for the model can be calculated from the trips in motion table using the same approach. The results are shown in the table below.

Table 69: Model Peak Capacity Factors (PCFs)

Period	From	To	Length (hrs)	PHF	Model Capacity (hrs)
AM	07:00:00	08:15:00	1.25	0.85	1.18
MD	08:15:00	16:30:00	8.25	0.16	6.20
PM	16:30:00	17:30:00	1.00	1.00	1.00
NT	17:30:00	07:00:00	13.50	0.27	3.74

A general guideline to follow¹² suggests the model should provide 8-12 hours of capacity per day. The PCFs calculated directly from the survey data provide 12.12. This is slightly higher, but still acceptable.

¹²

https://www.fhwa.dot.gov/planning/tmip/publications/other_reports/validation_and_reasonableness_2010/ch08.cfm

13 ASSIGNMENT VALIDATION

The final step in the travel demand model is trip assignment. This is the process of assigning the trips between TAZs to the individual links in the highway network and the transit network. This step is performed iteratively with overall model calibration and validation. When overall model calibration and validation is achieved, as measured by established performance measures, the trip assignment step provides the data needed for: 1) testing alternative transportation plans; 2) establishing priorities between different transportation investment strategies; 3) analyzing alternative locations for roadway improvements; and 4) forecasting design volumes needed to adequately design and construct new roadway facilities. The reliability of the output from this step is dependent upon the reliability of all the proceeding steps.

13.1 Highway Assignment

The algorithms used in traffic assignment attempt to replicate the process of choosing the best path between a given origin and destination. For the NRV model, the algorithm used for assignment is an equilibrium assignment. This widely accepted best practice approach produces link loadings by optimally seeking user-equilibrium path loadings reflecting user path choices as influenced by congestion on the network. During the assignment process, the trip table is assigned to the highway network over multiple iterations. At the end of each iteration, link travel times are recalculated using the total link demand and compared to the link travel times of the previous iteration. The aggregate change of link travel times between the current iteration and the previous is compared against the convergence criteria. Thus, the number of iterations is determined by a user defined closure parameter (set to 0.000001 for the NRV model) or for a maximum number of iterations (set to 500 for the NRV model). The final assignment represents an optimum combination of previous assignments using the Frank-Wolfe algorithm.

For a given iteration, the volume-delay function is used to update the link speeds based on the previous iteration's vehicle demand and the link capacity. The conical function is used for the NRV model. The values for alpha in the function are shown in

Table 70.

Highway trip assignment is performed separately for the AM peak period, Midday period, the PM peak period, and the Night period. At the end of the assignment procedure, the time period assignments are summed to produce a daily traffic assignment.

Table 70: NRV Volume-Delay Function Alpha

AreaType	HCMTType	Alpha
Urban	Freeway	10
Urban	MLHighway	8
Urban	TLHighway	6
Urban	PrArterial	6
Urban	MinArterial	6
Urban	Collector	4
Urban	Local	2
Urban	CC	10
Suburban	Freeway	10
Suburban	MLHighway	8
Suburban	TLHighway	6
Suburban	PrArterial	6
Suburban	MinArterial	6
Suburban	Collector	4
Suburban	Local	2
Suburban	CC	10
Rural	Freeway	10
Rural	MLHighway	8
Rural	TLHighway	6
Rural	PrArterial	6
Rural	MinArterial	6
Rural	Collector	4
Rural	Local	2
Rural	CC	10

13.2 Final Trip Assignment Results

There are three basic comparisons made when comparing actual traffic counts to estimated trip assignment results: 1) global measures of vehicle miles traveled (VMT) and average speed by facility type, 2) comparisons by screenline, volume group, and facility type, and 3) link level comparisons.

Following an iterative calibration and validation process acceptable performance measures were achieved. This section describes the final highway assignment results.

13.2.1 VEHICLE MILES TRAVELED (VMT)

At the regional level, comparisons of VMT provide useful information on how well the model understands the magnitude and geographic distribution of travel. Typically, estimated regional study area VMT should be within 5% of observed VMT. Table 71 summarizes calibrated base year VMT results for the NRV Model¹³. Overall regional VMT is within the acceptable 5% range and VMT by all facility types is also within an acceptable range of $\pm 10\%$. Given the limited count distribution by facility type these results are very reasonable for NRV.

Table 71: VMT by facility type

Facility Type	Observed VMT	Estimated VMT	% Deviation
Freeway	591,104	550,128	-6.93
MLHighway	123,206	127,893	3.8
TLHighway	48,188	44,429	-7.8
PrArterial	89,252	81,115	-9.12
MinArterial	34,055	30,322	-10.96
Collector	126,144	129,525	2.68
Local	16,672	17,515	5.06
Total	1,028,621	980,928	-4.64

13.2.2 SCREENLINE AND CUTLINE COMPARISONS

The NRV model utilizes three screenlines as shown in Figure 33.

- Screenline 1 - generally separates the communities of Blacksburg and Christiansburg, capturing the north-south flow between them.
- Screenline 2 - generally separates the communities of Radford and Blacksburg/Christiansburg, capturing the east-west flow between them.
- Screenline 3 – runs east of the major cities trying to get the major flows into the region from the east.

¹³ VMT tables in this section does not represent the VMT for the entire network, just the links that have counts so that validation could occur. Therefore results in this section will not be the same as the model assignment report produced by the model after every run.

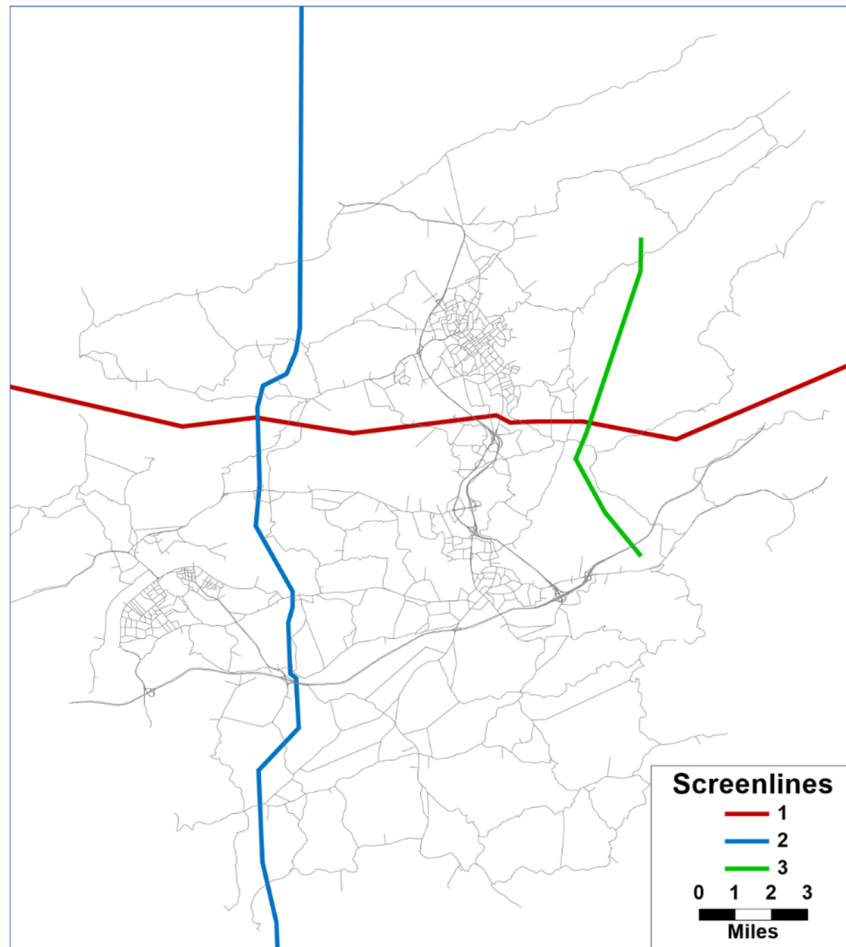


Figure 33: NRV Screenlines

The screenlines shown in Figure 33 were selected based on available traffic count data and a desire to capture major flows in the region. One of the challenges with the screenlines is that the traffic count coverage was not sufficient to cover all of the roadways crossing the screenline boundary. This impacts the reporting of the screenline statistics as all directional flows on links may not be captured.

When properly developed, screenline comparisons provide a good check on the need for possible adjustments to the trip distribution in the region through adjustments to the destination choice models. If all screenlines are either high or low, this may indicate a need to make global adjustments to daily travel. If some screenlines are high while others are low, then this may be an indication that the model does not fully understand the travel patterns in the region leading to further investigations into the destination choice model and even the underlying transportation network. Following the investigations and modifications summarized previously, the final assignment shows screenline results within an acceptable range, as shown in Table 72.

Table 72: Screenline Comparison

Screenline	Observed Flow	Model Flow	Percent Deviation	Target
1	74,900	76,683	2.38	10
2	74,910	78,566	4.88	10
3	49,300	51,040	3.53	10
Total	199,110	206,289	3.61	10

13.2.3 VOLUME GROUP AND FACILITY TYPE COMPARISONS

At the link level, several comparisons were made including link assignment summaries by volume group and difference between estimated and observed through the calculation of the percent root mean square error (%RMSE) by facility type and volume group. The %RMSE is a measure of the variance between the observed and modeled volumes (numerator), normalized by the average of the observed data (denominator) and is expressed as a percent. The formula is as follows:

$$\%RMSE = \frac{(\sum_j (Model_j - Count_j)^2 / (Number\ of\ Counts - 1))^{0.5}}{\sum_j Count_j / Number\ of\ Counts} * 100$$

Since the differences between estimated and observed are squared before they are averaged, the RMSE gives a relatively high weight to large errors. This means the %RMSE is most useful when large errors are particularly undesirable. Targets for the regional %RMSE are in the ranges of 30-40%, depending upon the number of low-volume roadway segments included in the count sample. The %RMSE should decrease as volumes increase - thus for facility types with high volumes such as freeways, the %RMSE is expected to be lower (estimated should better match observed) than %RMSE measured on facility types with low volumes such as collectors.

The %RMSE by facility type and volume group is summarized in Table 73 and

Table 74¹⁴. By facility type, all facilities are well within the target range, and the overall %RMSE is 33% which is below the 40% target. The volume group comparison also shows that all targets have been met, though the lower volume facilities do show more error as expected.

¹⁴ For NRV, the RMSE for divided facilities was calculated as two individual links instead of one link with both directions summed together.

Table 73: Percent Root Mean Square Error (RMSE) by Facility Group

Facility Type	Number of Links	% RMSE	Target
Freeway	14	12.98	25
MLHighway	26	29.96	35
TLHighway	12	35.94	50
PrArterial	28	21.46	50
MinArterial	23	37.71	65
Collector	106	52.28	100
Local	20	61.82	100
Total	229	33.46	35-40

Table 74: Percent Root Mean Square Error (RMSE) by Volume Group

Volume Group	Number of Links	% RMSE	Target
0 - 5000	123	52.7	100
5000 - 10000	51	35.02	45
10000 - 15000	29	28.16	35
15000 - 20000	11	21.99	30
20000 - 40000	13	18.68	25
40000 - 60000	2	1.21	20
Total	229	33.46	40

A look at RMSE by area type is performed to observe obvious issues with flows at the various density types. Although there are no guidelines, generally the same rule of 40% is observed. In Table 75, the suburban area type is slightly above the guideline but does not suggest an issue given how other measures discussed in this section compare to targets. This is likely an issue with the number of links that are suburban and that also have count information.

Table 75: Percent Root Mean Square Error (RMSE) by Area Type

Area Type	Number of Links	% RMSE
Urban	96	33.28
Suburban	31	41.1
Rural	102	23.2
Total	229	33.46

13.2.4 COUNT TO FLOW COMPARISONS

Table 76 provides a summary of the assignment results by comparing the daily count to the daily flow by volume group. The percent difference is calculated and compared to recommended target values. As the volume increases, the calculated percent difference should decrease as recommended by the target values. All volume groups are well within the acceptable error.

Table 77 provides a summary of the assignment results by comparing the daily count to the daily flow by facility type. Although no specific targets are included, generally within +/- 10% is reasonable for collectors and above. The results for NRV are reasonable by facility type.

Table 76: Assignment Summary by Volume Group

Volume Group	Observed Flow	Model Flow	Percent Deviation	Target
0 - 5000	224,260	216,013	-3.68	100
5000 - 10000	369,915	353,741	-4.37	45
10000 - 15000	361,434	327,618	-9.36	35
15000 - 20000	198,000	177,516	-10.35	30
20000 - 40000	300,700	276,997	-7.88	25
40000 - 60000	90,522	91,071	0.61	20
Total	1,544,831	1,442,957	-6.59	10

Table 77: Assignment Summary by Facility Type

Facility Group	Observed Flow	Model Flow	Percent Deviation
Freeway	324,522	305,529	-5.85
MLHighway	250,434	248,825	-0.64
TLHighway	78,400	77,364	-1.32
PrArterial	333,400	299,413	-10.19
MinArterial	255,700	231,902	-9.31
Collector	283,781	258,864	-8.78
Local	18,594	21,061	13.27
Total	1,544,831	1,442,957	-6.59

Checks were also performed to confirm that 75% of freeway links were within +/- 20% of the traffic counts, and that 50% were within +/- 10% of the traffic counts. For links with volume greater than or equal to 10,000 vehicles per day, checks were performed to confirm that 75% of the links were within +/- 30% of the traffic counts, and that 50% within +/- 15% of the traffic counts. NRV meets all of these

targets except for the 20% error for freeways. Although it is close to this target at 71%, the error can be contributed to one link that has a path diversion issue that the model cannot see without manually adjusting turn penalties or speeds. Neither of which seemed like the best option to fix the spot issue¹⁵. See Table 78 for the results of the individual target comparisons.

Table 78: Individual Link Targets

Test	Percent passing	Target
Links greater than 10k volume under 30% error	84.62	75
Links greater than 10k volume under 15% error	67.31	50
Freeway links under 20% error	71.43	75
Freeway links under 10% error	71.43	50

Finally, the observed and estimated daily volumes by link were plotted and the r-squared value was calculated as a final indication of the degree to which the counts and estimated volumes match. The r-squared value is, in part, a function of the number of observations and therefore it is difficult to establish a guideline for acceptable values of r-squared. In general, a high r-squared value is desirable.

Figure 34 shows a scatter plot of observed versus estimated volumes by link and facility type. The diagonal line on the graph is defined by the set of points where the estimated count equals the observed count. If all counts fell along this line the r-squared value would be equal to one. The highway traffic assignment for the NRV model shows a tight fit between the observed and estimated values for the region.

Additional plots by area type and facility type are shown in Figure 35 and Figure 36. These are also reasonable with multi-lane highways having a value slightly outside normal ranges. Additional review determined that forcing the model manually on the edges of the boundary was not the logical method and that loadings and network connectivity also contributed to a few locations.

¹⁵ Although the manual adjustments can fix the issue, it is recommended that this occur only in the use of the model for a project level forecast.

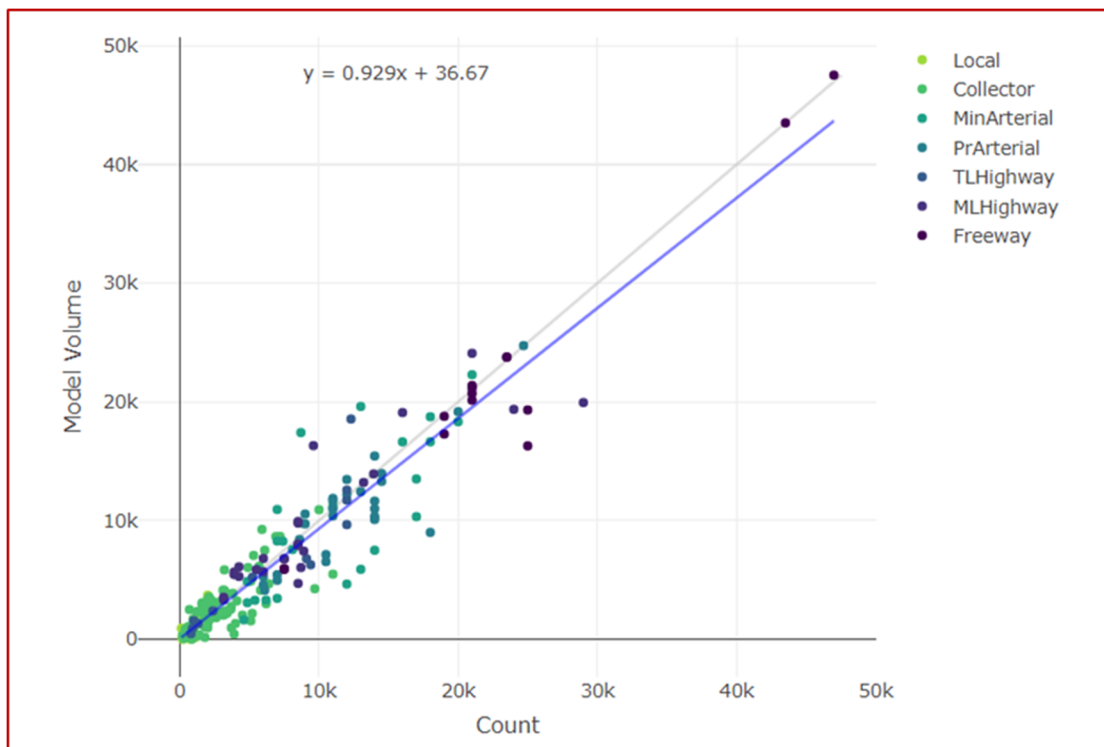


Figure 34: Scatter Plot of Observed Count and Estimated Flow

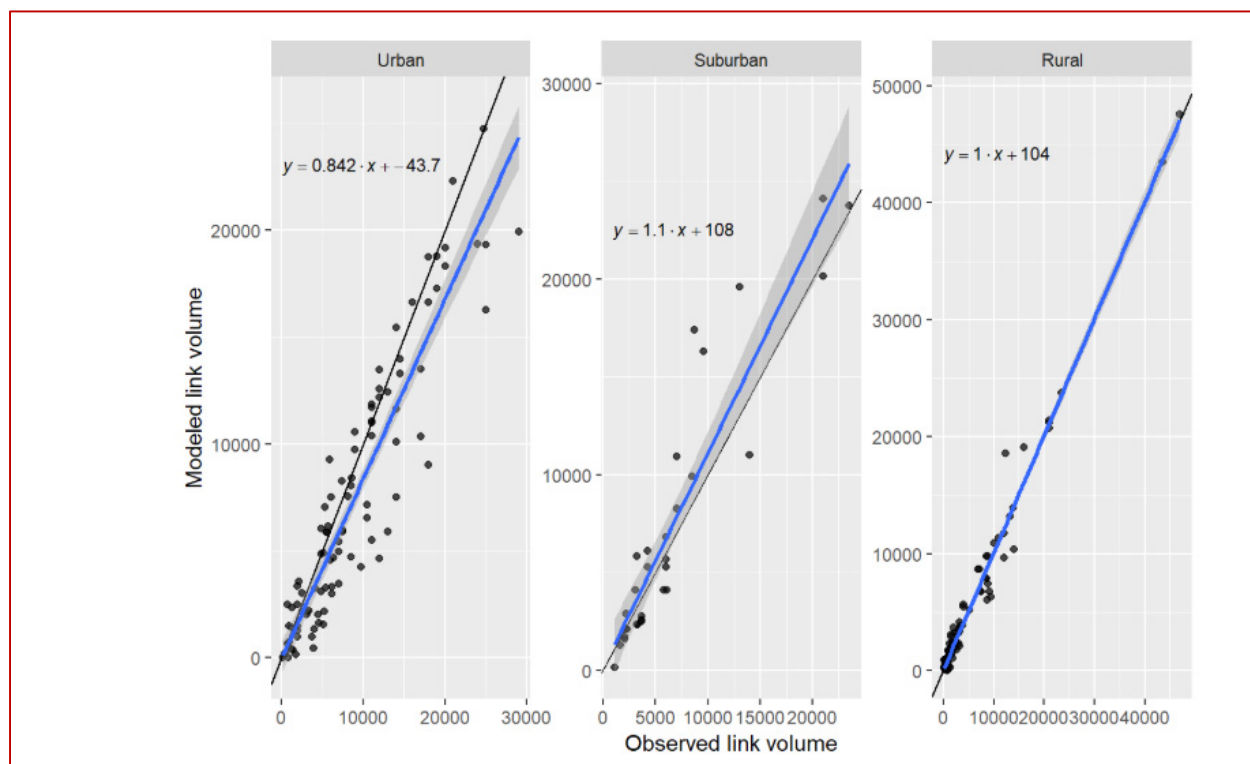


Figure 35: Scatter Plot of Observed Count and Estimated Flow by Area Type

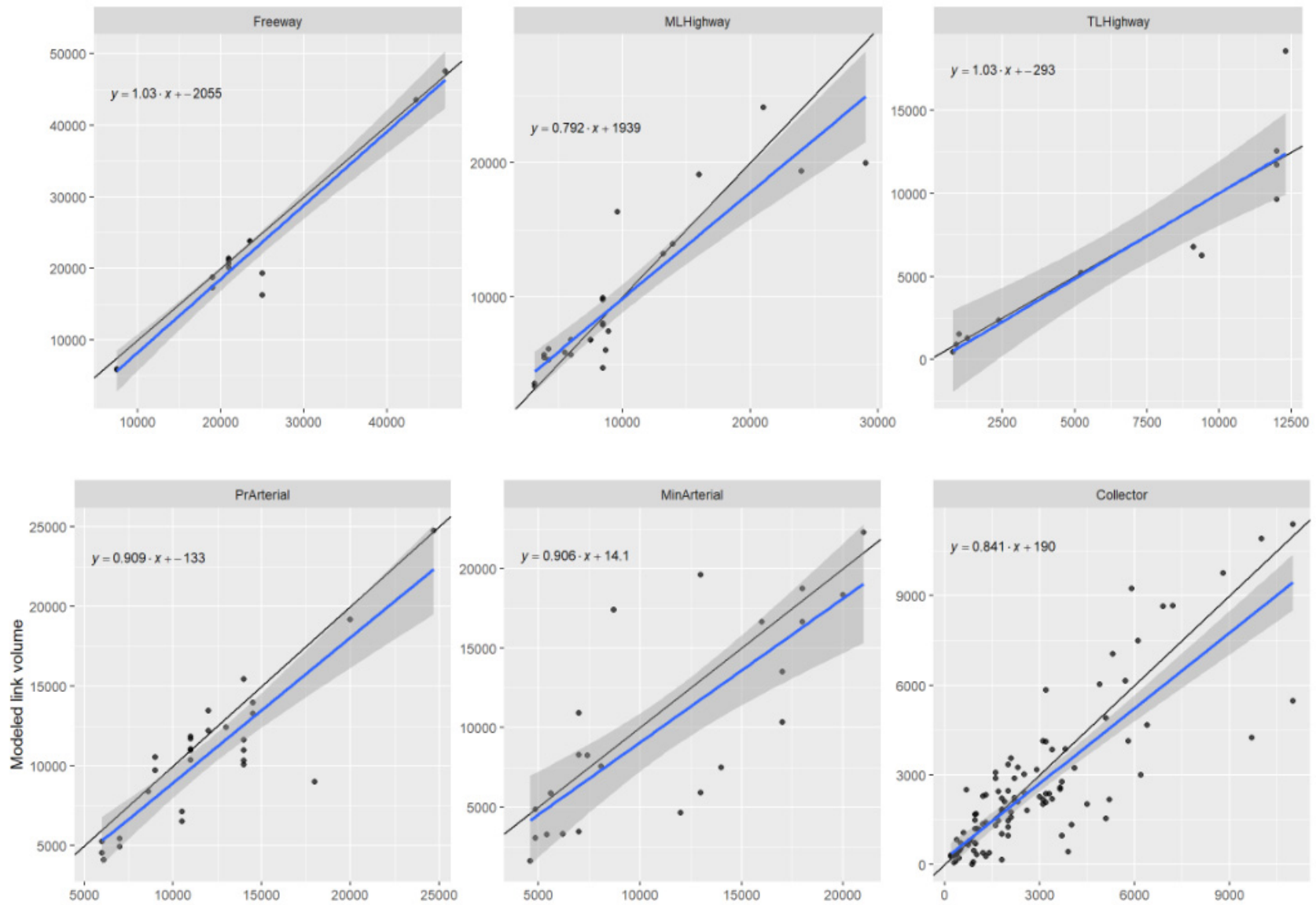


Figure 36: Scatter Plot of Observed Count and Estimated Flow by Facility Type

Figure 37 is a plot of the traffic count against the percent deviation of the modeled volume as compared to the maximum desirable deviation curve by facility type. The grey area represents the corresponding maximum desired deviation range for each facility type. These graphs are useful in determining which links exceed the maximum desirable deviation so that validation efforts can focus accordingly. The graphs show that more deviation is acceptable for lower volume/lower facility type links, whereas estimates of high volume/high facility type links should deviate less from the observed flows. As shown, most of the links fall within their corresponding maximum desired deviation range. The results show a good fit of the model estimated flows against traffic counts, especially for the higher volume/high facility type locations. It is expected that some locations will be outside the maximum desirable deviation curve, as it is extremely important not to over calibrate a model just to get all of the link flows to match the traffic counts. One reason is that all traffic counts include some degree of error, but the most important reason is that the model should not be forced to match traffic counts through individual link adjustments, as the model must remain sensitive to future changes in travel demand and traffic flow.

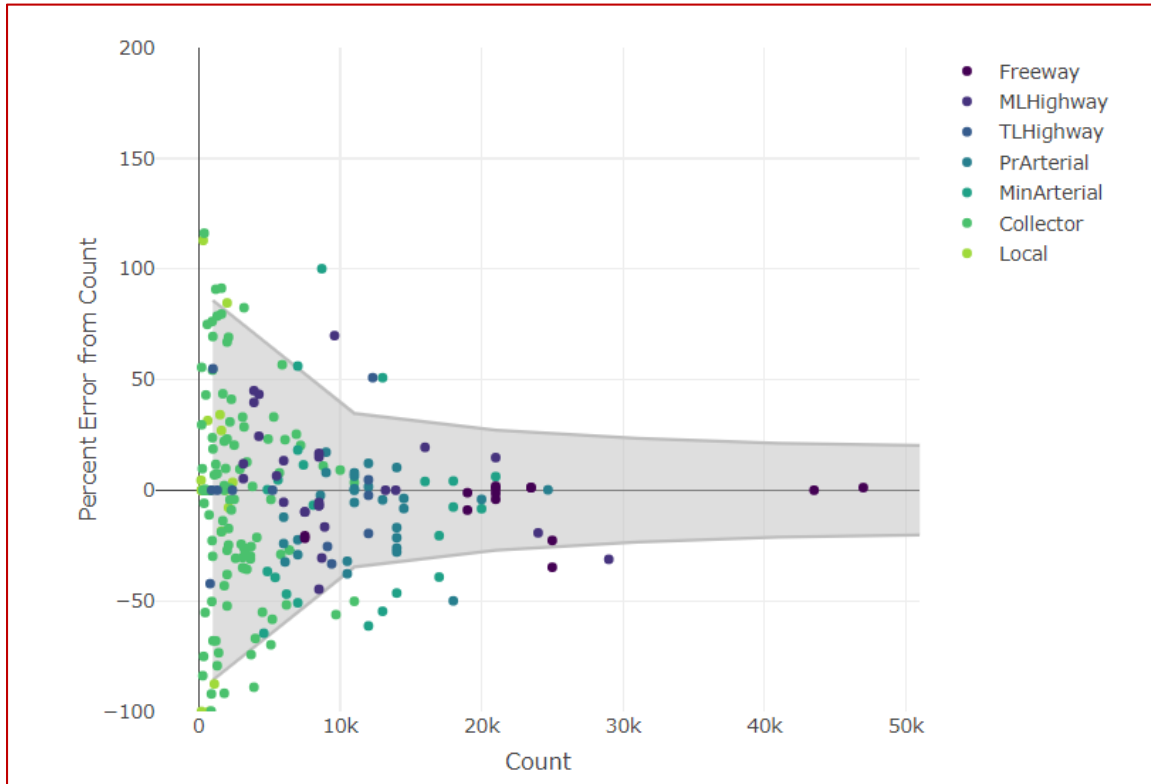


Figure 37: Plot of Traffic Count vs. the Percent Deviation of Modeled Flow by Facility Group

14 SENSITIVITY TESTING

Sensitivity checking is an important final check on model performance. The model is used to run various scenarios to ensure the response is reasonable.

Two main sensitivity tests were performed:

- Dropping a major roadway link
- Modifying the SE data in a TAZ

14.1 Link Drop

In Figure 38, the left image shows congestion on the base year network, while the right shows the model's reaction to the removal of a section of I-81. The model's reaction, utilizing latent capacity where available and creating congestion where it isn't, is the appropriate response.

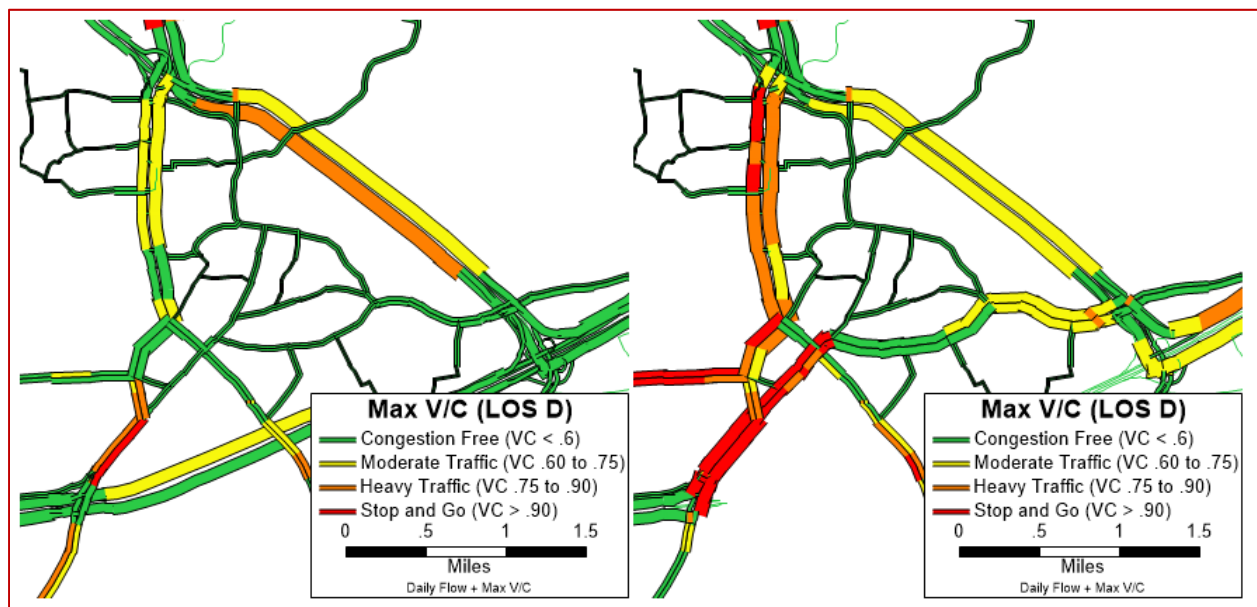


Figure 38: Link Drop Results

14.2 SE Change

Figure 39 shows the changes in the AM assignment when TAZ 110 experiences a tripling in the households and employment in the zone. Major changes in link volumes are localized, which indicates proper convergence of both highway assignment and model feedback. In general, the patterns of change make sense.

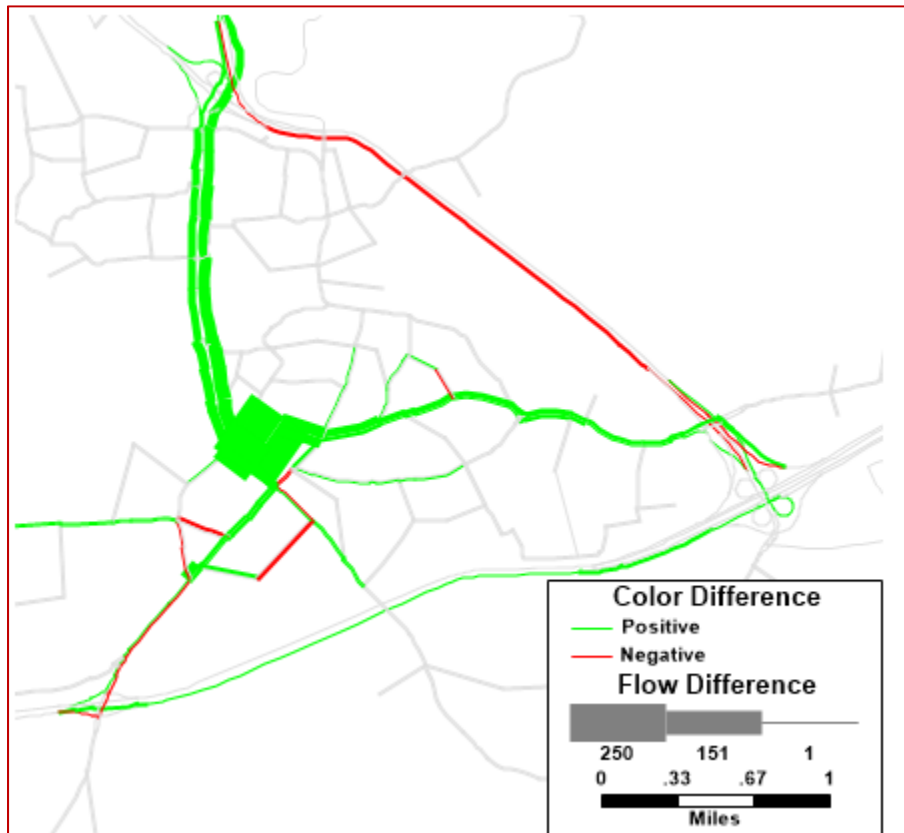


Figure 39: Results of SE Data Changes

APPENDIX A- NETWORK REVIEW

APPENDIX B- MODEL ATTRIBUTES FOR HIGHWAY NETWORK, TRANSIT ROUTES AND SOCIO-ECONOMIC DATA

APPENDIX C- MODE CHOICE COEFFICIENTS

Appendix A- Network Review

This appendix details the efforts to review, clean, and simplify the HERE roadway networks.

1.1 Network Corrections

The HERE shapefile network provided by VDOT on 11/29/2017 required moderate edits to render it useable for travel demand modeling. These edits can be broken down into five categories:

- Revision of network fields;
- Proper linking of count locations to the network;
- Correction of one way links and intersections;
- Simplification of network; and
- Creation of Centroids.

1.1.1 REVISION OF NETWORK FIELDS

The shapefile provided by VDOT contained a mixture of data fields used by HERE and VDOT. For the NRV model development a defined set of data fields are required, therefore data needed to be cross walked between the two schemas. This provided an opportunity to review and edit incorrect and/or incomplete fields.

1.1.2 FUNCTIONAL CLASS REVIEW

The functional class of the roadways needed to be verified as the shapefile came with conflicting functional class information between VDOT and HERE. The original VDOT functional classes are shown in Figure 1 and Figure 2. The maps highlight 1686 links missing a functional class value, including substantial portions of US 460 through Blacksburg. Figure 2 provides a closer look at the northbound lanes being classified as other, which is not acceptable for use in the travel model.

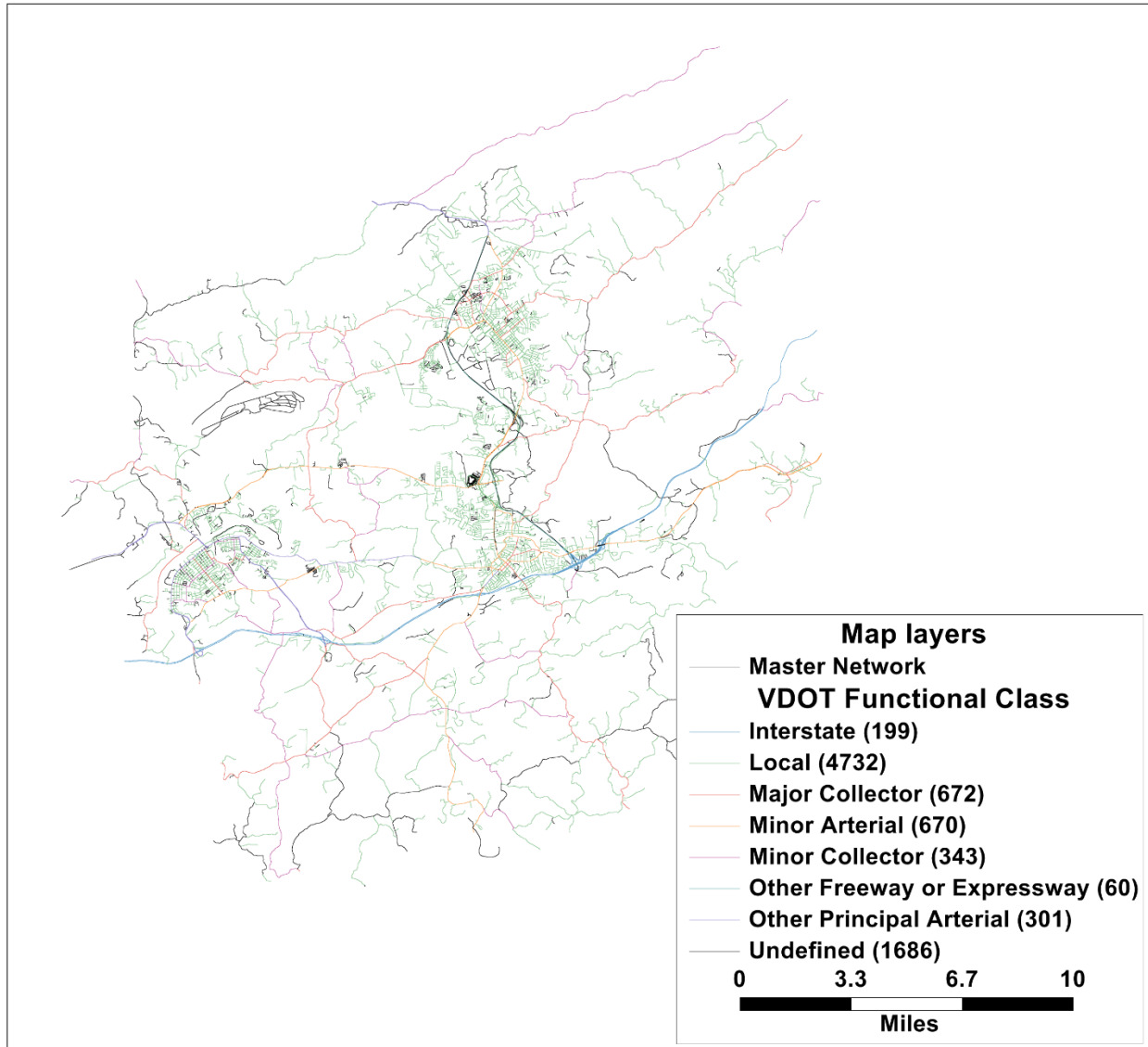


Figure 1: VDOT Functional Classification Values

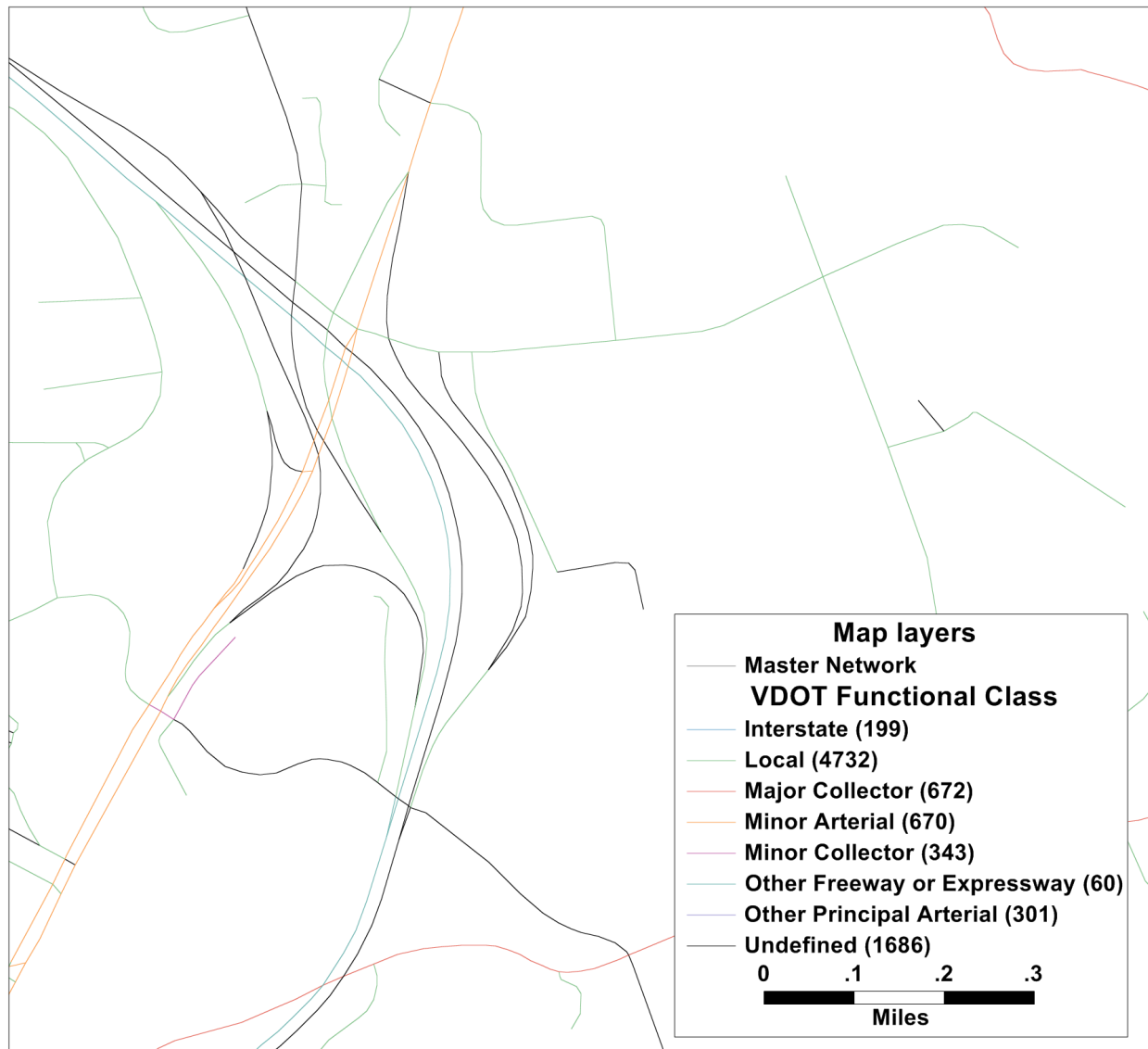


Figure 2: VDOT Functional Classification – Zoomed to Blacksburg

In contrast, the HERE dataset¹ behaves reasonably throughout the network. In Figure 3, the HERE functional classifications do not neatly correspond to familiar classifications of road types, but their functions are fairly intuitive, lying on a scale between Class 1 roads with “high volume, maximum speed traffic” and Class 5 roads “whose volume and traffic movement are below the level of any functional class”.² Figure 3 shows that the HERE functional classes classify the majority of streets as local Class 5s, there are no unclassified streets, and the major roads are all classified reasonably, though there are fewer distinctions than given by VDOT. Given the confidence in the HERE values, it was decided to use those values as the basis for the NRV model development.

¹ The HERE_FUNC_CLASS field provided was used for this analysis

² HERE (2014). NAVSTREETS Street Data Reference Manual v5.4.

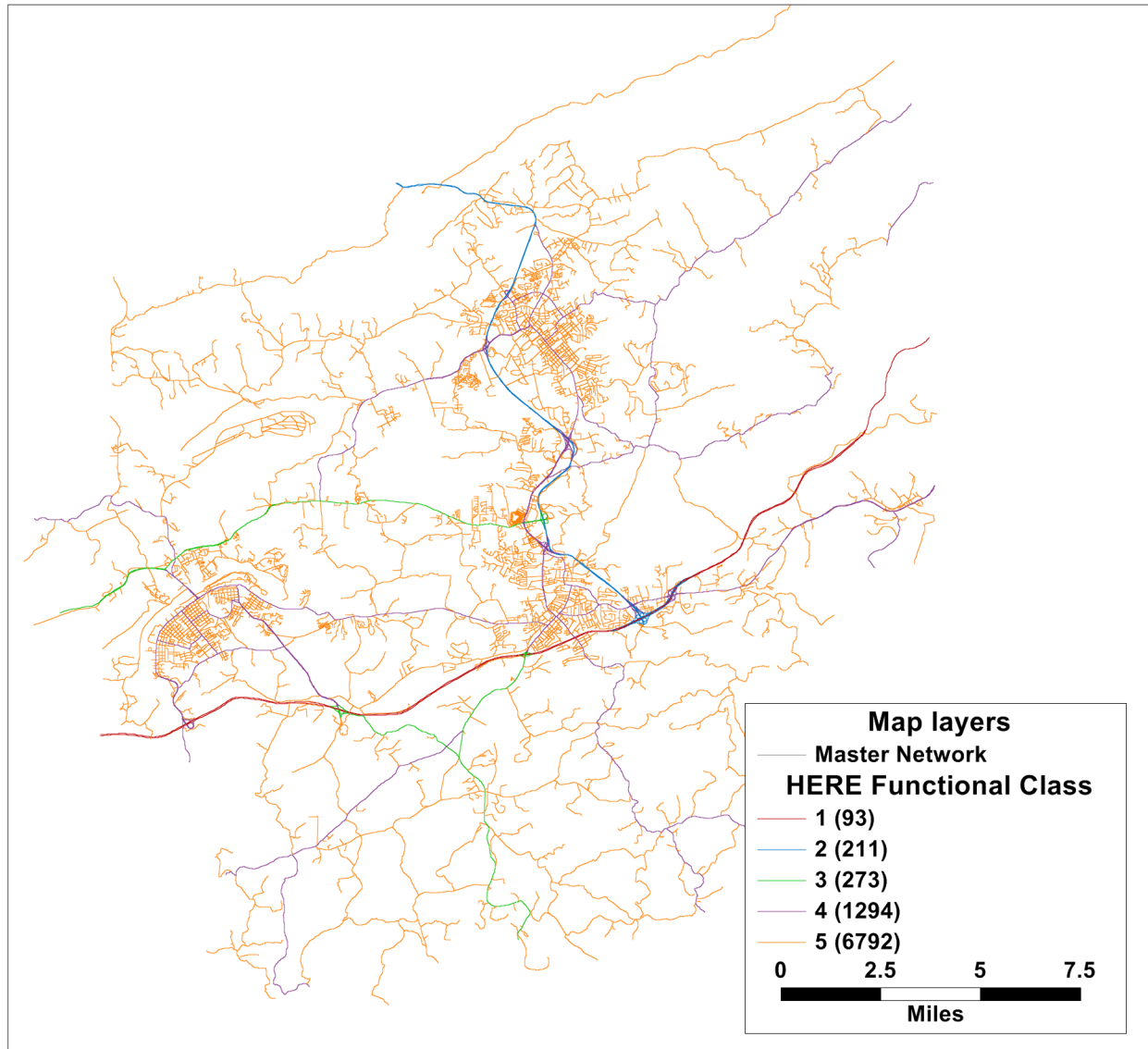


Figure 3: HERE Functional Classification Values

In this model, a cross walk between the HERE functional classes into a standard set of classes used by WSP models was developed. Table 1 shows the final values and descriptions for the functional class field for the NRV model.

HCMTType	Name
Freeway	Freeway
MLHighway	Multi-lane highway
TLHighway	Two lane highway
PrArterial	Principal arterial
MinArterial	Minor arterial
Collector	Collector
Local	Local
CC	Centroid Connector

Table 1: NRV Model Functional Classifications

1.2 Speed and Lane Review

Correct travel times and congestion levels in the NRV model depend (in large part) on an accurate representation of speed limits and the number of lanes. Network cleaning began with values provided by HERE and then the fields for network posted speeds and numbers of lanes were manually inspected and corrected where necessary. This work was completed by creating maps colored by speed and lane width. These maps were inspected visually for problems, which were then verified or corrected. As expected with most GIS databases the locations containing errors were ramps and other links near complicated highway interchanges. In many of these interchanges, the highway speed would wrongly be used on entry/exit ramps and arterial interchanges

With respect to a plot of speed limits (Figure 4), most of the network looked reasonable. Although there were 797 links with a coded speed limit of 0 mph, those links were all local streets that were eventually removed from the network and so did not pose a problem for network coding. On the remaining streets, there were few unexpected speed changes in the middle of a road segment. However, it was discovered that there were a few interchanges where the ramp's characteristics (including both posted speeds and lanes) are carried over into at least 1 link of the interstate. Figure 5 provides an example, where Link ID's 3231, 3236, and 3233 are all part of I-81 and should have the same posted speed of 70mph, but where Link 3236 has been wrongly coded with the speed attributes of the on-ramp (Link ID 3230). These issues were corrected in the final NRV network.

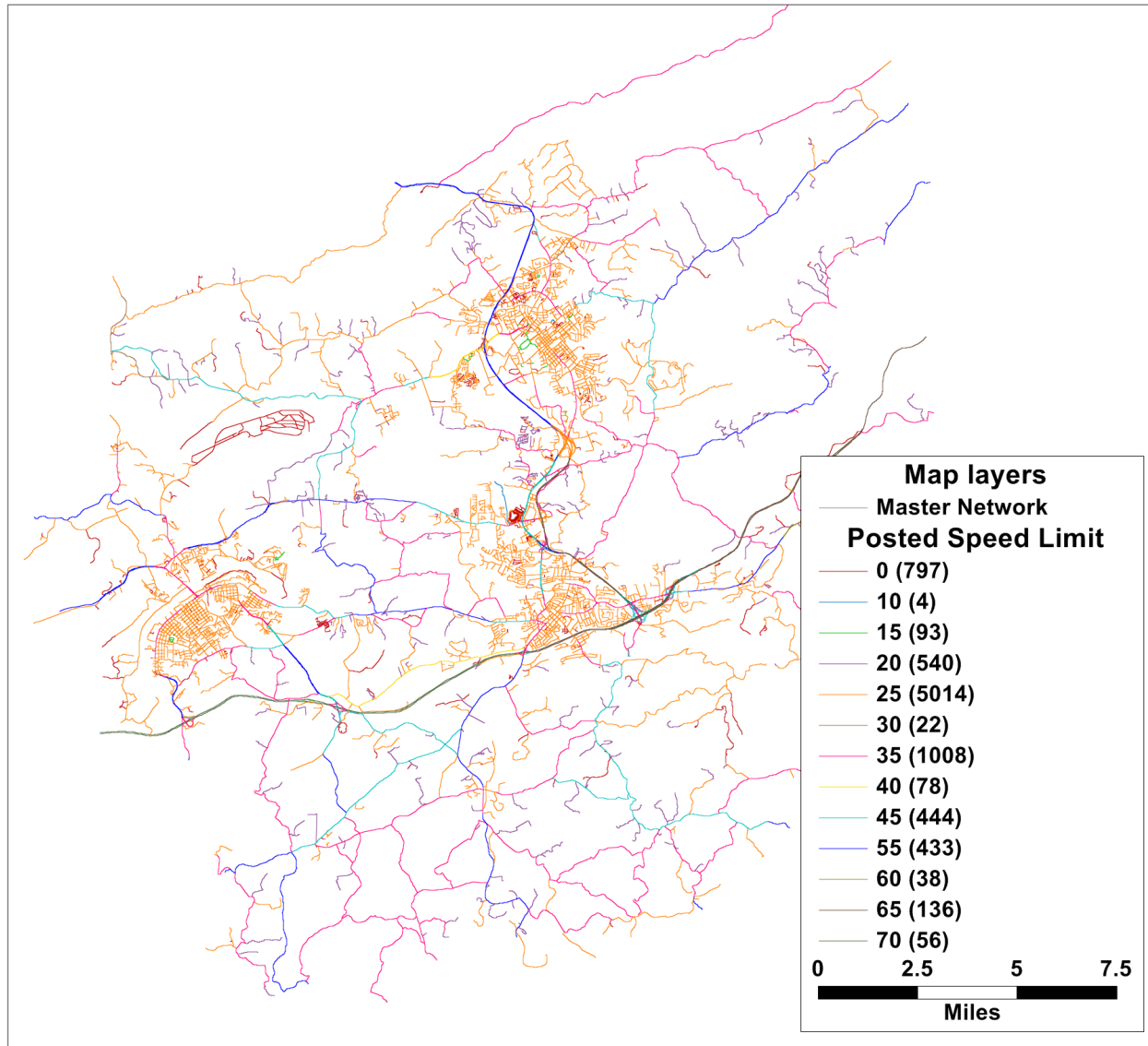


Figure 4: Plot of Posted Speeds

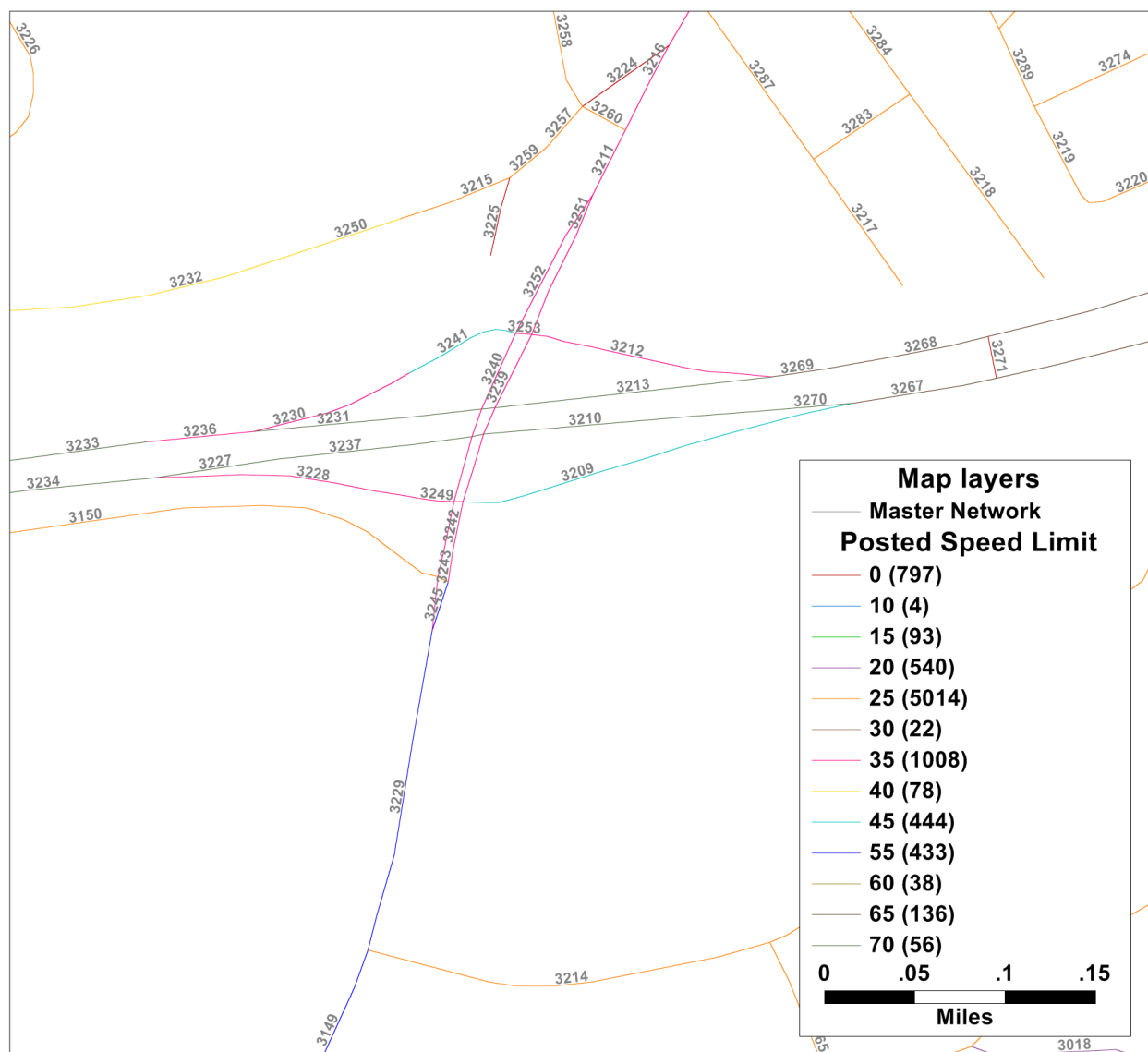


Figure 5: Example Interchange Speed Limit Issues

A review of the number of lanes data provided revealed more problems than the posted speed review. The network included 5,225 links with a value of 0 for the number of lanes, and improperly categorized southbound lanes on I-81 (from the start to Christiansburg) as being 2 lanes instead of 3. As can be seen in Figure 6, most of the 0-lane roads will likely not be utilized in the model network once simplified, but zero value link checks were still required to ensure that all roadways had a reasonable number of lanes value.

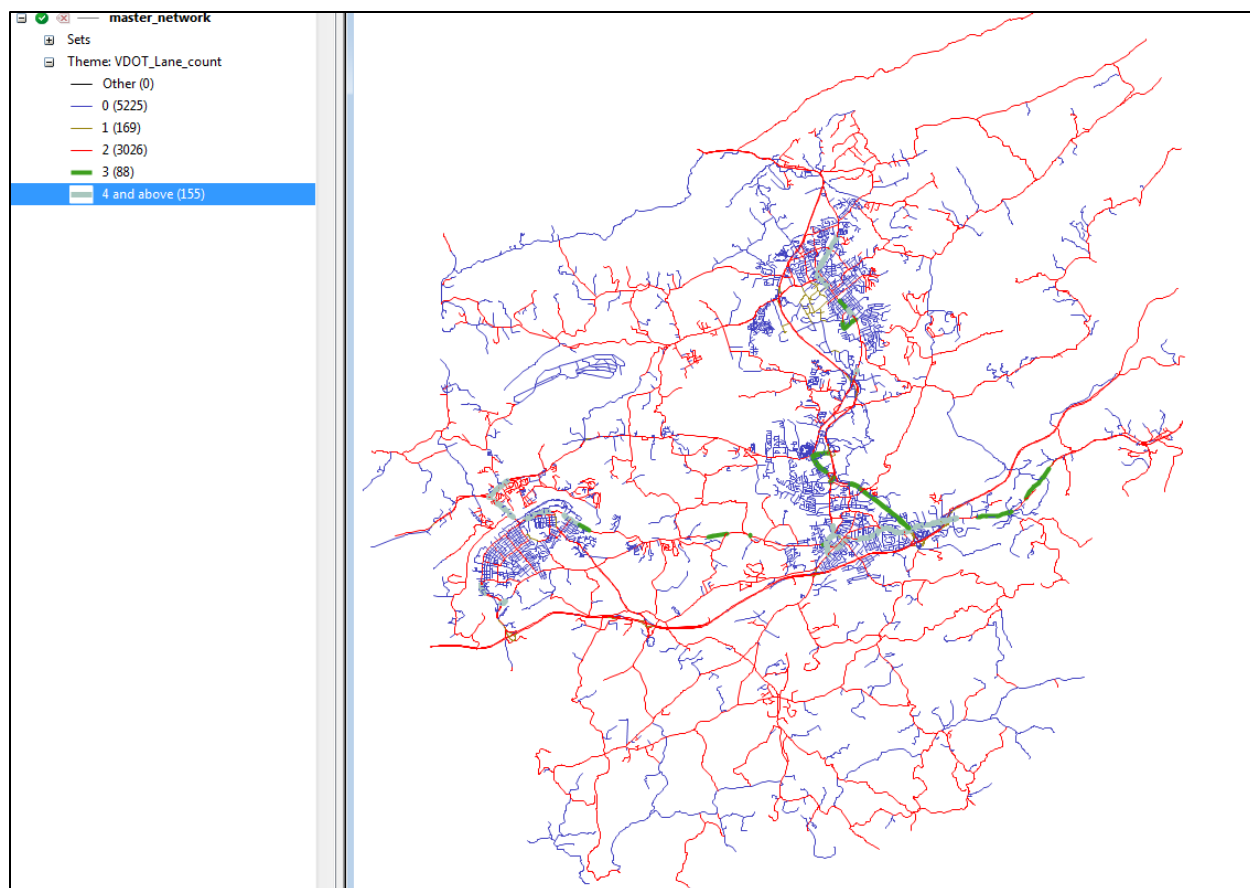


Figure 6: Number of Lanes Values

The number of lanes values were updated to reasonable values, with the expectation that network assignment may reveal additional links that need updating.

Proper linking of count locations to the network

The HERE network provided was preloaded with count ID numbers (designed as TMS Link ID) and Average Annual Weekday Traffic (AAWDT), however these ID numbers were not uniquely assigned to links on the network. A review of link IDs determined that of the 471 count IDs, 365 were assigned to more than one link. In one instance a single count ID was assigned to 70 links in the network.

Duplication of the counts on multiple links leads to issues in highway assignment so an effort to correct the count location issues required importing a point shapefile of VDOT count locations. The count location point layer was used to assign each count ID to unique links in the network through a GISDK script. The results of the tagging process were manually checked for reasonableness.

Correction of One Way Links and Intersections

In TransCAD, a roadway link is typically represented as single line with distinct attributes for the number of lanes in each direction. However, in some locations, the best representation of the network includes separate one way links. Interstate lanes, highway ramps, and roundabouts or other loops can be coded as one way to preserve length attributes and help visually orient users to the network.

Many of these one-way links were incorrectly coded in the original shapefile. The dataset contained a field called ONE_WAY that was intended to designate the directionality. As shown in Figure 7, the field has 3 values, “B” (bi-directional), “FT” (with segment arc directionality), “TF” (against segment arc directionality), with the latter two designating one-way links.

Even though the field indicated route directionality a review of the coding was performed to assure the values properly represented reality. It was discovered that several links were improperly marked as one way; these were corrected to ensure connectivity. Figure 8 displays an example in which there are properly coded one way links on US 11 near Radford, but these one-way links improperly extend to an area that should be bi-directional (as shown with the blue circles).

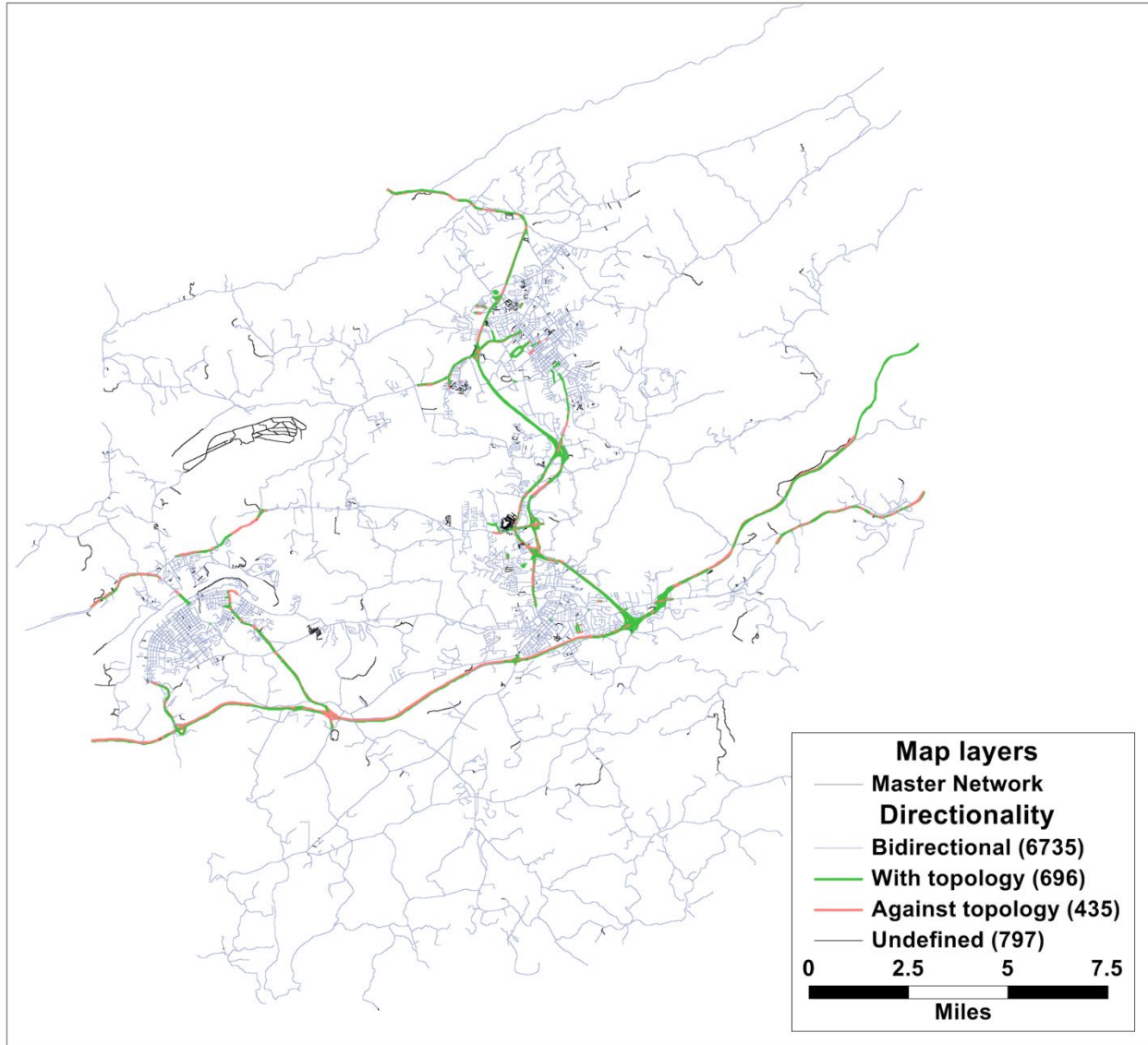


Figure 7: Initial Link Directionality Coding

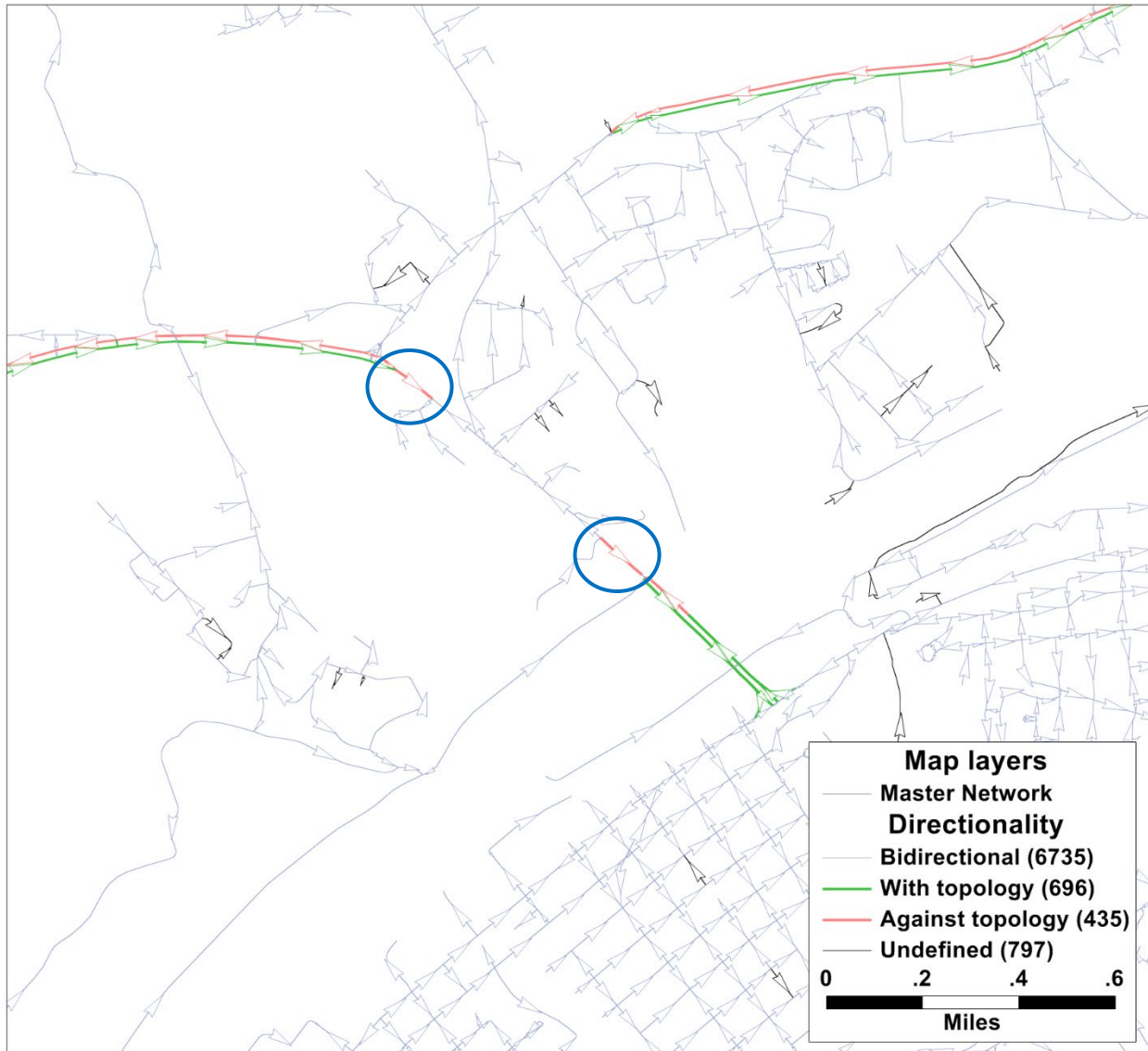


Figure 8: Example of Bidirectional Links Coded as One-Way (Circled in Blue)

Another issue were one-way links that were properly coded as one-way, but that flowed in the wrong direction. The picture below of the I-81/US 460 interchange in Christiansburg illustrates this problem. The directional arrows show the topology of the network. Blue links are bidirectional, green links show a flow of traffic with the arrows, and red links show a flow of traffic against the arrows. The red lane of I-81 (marked by the black star) indicates that the entirety of I-81 through this section is coded to flow south to north, which would break network continuity. This was corrected by marking all 1 way links as being FT.

In addition to the directionality and connection issues described there were 797 links with no value in the directionality field. Those links were updated with a value that represented the proper directionality of the roadway segment.

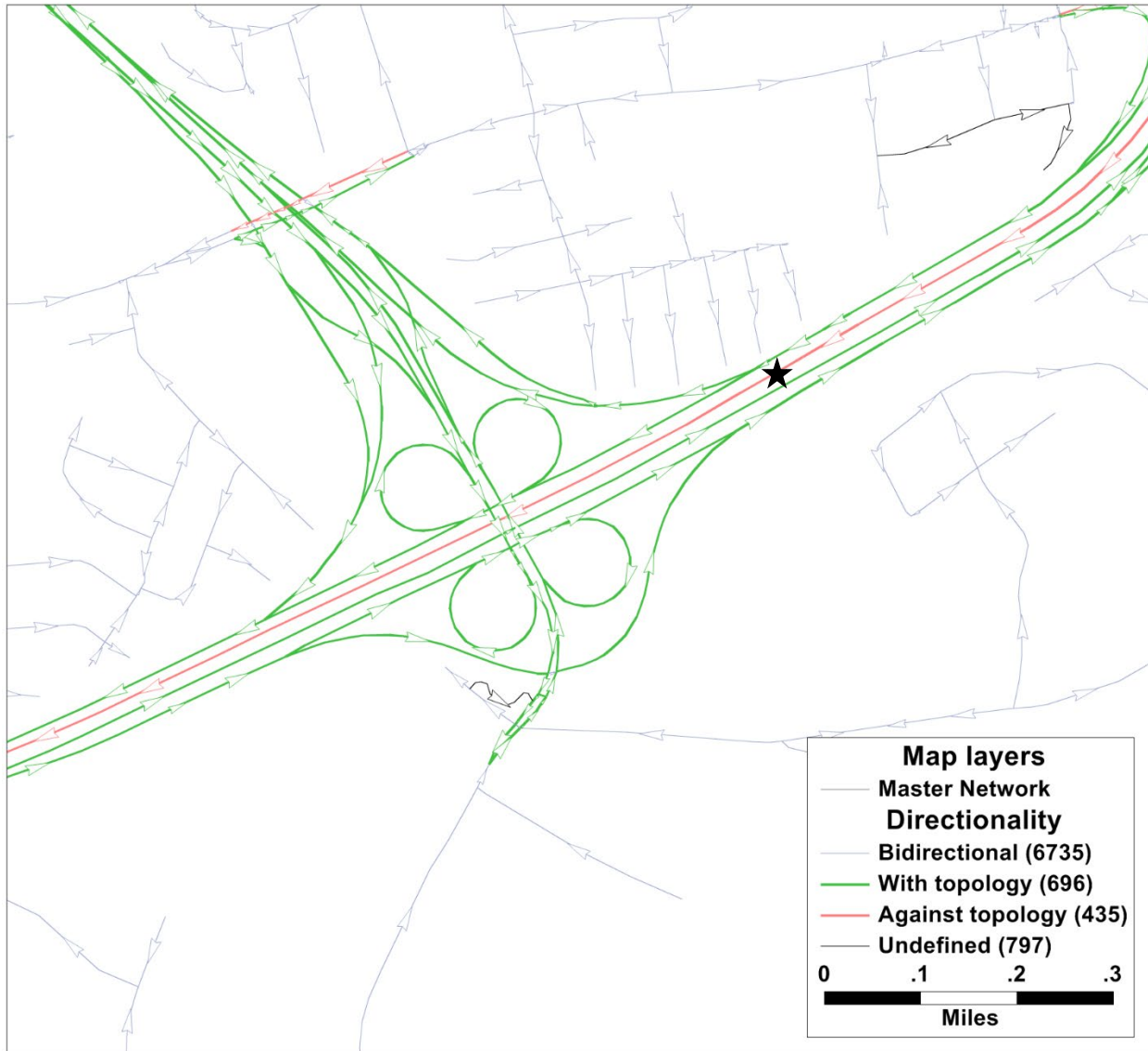


Figure 9: Example One-Way Links Coded in the Wrong Direction (Marked by Black Star)

Another problem in the network that required correction was the presence of improper turns. TransCAD determines whether vehicles are permitted to turn by linking together roadway segments at nodes (as shown by the black dots on Figure 10). Importing roadways that cross each other at different grades can confuse the network editor and generate routes that allow, for example, a right turn off the interstate rather than utilizing the cloverleaf ramps. The left side of Figure 10 highlights a route in which the coding allows for improper movements off the interstate, while the right side of the figure shows the result of that same movement after illogical connections were removed. These issues were inspected and addressed in the same process as the lane directionality.



Turn in Original Network

Turn in Final Network

Figure 10: Example of Improper Turning Movements: I-81 and US 460

Simplification of the Network

The original shapefile of the network was finely detailed and included 16,270 links as shown in Figure 11. Many of the links represent local neighborhood roads that are not congested, do not have reliable count data, and whose inclusion in the model would likely distort the ability to calibrate the model regionally. Typically, roadway links selected for inclusion in the model include Interstates, US Routes, Virginia state designated roadways and then local roadways that provide significant impact to the travel patterns in the region. Neighborhood roads and lower level collectors are not included as they are used for internal circulation that is not best represented in a regional travel model. Virtually all links located entirely inside a single TAZ were removed, which greatly simplifies the network and will allow the model to focus on the most important roads in the region.

Figure 12 highlights in red all roadways in the region that were removed from the original network file. This substantially reduced the number of roadway segments to be modeled.

Figure 13 through Figure 15 provide a closer look at the links removed from the three municipalities in the region.

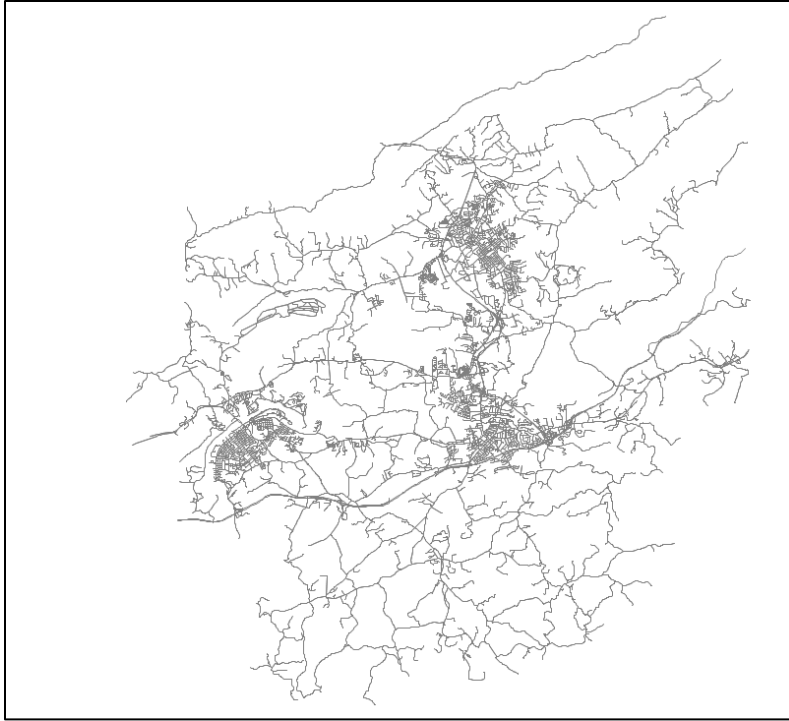


Figure 11: Initial Links Included in the NRV Network

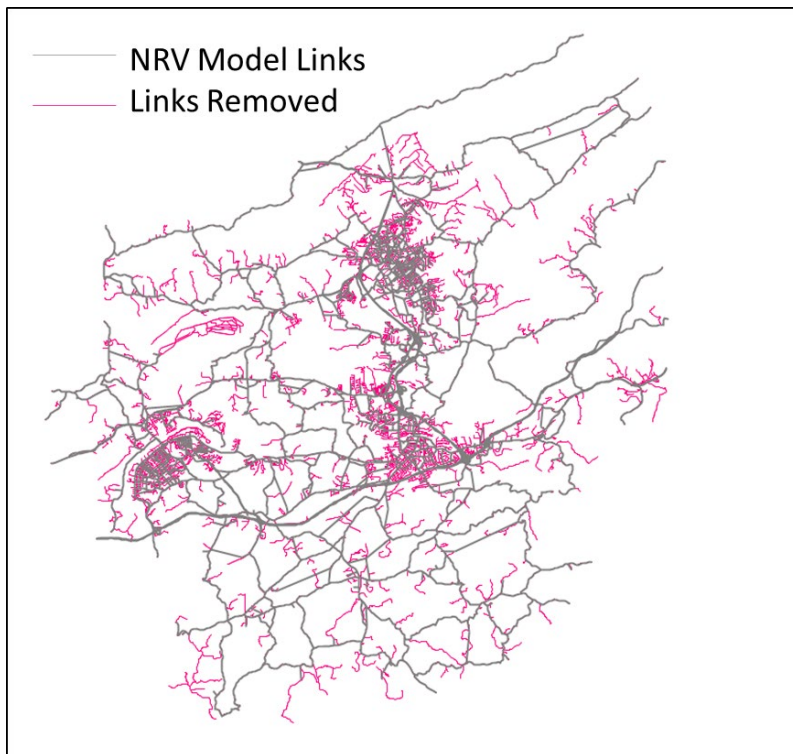


Figure 12: Links Flagged for Removal from the NRV Region

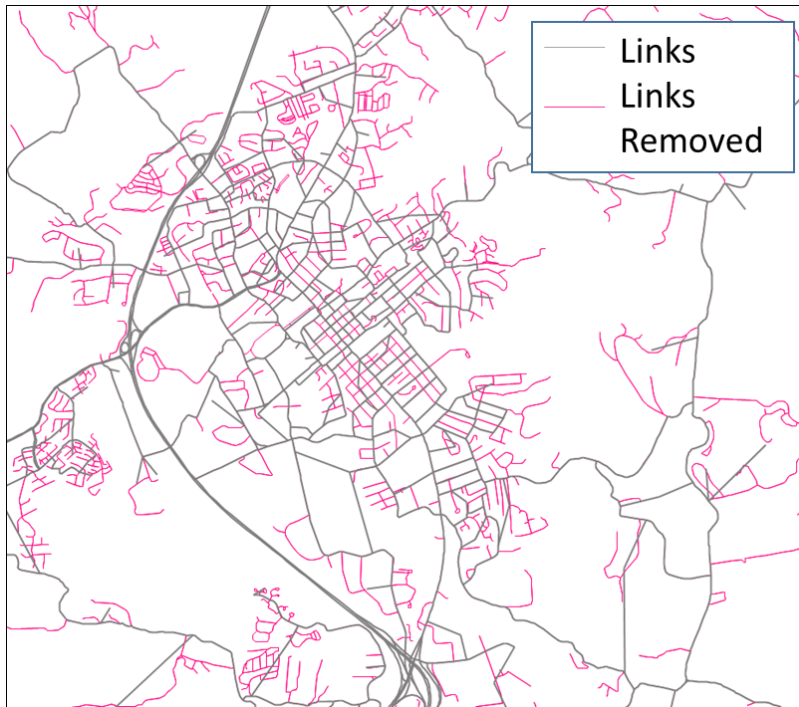


Figure 13: Links Removed in Blacksburg

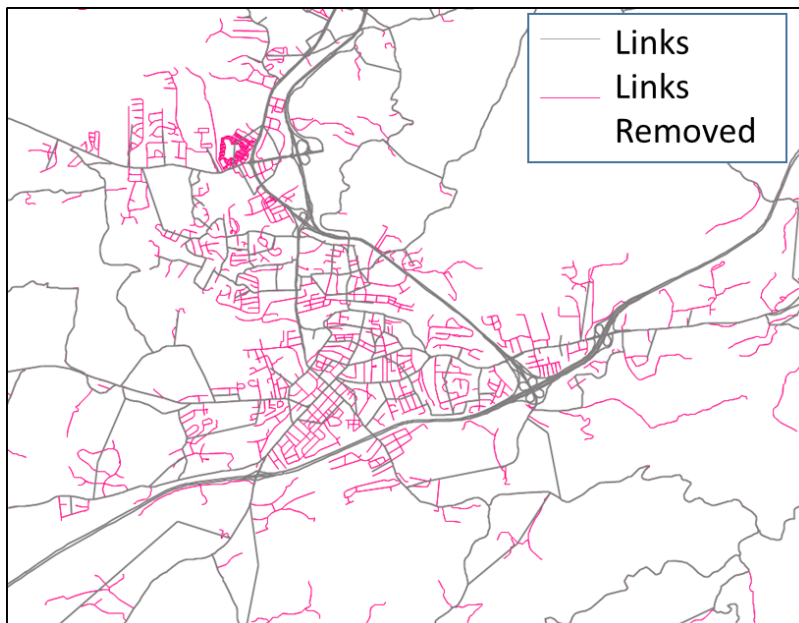


Figure 14: Links Removed in Christiansburg

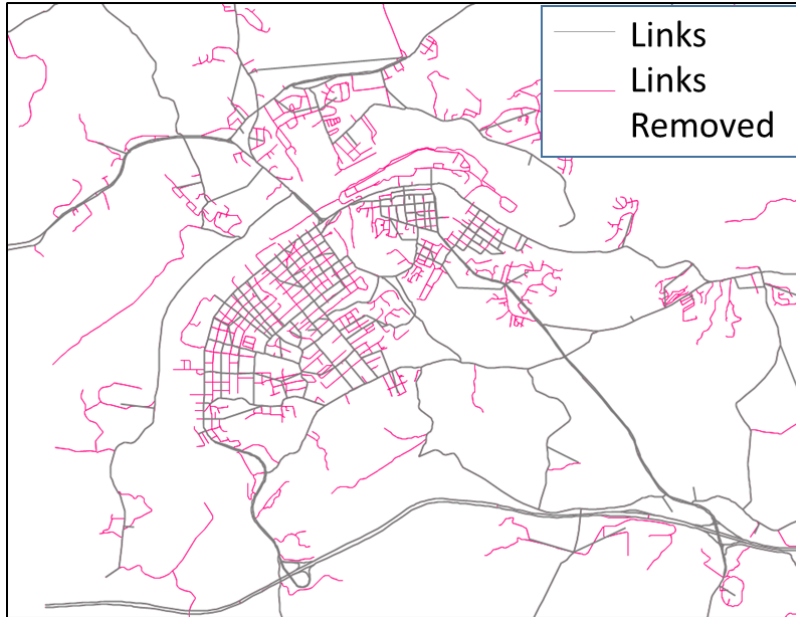


Figure 15: Links Removed in Radford

In addition to eliminating links that are not critical to model in the network, it is typical to clean up the number of links so that regions do not have to maintain attributes on redundant link segments in the network as they change over time. Figure 16 provides a look at the link segments along a key roadway. This roadway has multiple segments, designated by the blue nodes, and each of these segments have the same posted speeds and number of lanes. It is not necessary to have each segment and instead combining them into one segment is ideal for travel models. Therefore, the extra segments were cleaned and merged into a more manageable set of links as shown in Figure 17.

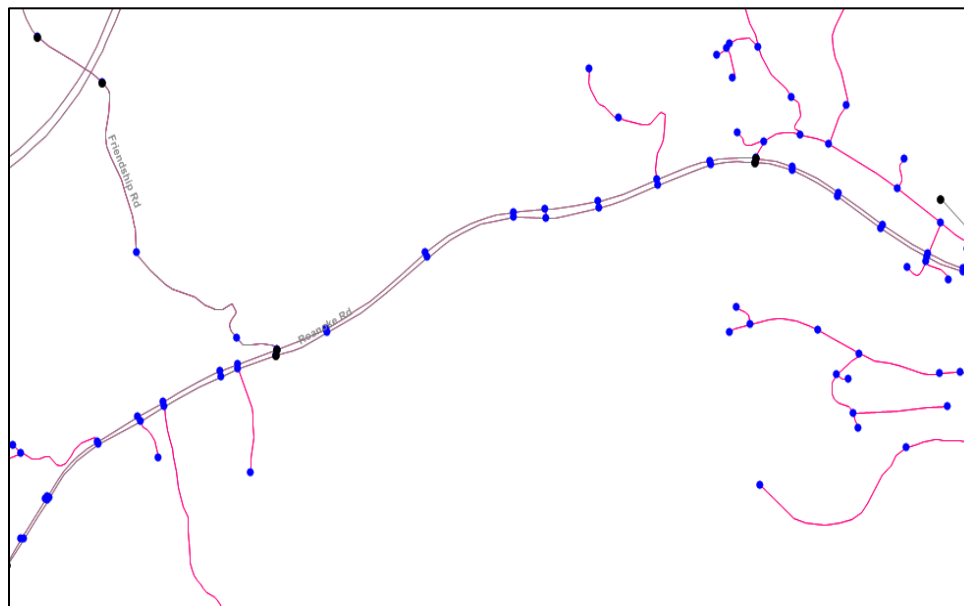


Figure 16: Example of Initial Link Segments on Key Roadway

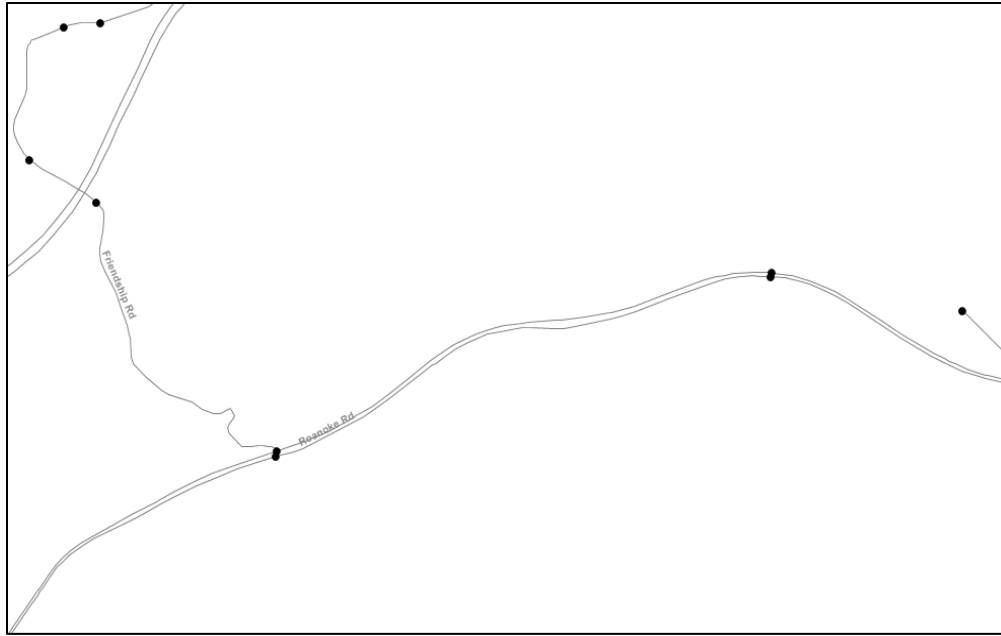


Figure 17: Example Cleaned Link Segments

The deletion of non-critical roadways in the region and the reduction of link segmentation led to an NRV network that includes only 3,000 links and reflects the key roadway connections in the region.

Centroid Connectors

The detailed local networks that were removed in the prior step are typically replaced by centroid connectors. Centroid connectors are lines that represent the connection of internal circulation roadways and neighborhood streets allow traffic to be loaded onto the regional highway network. They are manually added to all TAZs in the network and require judgement as to their position and connectivity based on the density of housing or economic activity in the area. Figure 18 displays the centroids, shown as red lines, for the NRV region. Figure 19 provides an example of a closer look at the centroid connectors in Blacksburg.

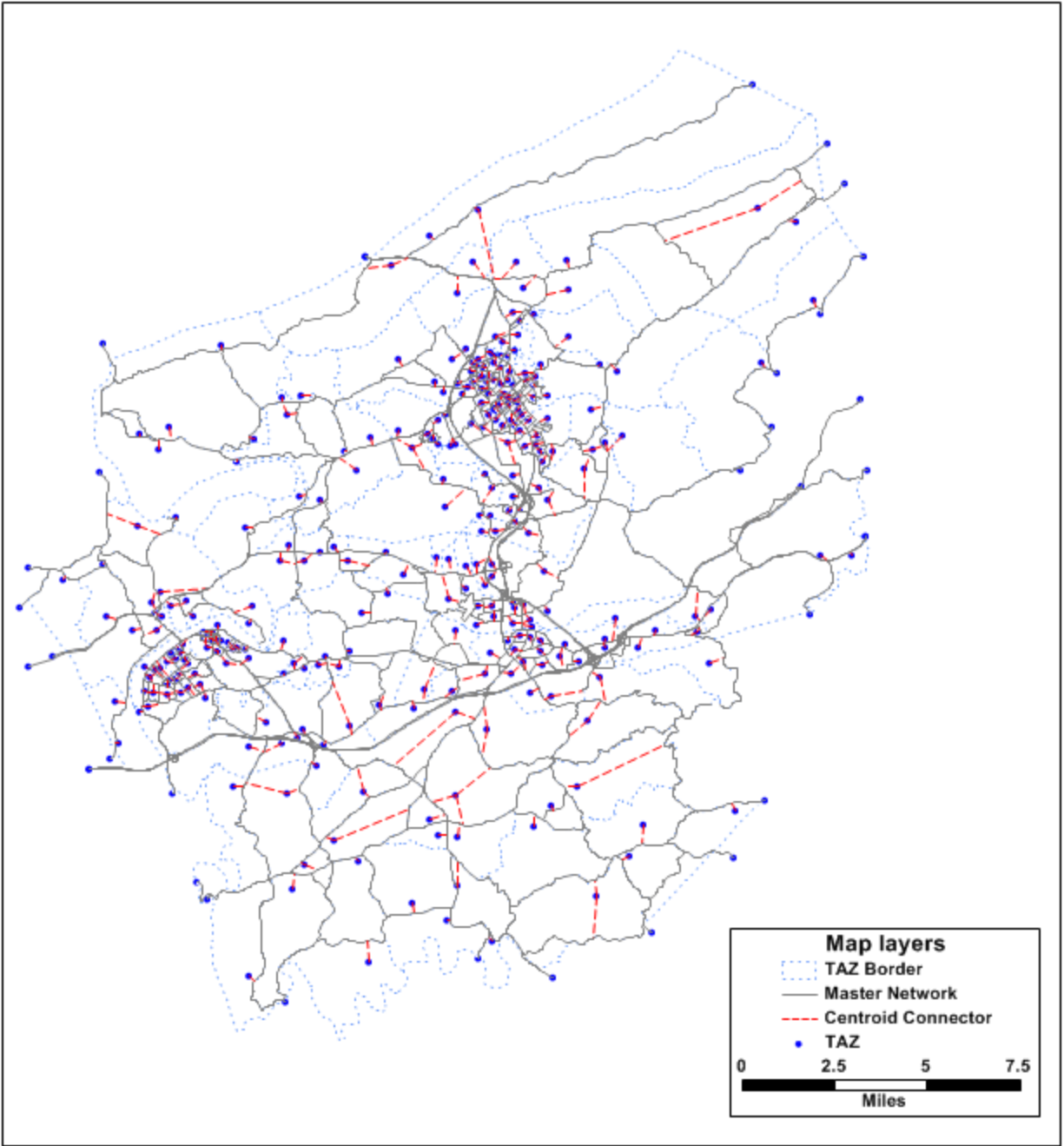


Figure 18: NRV Regional View of Centroid Connectors

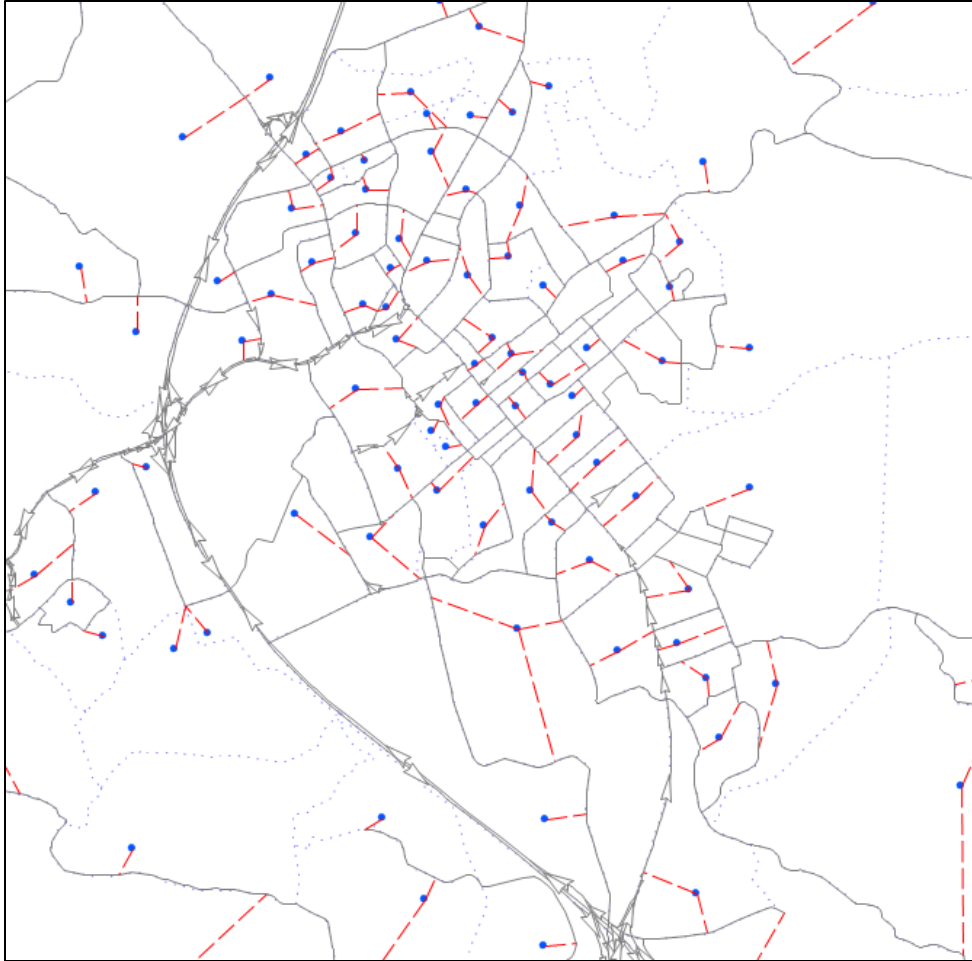


Figure 19: Blacksburg Centroid Connectors

Appendix B-Model Attributes for Highway Network, Transit Routes and Socio-economic Data

This appendix provides the details about the attributes required for the NRV model usage..

The directory structure for the NRV model is:

Folder	Subfolder	Description
master		
	area_type	Parameter files controlling area type
	assignment	Parameter files controlling assignment
	cv	Parameter files controlling commercial vehicles
	directionality	Parameter files controlling directionality and occupancy
	distribution	Parameter files controlling resident distribution
	external	Parameter files controlling external trips (EE and IEEI)
	generation	Parameter files controlling resident generation
	mode	Parameter files controlling resident mode choice
	networks	Master highway and transit networks
	sedata	Socio-economic tables
	skimming	Parameter files controlling network skimming (highway and transit)
	taz	TAZ geographic file
	tod	Parameter files controlling time of day
	university	Parameter files controlling university generation and distribution

Folder	Subfolder	Description
other		
	counts	Traffic count point file
src		
	bmp	Images needed by gui
	gisdk	gisdk source code
	R	R distribution/environment needed to run the model

For any scenario in the NRV model the structure will be:

Folder	Subfolder	Subfolder	Description
scenarios			
	Base_2016*		
		inputs	Exported from master/ (same folders)
		outputs/	
		assignment	Outputs of highway and transit assignment
		directionality	Outputs directionality and occupancy
		distribution	Outputs of distribution models
		external	Outputs of external models
		generation	Outputs of generation models
		mode	Outputs of mode choice models
		networks	Output highway and transit networks
		sedata	Output se data (with lots of extra zonal data appended)
		skims	Output of highway and transit skimming
		summary	Standard/Automated summaries produced
		taz	Output TAZ layer

* scenarios can be named anything but all scenarios will have this same folder structure

The data used for transit routes in the NRV model are:

Field Name	Description
Route_ID	Transcad Provided ID
Route_Name	Local Route Name
ProjID	Used if future project not in base year
Mode	Local Bus, Express Bus, LRT/BRT
Fare	Cost in dollars
AM_Headway	Average period headway for route
MD_Headway	Average period headway for route
PM_Headway	Average period headway for route
NT_Headway	Average period headway for route

The fields used in NRV model for the highway links are shown in the series of tables on the following pages. Values in blue are inputs required by the user to assure the model works, the values in green are calculated during the model operation.

Field Name	Description/Code	Additional Information
ID	TransCAD generated field	
Dir	TransCAD generated field	
Length	distance in miles	
Name	Name of road	
HCMTType	HCM Facility Type	Freeway=Freeway,MLHighway =Multi-Lane Highway,TLHighway=Two-Lane Highway,PrArterial=Principal Arterial,MinArterial=Minor Arterial,Collector=Collector,Local=Local,CC = Centroid
PostedSpeed	Posted Speed Limit	Centroid connector speeds are overwritten by the model based on area type
HCMMedian	HCM Median Type	Restrictive,NonRestrictive,None
ABLanes	Number of lanes in AB direction	
BALanes	Number of lanes in BA direction	
Terrain	Level,Rolling,Mountain	
HOV	How many passengers needed to use this link?	0 = no restriction,2 = HOV2+,3 = HOV3+
Walk	1: Walking allowed 2: Not allowed	
Drive	1: Driving allowed 2: Not allowed	
Transit	1: Transit allowed 2: Not allowed	
CountID	For one-way paired links the CountID is duplicated to allow them to be grouped.	
DailyCount	OCT_AWDT field from the count point file.	
SUTCount	SUT percentage applied to DailyCount	
MUTCount	MUT percentage applied to DailyCount	
Screenline	Screenline number. See the Screenline.dbd,in the "other" folder.	
p1ID	Project Field,Future Project ID number	
p1HCMTType	Project Field,HCM Facility Type	
p1PostedSpeed	Project Field,Posted Speed Limit	
p1ABLanes	Project Field,Project AB Lanes	
p1BALanes	Project Field,project BA Lanes	
p1HCMMedian	Project Field,HCM Median Type	
p1Terrain	Project Field,Level,Rolling,Mountain	
AreaType	Suburban, Rural and Urban calculated by model	
ramp	Is this link a ramp? Yes or no	
capd_phpl	daily capacity D for use in planning(per hr per lane)	
cape_phpl	daily capacity E for use in planning (per hr per lane)	

Values in blue are inputs required by the user to assure the model works, the values in green are calculated during the model operation.

Field Name	Description/Code	Additional Information
ABAMCapD	hourly los D capacity per lane	
BAAMCapD	hourly los D capacity per lane	
ABMDCapD	hourly los D capacity per lane	
BAMDCapD	hourly los D capacity per lane	
ABPMCpD	hourly los D capacity per lane	
BAPMCpD	hourly los D capacity per lane	
ABNTCapD	hourly los D capacity per lane	
BANTCapD	hourly los D capacity per lane	
ABAMCapE	hourly los E capacity per lane	
BAAMCapE	hourly los E capacity per lane	
ABMDCapE	hourly los E capacity per lane	
BAMDCapE	hourly los E capacity per lane	
ABPMCpE	hourly los E capacity per lane	
BAPMCpE	hourly los E capacity per lane	
ABNTCapE	hourly los E capacity per lane	
BANTCapE	hourly los E capacity per lane	
FFSpeed	Free flow travel speed	
FFTime	Free flow travel time	
Alpha	VDF alpha value	
WalkTime	Length / 3 mph	
Mode	Marks all links with a 1 (nontransit mode)	
InitCongSpeed	Init guess at cong speed	
InitCongTime	Init guess at cong time	
AB_Flow_PCE_AM	AB Flow passed on passenger car equivalents	
BA_Flow_PCE_AM	BA Flow passed on passenger car equivalents	
Tot_Flow_PCE_AM	Total Flow passed on passenger car equivalents	
AB_Time_AM	AM Time by direction	
BA_Time_AM	AM Time by direction	
Max_Time_AM	Maximum AM Time between both directions	
AB_VOCE_AM	V/C based on LOS E	
BA_VOCE_AM	V/C based on LOS E	
Max_VOC_AM	Maximum VOC between both directions	
AB_VMT_AM	VMT by direction and time period	
BA_VMT_AM	VMT by direction and time period	
Tot_VMT_AM	Maximum VMT between both directions	
AB_VHT_AM	VHT by direction and time period	
BA_VHT_AM	VHT by direction and time period	
Tot_VHT_AM	Maximum VHT between both directions	

Values in blue are inputs required by the user to assure the model works, the values in green are calculated during the model operation.

Field Name	Description/Code	Additional Information
AB_Speed_AM	VHT by direction and time period	
BA_Speed_AM	VHT by direction and time period	
AB_VDF_AM	VDF by direction	
BA_VDF_AM	VDF by direction	
Max_VDF_AM	Maximum VDF between both directions	
AB_MSA_Flow_AM	MSA flow by direction and period	MSA=Method of Successive Averages
BA_MSA_Flow_AM	MSA flow by direction and period	MSA=Method of Successive Averages
AB_MSA_Cost_AM	MSA cost by direction and period	MSA=Method of Successive Averages
BA_MSA_Cost_AM	MSA cost by direction and period	MSA=Method of Successive Averages
AB_MSA_Time_AM	Congested MSA time fed back into skimming	MSA=Method of Successive Averages
BA_MSA_Time_AM	Congested MSA time fed back into skimming	MSA=Method of Successive Averages
AB_Flow_res_da_AM	Link flow by direction and period	resident drive alone
BA_Flow_res_da_AM	Link flow by direction and period	resident drive alone
AB_Flow_res_sr2_AM	Link flow by direction and period	resident shared ride 2
BA_Flow_res_sr2_AM	Link flow by direction and period	resident shared ride 2
AB_Flow_res_sr3_AM	Link flow by direction and period	resident shared ride 3+
BA_Flow_res_sr3_AM	Link flow by direction and period	resident shared ride 3+
AB_Flow_CV_AM	Link flow by direction and period	commercial vehicles
BA_Flow_CV_AM	Link flow by direction and period	commercial vehicles
AB_Flow_SUT_AM	Link flow by direction and period	single unit truck
BA_Flow_SUT_AM	Link flow by direction and period	single unit truck
AB_Flow_MUT_AM	Link flow by direction and period	multi-unit truck
BA_Flow_MUT_AM	Link flow by direction and period	multi-unit truck
AB_Flow_IEEI_AM	Link flow by direction and period	internal/external
BA_Flow_IEEI_AM	Link flow by direction and period	internal/external
AB_Flow_NHBNR_AM	Link flow by direction and period	non-home based non-resident
BA_Flow_NHBNR_AM	Link flow by direction and period	non-home based non-resident
AB_Flow_EE_AM	Link flow by direction and period	external to external
BA_Flow_EE_AM	Link flow by direction and period	external to external
AB_Flow_AM	Link flow by direction and period	
BA_Flow_AM	Link flow by direction and period	
Tot_Flow_AM	Total Link flow by period	

Values in blue are inputs required by the user to assure the model works, the values in green are calculated during the model operation.

Field Name	Description/Code	Additional Information
AB_Flow_PCE_MD	AB Flow passed on passenger car equivalents	
BA_Flow_PCE_MD	BA Flow passed on passenger car equivalents	
Tot_Flow_PCE_MD	Total Flow passed on passenger car equivalents	
AB_Time_MD	AM Time by direction	
BA_Time_MD	AM Time by direction	
Max_Time_MD	Maximum AM Time between both directions	
AB_VOCE_MD	V/C based on LOS E	
BA_VOCE_MD	V/C based on LOS E	
Max_VOC_MD	Maximum VOC between both directions	
AB_VMT_MD	VMT by direction and time period	
BA_VMT_MD	VMT by direction and time period	
Tot_VMT_MD	Maximum VMT between both directions	
AB_VHT_MD	VHT by direction and time period	
BA_VHT_MD	VHT by direction and time period	
Tot_VHT_MD	Maximum VHT between both directions	
AB_Speed_MD	VHT by direction and time period	
BA_Speed_MD	VHT by direction and time period	
AB_VDF_MD	VDF by direction	
BA_VDF_MD	VDF by direction	
Max_VDF_MD	Maximum VDF between both directions	
AB_MSA_Flow_MD	MSA flow by direction and period	MSA=Method of Successive Averages
BA_MSA_Flow_MD	MSA flow by direction and period	MSA=Method of Successive Averages
AB_MSA_Cost_MD	MSA cost by direction and period	MSA=Method of Successive Averages
BA_MSA_Cost_MD	MSA cost by direction and period	MSA=Method of Successive Averages
AB_MSA_Time_MD	Congested MSA time fed back into skimming	MSA=Method of Successive Averages
BA_MSA_Time_MD	Congested MSA time fed back into skimming	MSA=Method of Successive Averages
AB_Flow_res_da_MD	Link flow by direction and period	resident drive alone
BA_Flow_res_da_MD	Link flow by direction and period	resident drive alone
AB_Flow_res_sr2_MD	Link flow by direction and period	resident shared ride 2
BA_Flow_res_sr2_MD	Link flow by direction and period	resident shared ride 2
AB_Flow_res_sr3_MD	Link flow by direction and period	resident shared ride 3+
BA_Flow_res_sr3_MD	Link flow by direction and period	resident shared ride 3+
AB_Flow_CV_MD	Link flow by direction and period	commercial vehicles
BA_Flow_CV_MD	Link flow by direction and period	commercial vehicles

Values in blue are inputs required by the user to assure the model works, the values in green are calculated during the model operation.

Field Name	Description/Code	Additional Information
AB_Flow_SUT_MD	Link flow by direction and period	single unit truck
BA_Flow_SUT_MD	Link flow by direction and period	single unit truck
AB_Flow_MUT_MD	Link flow by direction and period	multi-unit truck
BA_Flow_MUT_MD	Link flow by direction and period	multi-unit truck
AB_Flow_IEEI_MD	Link flow by direction and period	internal/external
BA_Flow_IEEI_MD	Link flow by direction and period	internal/external
AB_Flow_NHBNR_MD	Link flow by direction and period	non-home based non-resident
BA_Flow_NHBNR_MD	Link flow by direction and period	non-home based non-resident
AB_Flow_EE_MD	Link flow by direction and period	external to external
BA_Flow_EE_MD	Link flow by direction and period	external to external
AB_Flow_MD	Link flow by direction and period	
BA_Flow_MD	Link flow by direction and period	
Tot_Flow_MD	Total Link flow by period	
AB_Flow_PCE_PM	AB Flow passed on passenger car equivalents	
BA_Flow_PCE_PM	BA Flow passed on passenger car equivalents	
Tot_Flow_PCE_PM	Total Flow passed on passenger car equivalents	
AB_Time_PM	AM Time by direction	
BA_Time_PM	AM Time by direction	
Max_Time_PM	Maximum AM Time between both directions	
AB_VOCE_PM	V/C based on LOS E	
BA_VOCE_PM	V/C based on LOS E	
Max_VOC_PM	Maximum VOC between both directions	
AB_VMT_PM	VMT by direction and time period	
BA_VMT_PM	VMT by direction and time period	
Tot_VMT_PM	Maximum VMT between both directions	
AB_VHT_PM	VHT by direction and time period	
BA_VHT_PM	VHT by direction and time period	
Tot_VHT_PM	Maximum VHT between both directions	
AB_Speed_PM	VHT by direction and time period	
BA_Speed_PM	VHT by direction and time period	
AB_VDF_PM	VDF by direction	
BA_VDF_PM	VDF by direction	
Max_VDF_PM	Maximum VDF between both directions	
AB_MSA_Flow_PM	MSA flow by direction and period	MSA=Method of Successive Averages
BA_MSA_Flow_PM	MSA flow by direction and period	MSA=Method of Successive Averages
AB_MSA_Cost_PM	MSA cost by direction and period	MSA=Method of Successive Averages
BA_MSA_Cost_PM	MSA cost by direction and period	MSA=Method of Successive Averages

Values in blue are inputs required by the user to assure the model works, the values in green are calculated during the model operation.

Field Name	Description/Code	Additional Information
AB_MSA_Time_PM	Congested MSA time fed back into skimming	MSA=Method of Successive Averages
BA_MSA_Time_PM	Congested MSA time fed back into skimming	MSA=Method of Successive Averages
AB_Flow_res_da_PM	Link flow by direction and period	resident drive alone
BA_Flow_res_da_PM	Link flow by direction and period	resident drive alone
AB_Flow_res_sr2_PM	Link flow by direction and period	resident shared ride 2
BA_Flow_res_sr2_PM	Link flow by direction and period	resident shared ride 2
AB_Flow_res_sr3_PM	Link flow by direction and period	resident shared ride 3+
BA_Flow_res_sr3_PM	Link flow by direction and period	resident shared ride 3+
AB_Flow_CV_PM	Link flow by direction and period	commercial vehicles
BA_Flow_CV_PM	Link flow by direction and period	commercial vehicles
AB_Flow_SUT_PM	Link flow by direction and period	single unit truck
BA_Flow_SUT_PM	Link flow by direction and period	single unit truck
AB_Flow_MUT_PM	Link flow by direction and period	multi-unit truck
BA_Flow_MUT_PM	Link flow by direction and period	multi-unit truck
AB_Flow_IEEI_PM	Link flow by direction and period	internal/external
BA_Flow_IEEI_PM	Link flow by direction and period	internal/external
AB_Flow_NHBNR_PM	Link flow by direction and period	non-home based non-resident
BA_Flow_NHBNR_PM	Link flow by direction and period	non-home based non-resident
AB_Flow_EE_PM	Link flow by direction and period	external to external
BA_Flow_EE_PM	Link flow by direction and period	external to external
AB_Flow_PM	Link flow by direction and period	
BA_Flow_PM	Link flow by direction and period	
Tot_Flow_PM	Total Link flow by period	
AB_Flow_PCE_NT	AB Flow passed on passenger car equivalents	
BA_Flow_PCE_NT	BA Flow passed on passenger car equivalents	
Tot_Flow_PCE_NT	Total Flow passed on passenger car equivalents	
AB_Time_NT	AM Time by direction	
BA_Time_NT	AM Time by direction	
Max_Time_NT	Maximum AM Time between both directions	
AB_VOCE_NT	V/C based on LOS E	
BA_VOCE_NT	V/C based on LOS E	
Max_VOC_NT	Maximum VOC between both directions	
AB_VMT_NT	VMT by direction and time period	
BA_VMT_NT	VMT by direction and time period	
Tot_VMT_NT	Maximum VMT between both directions	
AB_VHT_NT	VHT by direction and time period	
BA_VHT_NT	VHT by direction and time period	
Tot_VHT_NT	Maximum VHT between both directions	

Values in blue are inputs required by the user to assure the model works, the values in green are calculated during the model operation.

Field Name	Description/Code	Additional Information
AB_Speed_NT	VHT by direction and time period	
BA_Speed_NT	VHT by direction and time period	
AB_VDF_NT	VDF by direction	
BA_VDF_NT	VDF by direction	
Max_VDF_NT	Maximum VDF between both directions	
AB_MSA_Flow_NT	MSA flow by direction and period	MSA=Method of Successive Averages
BA_MSA_Flow_NT	MSA flow by direction and period	MSA=Method of Successive Averages
AB_MSA_Cost_NT	MSA cost by direction and period	MSA=Method of Successive Averages
BA_MSA_Cost_NT	MSA cost by direction and period	MSA=Method of Successive Averages
AB_MSA_Time_NT	Congested MSA time fed back into skimming	MSA=Method of Successive Averages
BA_MSA_Time_NT	Congested MSA time fed back into skimming	MSA=Method of Successive Averages
AB_Flow_res_da_NT	Link flow by direction and period	resident drive alone
BA_Flow_res_da_NT	Link flow by direction and period	resident drive alone
AB_Flow_res_sr2_NT	Link flow by direction and period	resident shared ride 2
BA_Flow_res_sr2_NT	Link flow by direction and period	resident shared ride 2
AB_Flow_res_sr3_NT	Link flow by direction and period	resident shared ride 3+
BA_Flow_res_sr3_NT	Link flow by direction and period	resident shared ride 3+
AB_Flow_CV_NT	Link flow by direction and period	commercial vehicles
BA_Flow_CV_NT	Link flow by direction and period	commercial vehicles
AB_Flow_SUT_NT	Link flow by direction and period	single unit truck
BA_Flow_SUT_NT	Link flow by direction and period	single unit truck
AB_Flow_MUT_NT	Link flow by direction and period	multi-unit truck
BA_Flow_MUT_NT	Link flow by direction and period	multi-unit truck
AB_Flow_IEEI_NT	Link flow by direction and period	internal/external
BA_Flow_IEEI_NT	Link flow by direction and period	internal/external
AB_Flow_NHBNR_NT	Link flow by direction and period	non-home based non-resident
BA_Flow_NHBNR_NT	Link flow by direction and period	non-home based non-resident
AB_Flow_EE_NT	Link flow by direction and period	external to external
BA_Flow_EE_NT	Link flow by direction and period	external to external
AB_Flow_NT	Link flow by direction and period	
BA_Flow_NT	Link flow by direction and period	
Tot_Flow_NT	Total Link flow by period	
AB_Delay_AM	(CongTime - FFTime) * Flow / 60	
BA_Delay_AM	(CongTime - FFTime) * Flow / 60	
AB_Delay_MD	(CongTime - FFTime) * Flow / 60	
BA_Delay_MD	(CongTime - FFTime) * Flow / 60	
AB_Delay_PM	(CongTime - FFTime) * Flow / 60	
BA_Delay_PM	(CongTime - FFTime) * Flow / 60	

Values in blue are inputs required by the user to assure the model works, the values in green are calculated during the model operation.

Field Name	Description/Code	Additional Information
AB_Delay_NT	(CongTime - FFTime) * Flow / 60	
BA_Delay_NT	(CongTime - FFTime) * Flow / 60	
AB_VOCD_AM	V/C based on LOS D	
BA_VOCD_AM	V/C based on LOS D	
AB_VOCD_MD	V/C based on LOS D	
BA_VOCD_MD	V/C based on LOS D	
AB_VOCD_PM	V/C based on LOS D	
BA_VOCD_PM	V/C based on LOS D	
AB_VOCD_NT	V/C based on LOS D	
BA_VOCD_NT	V/C based on LOS D	
AB_Flow_res_da_Daily	Daily AB Flow_res_da	
BA_Flow_res_da_Daily	Daily BA Flow_res_da	
AB_Flow_res_sr2_Daily	Daily AB Flow_res_sr2	
BA_Flow_res_sr2_Daily	Daily BA Flow_res_sr2	
AB_Flow_res_sr3_Daily	Daily AB Flow_res_sr3	
BA_Flow_res_sr3_Daily	Daily BA Flow_res_sr3	
AB_Flow_CV_Daily	Daily AB Flow_CV	
BA_Flow_CV_Daily	Daily BA Flow_CV	
AB_Flow_SUT_Daily	Daily AB Flow_SUT	
BA_Flow_SUT_Daily	Daily BA Flow_SUT	
AB_Flow_MUT_Daily	Daily AB Flow_MUT	
BA_Flow_MUT_Daily	Daily BA Flow_MUT	
AB_Flow_IEEI_Daily	Daily AB Flow_IEEI	
BA_Flow_IEEI_Daily	Daily BA Flow_IEEI	
AB_Flow_NHBNR_Daily	Daily AB Flow_NHBNR	
BA_Flow_NHBNR_Daily	Daily BA Flow_NHBNR	
AB_Flow_EE_Daily	Daily AB Flow_EE	
BA_Flow_EE_Daily	Daily BA Flow_EE	
AB_Flow_Daily	Daily AB Flow	
BA_Flow_Daily	Daily BA Flow	
AB_VMT_Daily	Daily AB VMT	
BA_VMT_Daily	Daily BA VMT	
AB_VHT_Daily	Daily AB VHT	
BA_VHT_Daily	Daily BA VHT	
AB_Delay_Daily	Daily AB Delay	
BA_Delay_Daily	Daily BA Delay	

Values in blue are inputs required by the user to assure the model works, the values in green are calculated during the model operation.

Field Name	Description/Code	Additional Information
res_da_Flow_Daily	Daily res_da flow in both directions	
res_sr2_Flow_Daily	Daily res_sr2 flow in both directions	
res_sr3_Flow_Daily	Daily res_sr3 flow in both directions	
CV_Flow_Daily	Daily CV flow in both directions	
SUT_Flow_Daily	Daily SUT flow in both directions	
MUT_Flow_Daily	Daily MUT flow in both directions	
IEEI_Flow_Daily	Daily IEEI flow in both directions	
NHBNR_Flow_Daily	Daily NHBNR flow in both directions	
EE_Flow_Daily	Daily EE flow in both directions	
Flow_Daily	Daily flow in both directions	
VMT_Daily	Daily VMT in both directions	
VHT_Daily	Daily VHT in both directions	
Delay_Daily	Daily Delay in both directions	
AB_Speed_Daily	Slowest speed throughout day	
BA_Speed_Daily	Slowest speed throughout day	
AB_Time_Daily	Highest time throughout day	
BA_Time_Daily	Highest time throughout day	
AB_VOCE_Daily	Highest LOS E v/c throughout day	
BA_VOCE_Daily	Highest LOS E v/c throughout day	
AB_VOCD_Daily	Highest LOS D v/c throughout day	
BA_VOCD_Daily	Highest LOS D v/c throughout day	
NumCountLinks_All	Number of links with this count ID	
Count_All	Repeat of the count field	
Volume_All	Total Daily Link Flow	
diff_All	Volume - Count	
absdiff_All	abs(diff)	
pctdiff_All	diff / Count * 100	
MDD_All	Maximum Desirable Deviation	
ExceedMDD_All	If link exceeds MDD	
NumCountLinks_SUT	Number of links with this count ID	
Count_SUT	Repeat of the count field	
Volume_SUT	Total Daily Link Flow	
diff_SUT	Volume - Count	
absdiff_SUT	abs(diff)	
pctdiff_SUT	diff / Count * 100	
MDD_SUT	Maximum Desirable Deviation	
ExceedMDD_SUT	If link exceeds MDD	

Values in blue are inputs required by the user to assure the model works, the values in green are calculated during the model operation.

Field Name	Description/Code	Additional Information
NumCountLinks_MUT	Number of links with this count ID	
Count_MUT	Repeat of the count field	
Volume_MUT	Total Daily Link Flow	
diff_MUT	Volume - Count	
absdiff_MUT	abs(diff)	
pctdiff_MUT	diff / Count * 100	
MDD_MUT	Maximum Desirable Deviation	
ExceedMDD_MUT	If link exceeds MDD	
AB_TransitFlow	Link level transit flow by direction	
BA_TransitFlow	Link level transit flow by direction	
AB_NonTransitFlow	Link level non-transit flow by direction	
BA_NonTransitFlow	Link level non-transit flow by direction	
AB_Total_TransitFlow	Link level total transit flow by direction	
BA_Total_TransitFlow	Link level total transit flow by direction	
AB_Access_Walk_Flow	Access walking flow on link-used for transit project analysis	
BA_Access_Walk_Flow	Access walking flow on link-used for transit project analysis	
AB_Xfer_Walk_Flow	transfer walking flow on link-used for transit project analysis	
BA_Xfer_Walk_Flow	transfer walking flow on link-used for transit project analysis	
AB_Egress_Walk_Flow	Egress walking flow on link-used for transit project analysis	
BA_Egress_Walk_Flow	Egress walking flow on link-used for transit project analysis	
AB_Walk_Flow	Total walk flow on link(for transit)	
BA_Walk_Flow	Total walk flow on link(for transit)	
AB_Drive_Flow	Total drive flow on link (for transit)	
BA_Drive_Flow	Total drive flow on link (for transit)	

The fields containing information on the socio-economic data for the NRV model are including on the remaining pages.

Field Name	Description
ID	Transcad Provided Identifier
Households	Occupied housing units
HHPopulation	Population living in households
GQPopulation	Not used by model. Population living in group quarters. Left in th SE data to make the difference between HHPopulation and TotalPopulation explicit.
Workers	Employed persons
Vehicles	Vehicles owned by households
MedianIncome	Median household income
Industry	Industrial Employment NAICS 11, 21, 22, 23, 31-33, 42, 48, 49, 56
Retail	Retail Employment NAICS 44, 45
Service	Service Employment NAICS 61, 62, 71, 72, 81
Office	Office Employment NAICS 51, 52, 53, 54, 55, 56, 92
K12Enrollment	K-12 School Enrollment The number of students attending
VTEnrollOn	On-campus enrollment for VT
VTEnrollOff	Off-campus enrollment for VT This is used for HBU trip attractions, and is evenly distributed across the VT zones.
VTStudOff	Off-campus student households for VT
RadEnrollOn	On-campus enrollment for Radford
RadEnrollOff	Off-campus enrollment for Radford
RadStudOff	Off-campus student households for Radford
InternalZone	Is the zone considered inside the model boundary (non-external)
TotEmp	Total Employment Created during area type calculation
Density	Used to calculate initial AT
AreaType	Area type of the zone
ATSmoothed	Whether or not the area type was smoothed
avg_size	HHPopulation / Households Capped within min/max of marginal table
avg_work	Workers / Households Capped within min/max of marginal table
avg_veh	Vehicles / Households Capped within min/max of marginal table
inc_ratio	MedianIncome / 45607 (the regional median income) Capped within min/max of marginal table
siz1	number of 1 person HHs
siz2	number of 2 person HHs
siz3	number of 3 person HHs
siz4	number of 4 person HHs
wrk0	number of 0 worker HHs
wrk1	number of 1 worker HHs
wrk2	number of 2 worker HHs
wrk3	number of 3 worker HHs

NRV SE data fields con't

Field Name	Description
veh0	number of 0 vehicle HHs
veh1	number of 1 vehicle HHs
veh2	number of 2 vehicle HHs
veh3	number of 3 vehicle HHs
incH	number of High income HHs
incL	number of Low income HHs
incMH	Medium high income level
incML	Medium low income level
HBO_ihvi	HBO productions, market segment "income: high, vehicle: insufficient"
HBO_ihvs	HBO productions, market segment "income: high, vehicle: sufficient"
HBO_ilvi	HBO productions, market segment "income: low, vehicle: insufficient"
HBO_ilvs	HBO productions, market segment "income: high, vehicle: sufficient"
HBO_v0	HBO productions, market segment vehicle: zero
HBSC	HB School productions
HBSH	HB Shop productions
HBW_ihvi	HBW productions, market segment "income: high, vehicle: insufficient"
HBW_ihvs	HBW productions, market segment "income: high, vehicle: sufficient"
HBW_ilvi	HBW productions, market segment "income: low, vehicle: insufficient"
HBW_ilvs	HBW productions, market segment "income: high, vehicle: sufficient"
HBW_v0	HBW productions, market segment vehicle: zero
NHBO	NHBO productions
NHBW	NHBW productions
HBW_tot	HBW productions all markets
HBO_tot	HBO productions all markets
HBWa_v0	Generation Daily attractions See attr_rates.csv for rates
HBWa_ilvi	Generation Daily attractions See attr_rates.csv for rates
HBWa_ilvs	Generation Daily attractions See attr_rates.csv for rates
HBWa_ihvi	Generation Daily attractions See attr_rates.csv for rates
HBWa_ihvs	Generation Daily attractions See attr_rates.csv for rates
HBOa_v0	Generation Daily attractions See attr_rates.csv for rates
HBOa_ilvi	Generation Daily attractions See attr_rates.csv for rates
HBOa_ilvs	Generation Daily attractions See attr_rates.csv for rates
HBOa_ihvi	Generation Daily attractions See attr_rates.csv for rates
HBOa_ihvs	Generation Daily attractions See attr_rates.csv for rates
HBSHa	Generation Daily attractions See attr_rates.csv for rates
HBSCa	Generation Daily attractions See attr_rates.csv for rates

NRV SE data fields con't

Field Name	Description
NHBWa	Generation Daily attractions See attr_rates.csv for rates
NHBOa	Generation Daily attractions See attr_rates.csv for rates
HBUV	University productions made by VT students
HBUR	University productions made by Radford students
HBUVa	University attractions made by VT students
HBURa	University attractions made by Radford students
sHBW	HBW productions from on-campus students
sHBO	HBO productions from on-campus students
sNHB	NHB productions from on-campus students
sHBWa	HBW attractions from on-campus students
sHBOa	HBO attractions from on-campus students
sNHBa	NHB attractions from on-campus students
CV	Commercial Vehicles Productions = Attractions See cv_generation.csv for rates
SUT	Commercial Vehicles Productions = Attractions See cv_generation.csv for rates
MUT	Commercial Vehicles Productions = Attractions See cv_generation.csv for rates
CVa	Commercial Vehicles Productions = Attractions See cv_generation.csv for rates
SUTa	Commercial Vehicles Productions = Attractions See cv_generation.csv for rates
MUTa	Commercial Vehicles Productions = Attractions See cv_generation.csv for rates
AWDT	AWDT at external station
IEEI	IEEI productions AWDT at station - EE trips
IEEIa	IX attractions See ix_generation.csv for rates
CV_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
CV_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
CV_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
CV_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
CVa_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
CVa_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
CVa_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
CVa_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBO_ihvi_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBO_ihvi_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBO_ihvi_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBO_ihvi_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBO_ihvs_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBO_ihvs_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBO_ihvs_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBO_ihvs_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBO_ilvi_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates

NRV SE data fields con't

[illegible]

NRV SE data fields con't

[illegible]

NRV SE data fields con't

[illegible]

NRV SE data fields con't

Field Name	Description
NHBOa_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
NHBOa_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
NHBW_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
NHBW_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
NHBW_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
NHBW_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
NHBWa_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
NHBWa_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
NHBWa_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
NHBWa_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
SUT_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
SUT_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
SUT_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
SUT_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
SUTa_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
SUTa_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
SUTa_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
SUTa_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBUV_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBUV_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBUV_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBUV_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBUR_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBUR_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBUR_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBUR_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBUVa_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBUVa_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBUVa_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBUVa_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBURa_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBURa_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBURa_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
HBURa_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBO_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBO_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBO_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBO_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates

NRV SE data fields con't

Field Name	Description
sHBW_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBW_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBW_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBW_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sNHB_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sNHB_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sNHB_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sNHB_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBOa_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBOa_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBOa_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBOa_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBWa_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBWa_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBWa_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sHBWa_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sNHBa_AM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sNHBa_MD	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sNHBa_NT	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
sNHBa_PM	Time of Day Generation fields split into time period See time_of_day_factors.csv for rates
d_HBW_v0_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ilvi_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ilvs_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ihvi_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ihvs_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_NHB_all_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_v0_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ilvi_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ilvs_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ihvi_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ihvs_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBSC_all_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_IEEI_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_CV_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_SUT_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_MUT_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency

NRV SE data fields con't

Field Name	Description
d_HBWa_v0_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBWa_ilvi_AM	number of BWa_ilvi_AM person HHs
d_HBWa_ilvs_AM	number of BWa_ilvs_AM person HHs
d_HBWa_ihvi_AM	number of BWa_ihvi_AM person HHs
d_HBWa_ihvs_AM	number of BWa_ihvs_AM person HHs
d_NHBa_all_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_v0_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ilvi_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ilvs_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ihvi_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ihvs_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBSCa_all_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_IEEla_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_CVa_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_SUTa_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_MUTa_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBUV_all_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBUVa_all_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBUR_all_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBURa_all_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBW_all_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBWa_all_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBO_all_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBOa_all_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sNHB_all_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sNHBa_all_AM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
ieei_attr_AM	IEEI attractions after distribution
ei_attr_AM	EI attractions (a percent of IEEI attrs)
d_NHBNR_all_AM	NHB productions from non residents
d_NHBNRa_all_AM	NHB attractions from non residents
d_HBW_v0_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ilvi_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ilvs_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ihvi_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ihvs_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_NHB_all_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_v0_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ilvi_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency

NRV SE data fields con't

Field Name	Description
d_HBO_ilvs_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ihvi_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ihvs_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBSC_all_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_IEEI_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_CV_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_SUT_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_MUT_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBWa_v0_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBWa_ilvi_MD	number of BWa_ilvi_MD person HHs
d_HBWa_ilvs_MD	number of BWa_ilvs_MD person HHs
d_HBWa_ihvi_MD	number of BWa_ihvi_MD person HHs
d_HBWa_ihvs_MD	number of BWa_ihvs_MD person HHs
d_NHBa_all_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_v0_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ilvi_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ilvs_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ihvi_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ihvs_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBSCa_all_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_IEEIa_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_CVa_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_SUTa_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_MUTa_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBUV_all_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBUVa_all_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBUR_all_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBURa_all_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBW_all_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBWa_all_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBO_all_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBOa_all_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sNHB_all_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sNHBa_all_MD	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
ieei_attr_MD	IEEI attractions after distribution
ei_attr_MD	EI attractions (a percent of IEEI attr)
d_NHBNR_all_MD	NHB productions from non residents
d_NHBNRa_all_MD	NHB attractions from non residents

NRV SE data fields con't

Field Name	Description
d_HBW_ilvi_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ilvs_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ihvi_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ihvs_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_NHB_all_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_v0_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ilvi_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ilvs_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ihvi_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ihvs_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBSC_all_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_IEEI_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_CV_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_SUT_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_MUT_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBWa_v0_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBWa_ilvi_PM	number of BWa_ilvi_PM person HHs
d_HBWa_ilvs_PM	number of BWa_ilvs_PM person HHs
d_HBWa_ihvi_PM	number of BWa_ihvi_PM person HHs
d_HBWa_ihvs_PM	number of BWa_ihvs_PM person HHs
d_NHBa_all_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_v0_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ilvi_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ilvs_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ihvi_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ihvs_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBSCa_all_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_IEEIa_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_CVa_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_SUTa_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_MUTa_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBUV_all_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBUVa_all_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBUR_all_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBURa_all_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBW_all_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBWa_all_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBO_all_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency

NRV SE data fields con't

Field Name	Description
d_sNHB_all_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sNHBa_all_PM	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
ieei_attr_PM	IEEI attractions after distribution
ei_attr_PM	EI attractions (a percent of IEEI attrs)
d_NHBNR_all_PM	NHB productions from non residents
d_NHBNRa_all_PM	NHB attractions from non residents
d_HBW_v0_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ilvi_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ilvs_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ihvi_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBW_ihvs_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_NHB_all_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_v0_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ilvi_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ilvs_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ihvi_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBO_ihvs_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBSC_all_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_IEEI_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_CV_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_SUT_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_MUT_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBWa_v0_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBWa_ilvi_NT	number of BWa_ilvi_NT person HHs
d_HBWa_ilvs_NT	number of BWa_ilvs_NT person HHs
d_HBWa_ihvi_NT	number of BWa_ihvi_NT person HHs
d_HBWa_ihvs_NT	number of BWa_ihvs_NT person HHs
d_NHBa_all_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_v0_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ilvi_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ilvs_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ihvi_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBOa_ihvs_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBSCa_all_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_IEEIa_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_CVa_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_SUTa_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_MUTa_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency

NRV SE data fields con't

Field Name	Description
d_HBUVa_all_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBUR_all_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_HBURa_all_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBW_all_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBWa_all_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBO_all_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sHBOa_all_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sNHB_all_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
d_sNHBa_all_NT	Distribution Trips after purpose conversion See d_purp_conversion.csv for equivalency
ieei_attr_NT	IEEI attractions after distribution
ei_attr_NT	EI attractions (a percent of IEEI attrs)
d_NHBNR_all_NT	NHB productions from non residents
d_NHBNRa_all_NT	NHB attractions from non residents

Appendix C- Mode Choice Coefficients

purpose	segment	section	term	value
default	default	nest_coeff	auto	0.7
default	default	nest_coeff	transit	0.7
default	default	nest_coeff	nonmotorized	0.7
default	default	nest_coeff	sr	0.7
default	default	nest_coeff	walk_acc	0.7
default	default	nest_coeff	pnr_acc	0.7
default	default	nest_coeff	knr_acc	0.7
HBW	v0	asc	da	-999
HBW	v0	asc	sr2	0
HBW	v0	asc	sr3	-1.83471
HBW	v0	asc	walk	1.692374
HBW	v0	asc	bike	-0.111661
HBW	v0	asc	w_lb	-5.292398
HBW	v0	asc	pnr_lb	-5.292398
HBW	v0	asc	knr_lb	-5.292398
HBW	v0	asc	w_eb	-5.292398
HBW	v0	asc	pnr_eb	-5.292398
HBW	v0	asc	knr_eb	-5.292398
HBW	v0	asc	w_brt	0.1
HBW	v0	asc	pnr_brt	0.1
HBW	v0	asc	knr_brt	0.1
HBW	v0	asc	w_cr	0.1
HBW	v0	asc	pnr_cr	0.1
HBW	v0	asc	knr_cr	0.1
HBW	v0	coeff	auto_op_cost	-0.16315
HBW	v0	coeff	ivt_drive	-0.025
HBW	v0	coeff	nm_time1	-0.0625
HBW	v0	coeff	nm_time2	-0.09375
HBW	v0	coeff	fare	-0.423
HBW	v0	coeff	ivt_transit	-0.025
HBW	v0	coeff	dwelt_time	-0.025
HBW	v0	coeff	initial_wait1	-0.05625
HBW	v0	coeff	initial_wait2	-0.025
HBW	v0	coeff	xfer_wait_time	-0.0625
HBW	v0	coeff	access_walk_time	-0.0625
HBW	v0	coeff	egress_walk_time	-0.0625
HBW	v0	coeff	access_drive_time	-0.05625
HBW	v0	coeff	egress_drive_time	-0.05625
HBW	v0	coeff	xfer_walk_time	-0.0625

purpose	segment	section	term	value
HBW	ilvi	asc	da	0
HBW	ilvi	asc	sr2	-1.54361
HBW	ilvi	asc	sr3	-1.83471
HBW	ilvi	asc	walk	1.692374
HBW	ilvi	asc	bike	-0.111661
HBW	ilvi	asc	w_lb	-5.292398
HBW	ilvi	asc	pnr_lb	-5.292398
HBW	ilvi	asc	knr_lb	-5.292398
HBW	ilvi	asc	w_eb	-5.292398
HBW	ilvi	asc	pnr_eb	-5.292398
HBW	ilvi	asc	knr_eb	-5.292398
HBW	ilvi	asc	w_brt	0.1
HBW	ilvi	asc	pnr_brt	0.1
HBW	ilvi	asc	knr_brt	0.1
HBW	ilvi	asc	w_cr	0.1
HBW	ilvi	asc	pnr_cr	0.1
HBW	ilvi	asc	knr_cr	0.1
HBW	ilvi	coeff	auto_op_cost	-0.16315
HBW	ilvi	coeff	ivt_drive	-0.025
HBW	ilvi	coeff	nm_time1	-0.0625
HBW	ilvi	coeff	nm_time2	-0.09375
HBW	ilvi	coeff	fare	-0.228
HBW	ilvi	coeff	ivt_transit	-0.025
HBW	ilvi	coeff	dwell_time	-0.025
HBW	ilvi	coeff	initial_wait1	-0.05625
HBW	ilvi	coeff	initial_wait2	-0.025
HBW	ilvi	coeff	xfer_wait_time	-0.0625
HBW	ilvi	coeff	access_walk_time	-0.0625
HBW	ilvi	coeff	egress_walk_time	-0.0625
HBW	ilvi	coeff	access_drive_time	-0.05625
HBW	ilvi	coeff	egress_drive_time	-0.05625
HBW	ilvi	coeff	xfer_walk_time	-0.0625
HBW	ilvs	asc	da	0
HBW	ilvs	asc	sr2	-1.54361
HBW	ilvs	asc	sr3	-1.83471
HBW	ilvs	asc	walk	1.692374
HBW	ilvs	asc	bike	-0.111661
HBW	ilvs	asc	w_lb	-5.292398
HBW	ilvs	asc	pnr_lb	-5.292398
HBW	ilvs	asc	knr_lb	-5.292398
HBW	ilvs	asc	w_eb	-5.292398
HBW	ilvs	asc	pnr_eb	-5.292398

purpose	segment	section	term	value
HBW	ilvs	asc	knr_eb	-5.292398
HBW	ilvs	asc	w_brt	0.1
HBW	ilvs	asc	pnr_brt	0.1
HBW	ilvs	asc	knr_brt	0.1
HBW	ilvs	asc	w_cr	0.1
HBW	ilvs	asc	pnr_cr	0.1
HBW	ilvs	asc	knr_cr	0.1
HBW	ilvs	coeff	auto_op_cost	-0.16315
HBW	ilvs	coeff	ivt_drive	-0.025
HBW	ilvs	coeff	nm_time1	-0.0625
HBW	ilvs	coeff	nm_time2	-0.09375
HBW	ilvs	coeff	fare	-0.26
HBW	ilvs	coeff	ivt_transit	-0.025
HBW	ilvs	coeff	dwell_time	-0.025
HBW	ilvs	coeff	initial_wait1	-0.05625
HBW	ilvs	coeff	initial_wait2	-0.025
HBW	ilvs	coeff	xfer_wait_time	-0.0625
HBW	ilvs	coeff	access_walk_time	-0.0625
HBW	ilvs	coeff	egress_walk_time	-0.0625
HBW	ilvs	coeff	access_drive_time	-0.05625
HBW	ilvs	coeff	egress_drive_time	-0.05625
HBW	ilvs	coeff	xfer_walk_time	-0.0625
HBW	ihvi	asc	da	0
HBW	ihvi	asc	sr2	-1.726926
HBW	ihvi	asc	sr3	-2.147934
HBW	ihvi	asc	walk	0.435762
HBW	ihvi	asc	bike	-2.804738
HBW	ihvi	asc	w_lb	- 10.189919
HBW	ihvi	asc	pnr_lb	- 10.189919
HBW	ihvi	asc	knr_lb	- 10.189919
HBW	ihvi	asc	w_eb	- 10.189919
HBW	ihvi	asc	pnr_eb	- 10.189919
HBW	ihvi	asc	knr_eb	- 10.189919
HBW	ihvi	asc	w_brt	0.1
HBW	ihvi	asc	pnr_brt	0.1
HBW	ihvi	asc	knr_brt	0.1
HBW	ihvi	asc	w_cr	0.1

purpose	segment	section	term	value
HBW	ihvi	asc	pnr_cr	0.1
HBW	ihvi	asc	knr_cr	0.1
HBW	ihvi	coeff	auto_op_cost	-0.09023
HBW	ihvi	coeff	ivt_drive	-0.025
HBW	ihvi	coeff	nm_time1	-0.0625
HBW	ihvi	coeff	nm_time2	-0.09375
HBW	ihvi	coeff	fare	-0.076
HBW	ihvi	coeff	ivt_transit	-0.025
HBW	ihvi	coeff	dwel_time	-0.025
HBW	ihvi	coeff	initial_wait1	-0.05625
HBW	ihvi	coeff	initial_wait2	-0.025
HBW	ihvi	coeff	xfer_wait_time	-0.0625
HBW	ihvi	coeff	access_walk_time	-0.0625
HBW	ihvi	coeff	egress_walk_time	-0.0625
HBW	ihvi	coeff	access_drive_time	-0.05625
HBW	ihvi	coeff	egress_drive_time	-0.05625
HBW	ihvi	coeff	xfer_walk_time	-0.0625
HBW	ihvs	asc	da	0
HBW	ihvs	asc	sr2	-1.726926
HBW	ihvs	asc	sr3	-2.147934
HBW	ihvs	asc	walk	0.435762
HBW	ihvs	asc	bike	-2.804738
HBW	ihvs	asc	w_lb	- 10.189919
HBW	ihvs	asc	pnr_lb	- 10.189919
HBW	ihvs	asc	knr_lb	- 10.189919
HBW	ihvs	asc	w_eb	- 10.189919
HBW	ihvs	asc	pnr_eb	- 10.189919
HBW	ihvs	asc	knr_eb	- 10.189919
HBW	ihvs	asc	w_brt	0.1
HBW	ihvs	asc	pnr_brt	0.1
HBW	ihvs	asc	knr_brt	0.1
HBW	ihvs	asc	w_cr	0.1
HBW	ihvs	asc	pnr_cr	0.1
HBW	ihvs	asc	knr_cr	0.1
HBW	ihvs	coeff	auto_op_cost	-0.09023
HBW	ihvs	coeff	ivt_drive	-0.025
HBW	ihvs	coeff	nm_time1	-0.0625

purpose	segment	section	term	value
HBW	ihvs	coeff	nm_time2	-0.09375
HBW	ihvs	coeff	fare	-0.087
HBW	ihvs	coeff	ivt_transit	-0.025
HBW	ihvs	coeff	dwel_time	-0.025
HBW	ihvs	coeff	initial_wait1	-0.05625
HBW	ihvs	coeff	initial_wait2	-0.025
HBW	ihvs	coeff	xfer_wait_time	-0.0625
HBW	ihvs	coeff	access_walk_time	-0.0625
HBW	ihvs	coeff	egress_walk_time	-0.0625
HBW	ihvs	coeff	access_drive_time	-0.05625
HBW	ihvs	coeff	egress_drive_time	-0.05625
HBW	ihvs	coeff	xfer_walk_time	-0.0625
HBO	v0	asc	da	-999
HBO	v0	asc	sr2	0
HBO	v0	asc	sr3	-0.435021
HBO	v0	asc	walk	1.533725
HBO	v0	asc	bike	-1.159122
HBO	v0	asc	w_lb	-4.478728
HBO	v0	asc	pnr_lb	-4.478728
HBO	v0	asc	knr_lb	-4.478728
HBO	v0	asc	w_eb	-4.478728
HBO	v0	asc	pnr_eb	-4.478728
HBO	v0	asc	knr_eb	-4.478728
HBO	v0	asc	w_brt	0.1
HBO	v0	asc	pnr_brt	0.1
HBO	v0	asc	knr_brt	0.1
HBO	v0	asc	w_cr	0.1
HBO	v0	asc	pnr_cr	0.1
HBO	v0	asc	knr_cr	0.1
HBO	v0	coeff	auto_op_cost	-0.24472
HBO	v0	coeff	ivt_drive	-0.015
HBO	v0	coeff	nm_time1	-0.0625
HBO	v0	coeff	nm_time2	-0.09375
HBO	v0	coeff	fare	-0.234
HBO	v0	coeff	ivt_transit	-0.025
HBO	v0	coeff	dwel_time	-0.025
HBO	v0	coeff	initial_wait1	-0.05625
HBO	v0	coeff	initial_wait2	-0.025
HBO	v0	coeff	xfer_wait_time	-0.0625
HBO	v0	coeff	access_walk_time	-0.0625
HBO	v0	coeff	egress_walk_time	-0.0625
HBO	v0	coeff	access_drive_time	-0.05625

purpose	segment	section	term	value
HBO	v0	coeff	egress_drive_time	-0.05625
HBO	v0	coeff	xfer_walk_time	-0.0625
HBO	ilvi	asc	da	0
HBO	ilvi	asc	sr2	-0.041662
HBO	ilvi	asc	sr3	-0.435021
HBO	ilvi	asc	walk	1.533725
HBO	ilvi	asc	bike	-1.159122
HBO	ilvi	asc	w_lb	-4.478728
HBO	ilvi	asc	pnr_lb	-4.478728
HBO	ilvi	asc	knr_lb	-4.478728
HBO	ilvi	asc	w_eb	-4.478728
HBO	ilvi	asc	pnr_eb	-4.478728
HBO	ilvi	asc	knr_eb	-4.478728
HBO	ilvi	asc	w_brt	0.1
HBO	ilvi	asc	pnr_brt	0.1
HBO	ilvi	asc	knr_brt	0.1
HBO	ilvi	asc	w_cr	0.1
HBO	ilvi	asc	pnr_cr	0.1
HBO	ilvi	asc	knr_cr	0.1
HBO	ilvi	coeff	auto_op_cost	-0.24472
HBO	ilvi	coeff	ivt_drive	-0.015
HBO	ilvi	coeff	nm_time1	-0.0375
HBO	ilvi	coeff	nm_time2	-0.05625
HBO	ilvi	coeff	fare	-0.234
HBO	ilvi	coeff	ivt_transit	-0.015
HBO	ilvi	coeff	dwell_time	-0.015
HBO	ilvi	coeff	initial_wait1	-0.03375
HBO	ilvi	coeff	initial_wait2	-0.015
HBO	ilvi	coeff	xfer_wait_time	-0.0375
HBO	ilvi	coeff	access_walk_time	-0.0375
HBO	ilvi	coeff	egress_walk_time	-0.0375
HBO	ilvi	coeff	access_drive_time	-0.03375
HBO	ilvi	coeff	egress_drive_time	-0.03375
HBO	ilvi	coeff	xfer_walk_time	-0.0375
HBO	ilvs	asc	da	0
HBO	ilvs	asc	sr2	-0.041662
HBO	ilvs	asc	sr3	-0.435021
HBO	ilvs	asc	walk	1.533725
HBO	ilvs	asc	bike	-1.159122
HBO	ilvs	asc	w_lb	-4.478728
HBO	ilvs	asc	pnr_lb	-4.478728
HBO	ilvs	asc	knr_lb	-4.478728

purpose	segment	section	term	value
HBO	ilvs	asc	w_eb	-4.478728
HBO	ilvs	asc	pnr_eb	-4.478728
HBO	ilvs	asc	knr_eb	-4.478728
HBO	ilvs	asc	w_brt	0.1
HBO	ilvs	asc	pnr_brt	0.1
HBO	ilvs	asc	knr_brt	0.1
HBO	ilvs	asc	w_cr	0.1
HBO	ilvs	asc	pnr_cr	0.1
HBO	ilvs	asc	knr_cr	0.1
HBO	ilvs	coeff	auto_op_cost	-0.24472
HBO	ilvs	coeff	ivt_drive	-0.015
HBO	ilvs	coeff	nm_time1	-0.0375
HBO	ilvs	coeff	nm_time2	-0.05625
HBO	ilvs	coeff	fare	-0.234
HBO	ilvs	coeff	ivt_transit	-0.015
HBO	ilvs	coeff	dwell_time	-0.015
HBO	ilvs	coeff	initial_wait1	-0.03375
HBO	ilvs	coeff	initial_wait2	-0.015
HBO	ilvs	coeff	xfer_wait_time	-0.0375
HBO	ilvs	coeff	access_walk_time	-0.0375
HBO	ilvs	coeff	egress_walk_time	-0.0375
HBO	ilvs	coeff	access_drive_time	-0.03375
HBO	ilvs	coeff	egress_drive_time	-0.03375
HBO	ilvs	coeff	xfer_walk_time	-0.0375
HBO	ihvi	asc	da	0
HBO	ihvi	asc	sr2	-0.521704
HBO	ihvi	asc	sr3	-0.813913
HBO	ihvi	asc	walk	-0.240136
HBO	ihvi	asc	bike	-3.667266
HBO	ihvi	asc	w_lb	-9.495654
HBO	ihvi	asc	pnr_lb	-9.495654
HBO	ihvi	asc	knr_lb	-9.495654
HBO	ihvi	asc	w_eb	-9.495654
HBO	ihvi	asc	pnr_eb	-9.495654
HBO	ihvi	asc	knr_eb	-9.495654
HBO	ihvi	asc	w_brt	0.1
HBO	ihvi	asc	pnr_brt	0.1
HBO	ihvi	asc	knr_brt	0.1
HBO	ihvi	asc	w_cr	0.1
HBO	ihvi	asc	pnr_cr	0.1
HBO	ihvi	asc	knr_cr	0.1
HBO	ihvi	coeff	auto_op_cost	-0.13535

purpose	segment	section	term	value
HBO	ihvi	coeff	ivt_drive	-0.015
HBO	ihvi	coeff	nm_time1	-0.0375
HBO	ihvi	coeff	nm_time2	-0.05625
HBO	ihvi	coeff	fare	-0.234
HBO	ihvi	coeff	ivt_transit	-0.015
HBO	ihvi	coeff	dwell_time	-0.015
HBO	ihvi	coeff	initial_wait1	-0.03375
HBO	ihvi	coeff	initial_wait2	-0.015
HBO	ihvi	coeff	xfer_wait_time	-0.0375
HBO	ihvi	coeff	access_walk_time	-0.0375
HBO	ihvi	coeff	egress_walk_time	-0.0375
HBO	ihvi	coeff	access_drive_time	-0.03375
HBO	ihvi	coeff	egress_drive_time	-0.03375
HBO	ihvi	coeff	xfer_walk_time	-0.0375
HBO	ihvs	asc	da	0
HBO	ihvs	asc	sr2	-0.521704
HBO	ihvs	asc	sr3	-0.813913
HBO	ihvs	asc	walk	-0.240136
HBO	ihvs	asc	bike	-3.667266
HBO	ihvs	asc	w_lb	-9.495654
HBO	ihvs	asc	pnr_lb	-9.495654
HBO	ihvs	asc	knr_lb	-9.495654
HBO	ihvs	asc	w_eb	-9.495654
HBO	ihvs	asc	pnr_eb	-9.495654
HBO	ihvs	asc	knr_eb	-9.495654
HBO	ihvs	asc	w_brt	0.1
HBO	ihvs	asc	pnr_brt	0.1
HBO	ihvs	asc	knr_brt	0.1
HBO	ihvs	asc	w_cr	0.1
HBO	ihvs	asc	pnr_cr	0.1
HBO	ihvs	asc	knr_cr	0.1
HBO	ihvs	coeff	auto_op_cost	-0.13535
HBO	ihvs	coeff	ivt_drive	-0.015
HBO	ihvs	coeff	nm_time1	-0.0375
HBO	ihvs	coeff	nm_time2	-0.05625
HBO	ihvs	coeff	fare	-0.234
HBO	ihvs	coeff	ivt_transit	-0.015
HBO	ihvs	coeff	dwell_time	-0.015
HBO	ihvs	coeff	initial_wait1	-0.03375
HBO	ihvs	coeff	initial_wait2	-0.015
HBO	ihvs	coeff	xfer_wait_time	-0.0375
HBO	ihvs	coeff	access_walk_time	-0.0375

purpose	segment	section	term	value
HBO	ihvs	coeff	egress_walk_time	-0.0375
HBO	ihvs	coeff	access_drive_time	-0.03375
HBO	ihvs	coeff	egress_drive_time	-0.03375
HBO	ihvs	coeff	xfer_walk_time	-0.0375
HBSC	all	asc	da	0
HBSC	all	asc	sr2	0.685801
HBSC	all	asc	sr3	0.648515
HBSC	all	asc	walk	1.025883
HBSC	all	asc	bike	0
HBSC	all	asc	w_lb	-2.098817
HBSC	all	asc	pnr_lb	-2.098817
HBSC	all	asc	knr_lb	-2.098817
HBSC	all	asc	w_eb	-2.098817
HBSC	all	asc	pnr_eb	-2.098817
HBSC	all	asc	knr_eb	-2.098817
HBSC	all	asc	w_brt	0.1
HBSC	all	asc	pnr_brt	0.1
HBSC	all	asc	knr_brt	0.1
HBSC	all	asc	w_cr	0.1
HBSC	all	asc	pnr_cr	0.1
HBSC	all	asc	knr_cr	0.1
HBSC	all	coeff	auto_op_cost	-0.24472
HBSC	all	coeff	ivt_drive	-0.015
HBSC	all	coeff	nm_time1	-0.0375
HBSC	all	coeff	nm_time2	-0.05625
HBSC	all	coeff	fare	-0.234
HBSC	all	coeff	ivt_transit	-0.015
HBSC	all	coeff	dwell_time	-0.015
HBSC	all	coeff	initial_wait1	-0.03375
HBSC	all	coeff	initial_wait2	-0.015
HBSC	all	coeff	xfer_wait_time	-0.0375
HBSC	all	coeff	access_walk_time	-0.0375
HBSC	all	coeff	egress_walk_time	-0.0375
HBSC	all	coeff	access_drive_time	-0.03375
HBSC	all	coeff	egress_drive_time	-0.03375
HBSC	all	coeff	xfer_walk_time	-0.0375
NHB	all	asc	da	0
NHB	all	asc	sr2	-0.517486
NHB	all	asc	sr3	-0.809642
NHB	all	asc	walk	-0.317759
NHB	all	asc	bike	-2.992526
NHB	all	asc	w_lb	-5.466817

purpose	segment	section	term	value
NHB	all	asc	pnr_lb	-5.466817
NHB	all	asc	knr_lb	-5.466817
NHB	all	asc	w_eb	-5.466817
NHB	all	asc	pnr_eb	-5.466817
NHB	all	asc	knr_eb	-5.466817
NHB	all	asc	w_brt	0.1
NHB	all	asc	pnr_brt	0.1
NHB	all	asc	knr_brt	0.1
NHB	all	asc	w_cr	0.1
NHB	all	asc	pnr_cr	0.1
NHB	all	asc	knr_cr	0.1
NHB	all	coeff	auto_op_cost	-0.1834
NHB	all	coeff	ivt_drive	-0.02
NHB	all	coeff	nm_time1	-0.05
NHB	all	coeff	nm_time2	-0.075
NHB	all	coeff	fare	-0.25
NHB	all	coeff	ivt_transit	-0.02
NHB	all	coeff	dwell_time	-0.02
NHB	all	coeff	initial_wait1	-0.043
NHB	all	coeff	initial_wait2	-0.02
NHB	all	coeff	xfer_wait_time	-0.05
NHB	all	coeff	access_walk_time	-0.05
NHB	all	coeff	egress_walk_time	-0.05
NHB	all	coeff	access_drive_time	-0.043
NHB	all	coeff	egress_drive_time	-0.043
NHB	all	coeff	xfer_walk_time	-0.05
HBUV	all	asc	da	0
HBUV	all	asc	sr2	-2.128502
HBUV	all	asc	sr3	-2.277431
HBUV	all	asc	walk	2.675274
HBUV	all	asc	bike	0.056401
HBUV	all	asc	w_lb	0.163659
HBUV	all	asc	pnr_lb	0.163659
HBUV	all	asc	knr_lb	0.163659
HBUV	all	asc	w_eb	0.163659
HBUV	all	asc	pnr_eb	0.163659
HBUV	all	asc	knr_eb	0.163659
HBUV	all	asc	w_brt	0.1
HBUV	all	asc	pnr_brt	0.1
HBUV	all	asc	knr_brt	0.1
HBUV	all	asc	w_cr	0.1
HBUV	all	asc	pnr_cr	0.1

purpose	segment	section	term	value
HBUV	all	asc	knr_cr	0.1
HBUV	all	coeff	auto_op_cost	-0.16315
HBUV	all	coeff	ivt_drive	-0.025
HBUV	all	coeff	nm_time1	-0.0625
HBUV	all	coeff	nm_time2	-0.09375
HBUV	all	coeff	fare	0
HBUV	all	coeff	ivt_transit	-0.025
HBUV	all	coeff	dwel_time	-0.025
HBUV	all	coeff	initial_wait1	-0.05625
HBUV	all	coeff	initial_wait2	-0.025
HBUV	all	coeff	xfer_wait_time	-0.0625
HBUV	all	coeff	access_walk_time	-0.0625
HBUV	all	coeff	egress_walk_time	-0.0625
HBUV	all	coeff	access_drive_time	-0.05625
HBUV	all	coeff	egress_drive_time	-0.05625
HBUV	all	coeff	xfer_walk_time	-0.0625
HBUR	all	asc	da	0
HBUR	all	asc	sr2	-2.121342
HBUR	all	asc	sr3	-2.259954
HBUR	all	asc	walk	1.084336
HBUR	all	asc	bike	-0.843424
HBUR	all	asc	w_lb	1.084727
HBUR	all	asc	pnr_lb	1.084727
HBUR	all	asc	knr_lb	1.084727
HBUR	all	asc	w_eb	1.084727
HBUR	all	asc	pnr_eb	1.084727
HBUR	all	asc	knr_eb	1.084727
HBUR	all	asc	w_brt	0.1
HBUR	all	asc	pnr_brt	0.1
HBUR	all	asc	knr_brt	0.1
HBUR	all	asc	w_cr	0.1
HBUR	all	asc	pnr_cr	0.1
HBUR	all	asc	knr_cr	0.1
HBUR	all	coeff	auto_op_cost	-0.16315
HBUR	all	coeff	ivt_drive	-0.025
HBUR	all	coeff	nm_time1	-0.0625
HBUR	all	coeff	nm_time2	-0.09375
HBUR	all	coeff	fare	0
HBUR	all	coeff	ivt_transit	-0.025
HBUR	all	coeff	dwel_time	-0.025
HBUR	all	coeff	initial_wait1	-0.05625
HBUR	all	coeff	initial_wait2	-0.025

purpose	segment	section	term	value
HBUR	all	coeff	xfer_wait_time	-0.0625
HBUR	all	coeff	access_walk_time	-0.0625
HBUR	all	coeff	egress_walk_time	-0.0625
HBUR	all	coeff	access_drive_time	-0.05625
HBUR	all	coeff	egress_drive_time	-0.05625
HBUR	all	coeff	xfer_walk_time	-0.0625
sHBW	all	asc	da	0
sHBW	all	asc	sr2	-1.047252
sHBW	all	asc	sr3	-0.959819
sHBW	all	asc	walk	9.893148
sHBW	all	asc	bike	2.894663
sHBW	all	asc	w_lb	-0.205358
sHBW	all	asc	pnr_lb	-0.205358
sHBW	all	asc	knr_lb	-0.205358
sHBW	all	asc	w_eb	-0.205358
sHBW	all	asc	pnr_eb	-0.205358
sHBW	all	asc	knr_eb	-0.205358
sHBW	all	asc	w_brt	0.1
sHBW	all	asc	pnr_brt	0.1
sHBW	all	asc	knr_brt	0.1
sHBW	all	asc	w_cr	0.1
sHBW	all	asc	pnr_cr	0.1
sHBW	all	asc	knr_cr	0.1
sHBW	all	coeff	auto_op_cost	-0.16315
sHBW	all	coeff	ivt_drive	-0.025
sHBW	all	coeff	nm_time1	-0.0625
sHBW	all	coeff	nm_time2	-0.09375
sHBW	all	coeff	fare	0
sHBW	all	coeff	ivt_transit	-0.025
sHBW	all	coeff	dwell_time	-0.025
sHBW	all	coeff	initial_wait1	-0.05625
sHBW	all	coeff	initial_wait2	-0.025
sHBW	all	coeff	xfer_wait_time	-0.0625
sHBW	all	coeff	access_walk_time	-0.0625
sHBW	all	coeff	egress_walk_time	-0.0625
sHBW	all	coeff	access_drive_time	-0.05625
sHBW	all	coeff	egress_drive_time	-0.05625
sHBW	all	coeff	xfer_walk_time	-0.0625
sHBO	all	asc	da	0
sHBO	all	asc	sr2	-1.047252
sHBO	all	asc	sr3	-0.959819
sHBO	all	asc	walk	9.893148

purpose	segment	section	term	value
sHBO	all	asc	bike	2.894663
sHBO	all	asc	w_lb	-0.205358
sHBO	all	asc	pnr_lb	-0.205358
sHBO	all	asc	knr_lb	-0.205358
sHBO	all	asc	w_eb	-0.205358
sHBO	all	asc	pnr_eb	-0.205358
sHBO	all	asc	knr_eb	-0.205358
sHBO	all	asc	w_brt	0.1
sHBO	all	asc	pnr_brt	0.1
sHBO	all	asc	knr_brt	0.1
sHBO	all	asc	w_cr	0.1
sHBO	all	asc	pnr_cr	0.1
sHBO	all	asc	knr_cr	0.1
sHBO	all	coeff	auto_op_cost	-0.24472
sHBO	all	coeff	ivt_drive	-0.015
sHBO	all	coeff	nm_time1	-0.0375
sHBO	all	coeff	nm_time2	-0.05625
sHBO	all	coeff	fare	0
sHBO	all	coeff	ivt_transit	-0.015
sHBO	all	coeff	dwell_time	-0.015
sHBO	all	coeff	initial_wait1	-0.03375
sHBO	all	coeff	initial_wait2	-0.015
sHBO	all	coeff	xfer_wait_time	-0.0375
sHBO	all	coeff	access_walk_time	-0.0375
sHBO	all	coeff	egress_walk_time	-0.0375
sHBO	all	coeff	access_drive_time	-0.03375
sHBO	all	coeff	egress_drive_time	-0.03375
sHBO	all	coeff	xfer_walk_time	-0.0375
sNHB	all	asc	da	0
sNHB	all	asc	sr2	-1.153713
sNHB	all	asc	sr3	-1.50426
sNHB	all	asc	walk	11.207421
sNHB	all	asc	bike	4.428222
sNHB	all	asc	w_lb	-0.243974
sNHB	all	asc	pnr_lb	-0.243974
sNHB	all	asc	knr_lb	-0.243974
sNHB	all	asc	w_eb	-0.243974
sNHB	all	asc	pnr_eb	-0.243974
sNHB	all	asc	knr_eb	-0.243974
sNHB	all	asc	w_brt	0.1
sNHB	all	asc	pnr_brt	0.1
sNHB	all	asc	knr_brt	0.1

purpose	segment	section	term	value
sNHB	all	asc	w_cr	0.1
sNHB	all	asc	pnr_cr	0.1
sNHB	all	asc	knr_cr	0.1
sNHB	all	coeff	auto_op_cost	-0.1834
sNHB	all	coeff	ivt_drive	-0.02
sNHB	all	coeff	nm_time1	-0.05
sNHB	all	coeff	nm_time2	-0.075
sNHB	all	coeff	fare	0
sNHB	all	coeff	ivt_transit	-0.02
sNHB	all	coeff	dwell_time	-0.02
sNHB	all	coeff	initial_wait1	-0.043
sNHB	all	coeff	initial_wait2	-0.02
sNHB	all	coeff	xfer_wait_time	-0.05
sNHB	all	coeff	access_walk_time	-0.05
sNHB	all	coeff	egress_walk_time	-0.05
sNHB	all	coeff	access_drive_time	-0.043
sNHB	all	coeff	egress_drive_time	-0.043
sNHB	all	coeff	xfer_walk_time	-0.05